# BRD: AI-Powered Indian Stock Market Intelligence Platform

Oct 11, 2026 · @lovely

## Document control

| Field | Value |
|---|---|
| Document | Business Requirements Document (BRD) |
| Product (working name) | Bharat Equity Intelligence Platform ("the Platform") |
| Version | 1.0, consolidated draft: six-pillar architecture, GitHub Actions execution, Gmail delivery, persistent CSV store |
| Status | Draft. Not approved for build until the decisions in section 23 are closed |
| Audience | Product, business analysis, engineering, quantitative research, data engineering, QA, compliance and legal |
| Regulatory research cut-off | 11 Oct 2026. Every regulatory claim cites its source; anything not cited is a design proposal |

**How to read this document.** Each statement is tagged with one of three labels so readers can tell evidence from intent:

- **[F] Verified fact**: a current rule or market fact, with a citation to an official or authoritative source.
- **[A] Business assumption**: something we believe to be true for planning, which must be confirmed. Assumptions are numbered A-01, A-02 and so on in section 5.
- **[D] Design decision**: a proposal from this BRD that the business may change. Decisions still open are listed as OD-01, OD-02 and so on in section 23.

Requirement IDs: FR-xxx (functional), NFR-xxx (non-functional), REG-xxx (regulatory control), TST-xxx (test), KPI-xx. The traceability matrix in Appendix D links business objectives to requirements, tests and KPIs.

**Nothing in this document is investment advice.** All prices, company names and figures in worked examples are hypothetical and labelled as such, unless a source is cited.

## Change log

Version 1.0 consolidates three inputs from the sponsor into this single document: the original BRD brief, the mandatory persistent-storage requirement, and the six-pillar consolidation specification (both 11 Oct 2026). No separate documents exist.

| Change | Sections affected | Effect on earlier content | Needs business approval? |
|---|---|---|---|
| Architecture reorganised into six mandatory pillars | 15, 8 (pillar mapping), D | Earlier layered architecture kept as component detail under the pillars | No (sponsor-mandated) |
| Initial deployment is a GitHub Actions pipeline; no cloud account or always-on server | 5, 15.5, 17, 22 | The 12 interactive screens and API move to a future web interface (Phase 4); report and email become the primary delivery | Yes: confirm deferral of the web interface (OD-21) |
| Three scheduled weekday runs and on-demand runs | 15.5, FR-150 to FR-159 | New | No |
| Research report emailed via Gmail to a configurable recipient | 15.6, 16.4, FR-160 to FR-164 | New | Yes: recipient list affects regulatory status (OD-22, 18.6) |
| Persistent CSV store, incremental updates, persisted across runners via a private Git data branch | 14.4, 14.4.9, FR-130 to FR-141 | Replaces the earlier warehouse as system of record | No |

| Change | Sections affected | Effect on earlier content | Needs business approval? |
|---|---|---|---|
| Model selection policy widened to evaluate gradient boosting, LSTM/GRU and Transformers when justified | 10.4, 10.9, FR-180 to FR-184 | Replaces "deep learning excluded from MVP" with "evaluated, deployed only if it passes the gate" | No |
| Engineering principles: no hardcoding, no mock data in production, no unverified claims | 5.6, 19.5 | Test fixtures limited to isolated tests | No |
| Regulatory analysis extended to emailed reports and personal versus shared use | 18.6 | Adds a lower-risk personal-use route | Yes (OD-01, OD-22) |
| New NFRs, tests, KPIs, risks, decisions and blockers | 17, 19, 20, 21, 23 | Additions only | No |

Verification status: this is a requirements document. Nothing described here has been implemented or tested yet; section 19.5 tracks test execution status.

# 1. Executive summary

We should build the Platform as a deterministic, source-cited research engine with an AI explanation layer on top, and we should not launch stock-specific price targets, stop-losses or "stocks to research today" lists to the public until the business holds a SEBI Research Analyst (RA) registration or legal counsel confirms another compliant route.

**What the Platform does.** It helps Indian retail and professional users research NSE and BSE listed companies through six analytical engines: fundamentals, technicals, market and macro context, news and corporate events, valuation and scenario forecasting, and risk (stop-loss, position sizing, options payoffs). Every number comes from deterministic code over licensed data. The language model explains, summarises and retrieves evidence; it never produces a number that the calculation layer did not produce.

**Why the regulatory route decides the product.** Three current rules shape what can ship:

- [F] Under the SEBI (Research Analysts) Regulations as amended on 16 Dec 2024, a research analyst is a person providing research services for consideration, recommendations must be corroborated by data and analysis, and clients must be told the extent of AI tool use (SEBI RA Regulations, amended 16 Dec 2024; SCC Online summary).
- [F] SEBI's RA FAQs of 23 Jul 2025 say research based on any methodology, including technical analysis, is not exempt, and that non-paying users still count as clients because consideration includes non-cash benefits (LKS summary of the FAQs).
- [F] An unregistered person engaged solely in education may not give future price indications and, from 1 Jul 2026, may use market price data only with a 30-day lag (SEBI circular of 8 May 2026, via TaxGuru; finfluencer FAQs of 29 Jan 2025, via Fox Mandal).

The business objective (forecasts, targets, stop-losses on named stocks with live prices) therefore cannot be met under an "education only" model. Section 18 sets out the options; decision OD-01 in section 23 must be closed before build of the forecasting and risk modules.

**MVP.** [D] NIFTY 100 cash equities (about 100 stocks), end-of-day plus licensed delayed intraday prices, five years of fundamentals, the fundamental scorecard, technical summary, valuation ranges with insufficient-data handling, a generic stop-loss and position-size calculator, an options payoff calculator for index options, the glossary in English and Hindi, full source citation, and the mandatory persistent CSV historical store with incremental updates (section 14.4). Forecast scenarios ship to an internal beta first, gated on the backtest acceptance criteria in section 19.

**How it runs (v1.0).** [D, sponsor-mandated] The first deployment is a Python research pipeline that runs on GitHub Actions three times each weekday (09:00, 13:00 and 16:00 IST) and on demand, with no cloud account and no always-on server. Each run loads the persistent CSV history from a private Git data branch, fetches only new or missing records, recomputes affected features, produces the research report, emails it through Gmail to a configured recipient, and commits the updated data and run records. The system is organised into six pillars: data ingestion, persistent storage and data quality, quantitative analytics, research and reporting, orchestration and delivery, and configuration, security and observability (section 15). An interactive web interface is deferred to a later phase.

**Biggest risks.** Regulatory classification (R-01), market data licensing cost and redistribution terms (R-02), model overconfidence presented as prediction (R-03), and LLM hallucination of figures (R-04). Mitigations are in section 21.

**Decisions needed now.** The regulatory operating model (OD-01), the licensed data vendor (OD-02), the monetisation model (OD-03), whether the MVP includes derivatives analytics beyond education (OD-05), which GitHub repository hosts the system (OD-28), and who may receive the emailed reports (OD-22). All 28 open decisions and the implementation blockers are in section 23.

## 2. Business problem and opportunity

Indian investors can get prices for free but cannot easily get research that is current, sourced, explained and honest about uncertainty; the opportunity is a research product whose every claim can be traced to a filing, a formula and a timestamp.

**Problems observed** (each is an [A] assumption until validated by user research in Phase 0):

| # | Problem | Who feels it | Consequence today |
|---|---|---|---|
| P1 | Research is scattered across exchange filings, annual reports, broker notes, news and social media | Retail investors, students | Hours per stock; decisions made on partial information |
| P2 | Unregistered tips and price targets with no stated assumptions or track record | Retail traders | Losses; no way to judge credibility. SEBI has acted on this through the finfluencer framework and PaRRVA (section 18) |
| P3 | Technical indicators shown as buy/sell signals without stated limits | Retail traders | False confidence, poor risk control |
| P4 | Derivatives used without understanding payoff, margin and expiry risk | F&O traders | Large losses; SEBI tightened index derivatives rules in Oct 2024 for investor protection (Zerodha summary of the 1 Oct 2024 circular) |

| # | Problem | Who feels it | Consequence today |
|---|---|---|---|
| P5 | Financial jargon, often only in English | First-time investors, Hindi-first users | Misread reports, misused metrics |
| P6 | No public record of how accurate past forecasts were | All users | Users cannot tell skill from luck |

**Opportunity.** [D] Differentiate on four things competitors rarely combine: (1) every figure cited to source and timestamp, (2) explicit scenarios and invalidation conditions instead of single targets, (3) a published forecast-accuracy ledger compared with naive baselines, and (4) bilingual explanation of every term in context.

**What we will not claim.** No guaranteed returns, no single-point "target price will be reached" language, no "AI predicts" marketing. Forecasts are model outputs with intervals and stated assumptions.

## 3. Business objectives and measurable outcomes

Success is measured on two separate scoreboards: product outcomes (are users served and is the business viable) and analytical integrity (are numbers correct, sourced, fresh and honestly calibrated). Predictive skill is never inferred from engagement.

| ID | Objective | Measurable outcome | Target (provisional) | Basis for target |
|---|---|---|---|---|
| BO-01 | Make company research faster | Median time for a test user to answer 5 standard research questions on one stock | 50% less than a baseline of the same users using public sources | [A] Baseline measured in Phase 0 usability study |
| BO-02 | Every material claim is traceable | Share of report claims with a resolvable citation and timestamp | 100% for numeric claims; 98% for narrative claims | [D] Enforced by FR-060 citation gate; narrative tolerance covers editorial summaries |

| ID | Objective | Measurable outcome | Target (provisional) | Basis for target |
|---|---|---|---|---|
| BO-03 | Calculations are correct | Pass rate of the golden-test suite of formulas | 100% before every release | [D] Deterministic code, so anything below 100% is a defect |
| BO-04 | Data is fresh and disclosed | Share of market-sensitive outputs within the freshness SLA for their tier | 99.5% of trading minutes | [A] Depends on vendor SLA (OD-02) |
| BO-05 | Forecasts are honest | Empirical coverage of 80% prediction intervals on out-of-sample data | Between 75% and 85% | [D] Standard calibration tolerance; see section 19 |
| BO-06 | Forecasts add information | Forecast error versus no-change baseline, same horizon and universe | Report it either way; publish a forecast publicly only if it beats the baseline on out-of-sample data with statistical significance | [D] Section 10.6 |
| BO-07 | Users understand terms | Correct answers on a 10-item comprehension check after using glossary help | 20 percentage-point uplift versus control | [A] Phase 2 A/B test |
| BO-08 | Operate compliantly | Open regulatory findings at audit; complaints resolved inside the regulatory window | Zero open findings; 100% within window | [F] RA complaints must be resolved within 7 working days per the Jan 2025 guidelines (FinSec Law tracker) |

| ID | Objective | Measurable outcome | Target (provisional) | Basis for target |
|---|---|---|---|---|
| BO-09 | Viable unit economics | Fully loaded cost per completed research report | Set after OD-02 and OD-03 | [A] Data and LLM costs unknown until vendor selection |

Objectives added in v1.0 for automated operation:

| ID | Objective | Measurable outcome | Target (provisional) | Basis for target |
|---|---|---|---|---|
| BO-10 | Research runs on schedule without servers | Share of scheduled weekday runs that start and complete successfully | 98% of scheduled runs per month | [A] GitHub schedule delays and drops under load are possible (15.5) |
| BO-11 | Authorised users can run research on demand | Manual runs complete with the same pipeline and outputs as scheduled runs | 100% parity in tests | FR-151 |
| BO-12 | Reports reach the recipient | Successful runs whose report email was accepted by Gmail SMTP | 99% | Submission status only; delivery to the inbox is not verifiable by SMTP (FR-162) |
| BO-13 | Every run is auditable and reproducible | Runs with complete records of data snapshot, model versions, predictions and status | 100% | FR-155, NFR-097 |
| BO-14 | Minimal redundant downloading | Historical ranges fetched more than once outside the look-back window | 0 | TST-HIST-J |

Targets marked provisional are reviewed at the end of Phase 1, once real baselines exist.

## 4. Stakeholders and user personas

Five personas drive requirements; the MVP serves the first three, and every persona sees the same evidence, with depth adjusted by view settings rather than different numbers.

### Stakeholders

| Stakeholder | Interest | Responsibility in this BRD |
|---|---|---|
| Business sponsor / founder | Product-market fit, revenue, regulatory standing | Closes open decisions in section 23 |
| Product manager | Scope, prioritisation, roadmap | Owns backlog (section 22) |
| Compliance officer (and RA principal officer if registered) | SEBI obligations, disclosures, complaints, audit | Approves REG-xxx controls, sign-off gate before launch |
| External Indian securities counsel | Legal classification of the service | Written opinion on OD-01 before public launch |
| Quantitative research lead | Methodology, validation, model risk | Owns sections 9 to 12 and backtest sign-off |
| Data engineering lead | Ingestion, quality, licensing compliance | Owns sections 14 and 15 |
| Engineering lead | Architecture, NFRs, security | Owns sections 15 and 17 |
| QA lead | Test strategy, golden datasets | Owns section 19 |
| Data vendors and exchanges | Licence compliance, attribution, usage reporting | Contract counterparties |
| SEBI / RAASB (if registered) | Investor protection, inspections | External oversight |

### Personas

| Persona | Profile [A] | Primary jobs | Key needs | Risk if served badly |
|---|---|---|---|---|
| Asha, long-term retail investor | 32, salaried, Tier-2 city, invests monthly in large caps, Hindi and English | Shortlist quality companies; check if a stock looks expensive | Scorecard, valuation range, plain-language reasons, Hindi explanations | Treats a fair-value estimate as a promise |
| Rohan, active swing trader | 27, trades cash and index options weekly | Find setups, set stops, size positions, understand options payoff | Technical summary, ATR stops, position sizer, payoff diagram, expiry rules | Over-leverages; ignores gap risk |
| Meera, finance student / learner | 21, preparing for NISM exams | Learn terms and methods with real examples | Glossary, worked examples, formulas | Confuses correlation with causation |
| Vikram, independent analyst or registered RA | 40, writes research for clients | Faster first draft of research, peer comps, source pack | Exportable tables, citations, audit trail | Copies unverified AI text into client reports |
| Internal research desk (if RA route chosen) | Platform's own analysts | Review and approve published research | Approval workflow, versioning, conflict logs | Unreviewed model output reaches clients |

### User journeys (v1.0 delivery model)

| Journey | Actor | Steps | Outcome |
|---|---|---|---|
| J1 Scheduled report | Report recipient (Asha, Rohan or the sponsor) | 1. Run starts at 09:00, 13:00 or 16:00 IST. 2. Pipeline updates data and research. 3. Email arrives with summary, findings, risks, data timestamp and model version, plus the detailed HTML report and CSV summary | Reads current, cited research without opening any app |
| J2 On-demand research | Authorised repository user | 1. Opens the repository's Actions tab. 2. Chooses "Run workflow" with optional symbols, mode, horizons and format. 3. Receives the report email | Research for chosen stocks between scheduled runs |
| J3 Failure notice | Recipient and operator | 1. A run fails validation or a provider is down. 2. A clearly labelled failure email is sent where possible; no stale report is sent as current. 3. Operator reads the run summary and logs | Nobody acts on stale data |
| J4 Full refresh or recovery | Operator (admin) | 1. Runs the workflow with full-refresh selected and a reason. 2. Safeguards back up current data. 3. Run is logged and audited | Controlled rebuild of history |

| Journey | Actor | Steps | Outcome |
|---|---|---|---|
| J5 Model update | Quant lead | 1. Retraining runs on schedule or drift trigger. 2. Validation report produced. 3. Approved model version promoted in the registry | Only validated models are used |
| J6 Audit | Compliance | Reads run records, prediction ledger and data revision logs for any date | Any report can be traced and reproduced |

# 5. Scope, exclusions, and assumptions

The Platform is a research and analytics product for Indian listed equities and exchange-traded equity derivatives; it does not execute trades, manage money, or give personalised advice in any release covered by this BRD.

## 5.1 In scope

- NSE and BSE listed equity shares; index and stock derivatives on NSE and BSE where licensed data exists (section 6).
- Research engines: fundamental, technical, macro and sector, news and events, valuation and forecasting, risk, derivatives analytics.
- Glossary and learning centre in English and Hindi.
- Watchlists and user-entered portfolios for risk analytics (no broker linkage in MVP).
- Forecast ledger and accuracy reporting.
- Compliance tooling: disclosures, audit logs, complaint handling, research approval workflow.

## 5.2 Out of scope (all phases unless reopened)

- Order placement, broker integration for execution, algorithmic trading. [F] Retail algo trading runs through brokers under SEBI's framework of 4 Feb 2025, effective 1 Oct 2025 (Taxmann), which would add obligations this BRD does not cover.
- Portfolio management, mutual fund distribution, insurance, credit.

- Personalised investment advice based on a client's risk profile (Investment Adviser territory) unless OD-01 selects that route.
- Currency, commodity and debt derivatives; crypto assets; unlisted shares; SME platform stocks in MVP.
- Social features (forums, user-published tips), which create finfluencer-association risk.

## 5.3 Assumptions

| ID | Assumption | Impact if wrong | Validation |
|---|---|---|---|
| A-01 | A licensed vendor can supply NSE and BSE prices, corporate actions and fundamentals with redistribution rights for display to end users at acceptable cost | MVP blocked or margins negative | Vendor RFP in Phase 0 (OD-02) |
| A-02 | The business will register as an RA (or partner with one) before showing stock-specific targets or stops publicly | Forecast and risk modules limited to internal or education mode | Legal opinion (OD-01) |
| A-03 | Users accept delayed (not tick-level real-time) prices for research use | Real-time licence needed; higher cost | User research Phase 0 |
| A-04 | Hindi is the first additional language; others later | Rework of glossary schema | Product decision OD-09 |
| A-05 | Users access on mobile web first; native apps later | UI priorities | Analytics after launch |
| A-06 | An LLM provider is available with Indian or contractually acceptable data residency | Architecture change | Security and DPDP review (OD-12) |

| ID | Assumption | Impact if wrong | Validation |
|---|---|---|---|
| A-07 | Five years of quarterly fundamentals and ten years of daily prices are enough for MVP valuation and backtests | Weaker validation | Quant review |
| A-08 | Users enter portfolios manually or by CSV | Lower engagement than broker sync | Phase 3 decision |

## 5.4 Ambiguities and conflicts found in the request

| ID | Ambiguity or conflict | Why it matters | Proposed resolution |
|---|---|---|---|
| C-01 | "Which stocks deserve research today" plus 1 to 12 month price scenarios versus "educational information" | Ranked lists and price indications on named stocks are likely research services; education-only persons may not give future price indications | Treat as regulated research; gate behind OD-01 |
| C-02 | "Position size based on the user's risk budget" | Personalised sizing may be read as personalised advice | MVP calculator uses only user-entered numbers, makes no instrument suggestion, and is labelled a calculator; legal review |
| C-03 | "Latest verified market price" and "data timestamp" versus education-only use | Education-only users may use price data only with a 30-day lag from 1 Jul 2026 | Live or delayed prices only under the regulated route |
| C-04 | "Probabilistic forecasts" and "confidence level" | Can read as prediction certainty | Show calibrated intervals with historical coverage; confidence describes evidence quality, not probability of profit |
| C-05 | "Market sentiment using legally accessible data" | Social media scraping is often barred by terms | Use licensed news and sentiment feeds only; no scraping |

| ID | Ambiguity or conflict | Why it matters | Proposed resolution |
|---|---|---|---|
| C-06 | "Futures and options for eligible securities" | Eligibility, lot sizes and expiry days change by circular | Read contract master daily; never hardcode (section 12) |
| C-07 | "Management quality indicators" | Subjective; risk of defamation if unsupported | Use only disclosed, factual indicators (auditor changes, pledges, related-party transactions, penalties) with citations |
| C-08 | "Hindi explanation where appropriate" | Translation of regulated disclosures must be accurate | Disclosures translated by a professional, reviewed by compliance |

## 5.5 Scope changes in v1.0

- **Added to scope:** GitHub Actions scheduled and manual execution; Gmail delivery of HTML reports with an optional CSV summary; persistent CSV datasets on a private Git data branch; failure notifications; run audit records.
- **Deferred (needs approval, OD-21):** the interactive web interface (screens in 16.2), the public API, accounts and roles for end users. Until then, "authorised users" means repository collaborators with permission to run workflows.
- **Still out of scope:** cloud infrastructure (AWS, Azure, GCP), always-on servers, order execution.

New assumptions:

| ID | Assumption | Impact if wrong | Validation |
|---|---|---|---|
| A-09 | The data repository is private, because licensed data must not be published | Licence breach if public | OD-23 |
| A-10 | The sender Gmail account has 2-Step Verification and can create an App Password | Email delivery blocked; alternative needed | Sponsor confirms (D-11) |

| ID | Assumption | Impact if wrong | Validation |
|---|---|---|---|
| A-11 | The MVP dataset (NIFTY 100 daily, 10 years, plus fundamentals) stays well under GitHub's recommended repository size | Migration needed earlier | Size estimate after first download; NFR-104 |
| A-12 | Runs complete within the GitHub Actions job time limit and monthly minute allowance | Missed runs or charges | Measure in Phase 1 (NFR-101, NFR-107) |
| A-13 | The chosen data provider exposes an API usable from GitHub-hosted runners (no IP allow-listing, or allow-listing GitHub ranges) | Provider unusable from Actions | Vendor RFP (OD-02) |

New conflicts found and resolved:

| ID | Conflict | Resolution |
|---|---|---|
| C-09 | Twelve interactive screens versus "no continuously running server" | Screens deferred to Phase 4; v1.0 delivers HTML email reports (OD-21) |
| C-10 | 09:00 IST run is before the 09:15 market open; 16:00 IST run may precede final end-of-day files | The 09:00 report covers the previous session plus overnight filings and news; the 13:00 report uses delayed intraday data marked incomplete; the 16:00 report labels prices provisional unless the provider confirms the final close (FR-153) |
| C-11 | "No mock data" versus the earlier offer to build against a mock provider | Mocks and fixtures only in isolated automated tests; production runs and reports use the licensed provider or stop with a blocker |
| C-12 | Emailing named-stock forecasts versus SEBI research rules | Recipients limited to the sponsor's own use until OD-01 and OD-22 are closed (18.6) |

| ID | Conflict | Resolution |
|---|---|---|
| C-13 | Git storage for data versus licence terms and repository size | Private repository; licence must allow storage (OD-02); thresholds and migration path in 14.4.9 |
| C-14 | Earlier "deep learning excluded from MVP" versus the new policy to evaluate LSTM, GRU and Transformers | Evaluate them in the validation track; deploy only if they pass the same gate as other models (10.9) |

### 5.6 Mandatory engineering principles

1. No hardcoded market prices, financial statements, company metrics, news, predictions, email addresses, credentials or user-specific settings; these come from configuration files, environment variables, GitHub Actions secrets or validated external data.
2. No mock market data, fabricated values, invented citations or synthetic records in production datasets or reports. Test fixtures and mocks are allowed only in isolated test runs and are written outside `data/`.
3. No placeholder presented as a finished feature; incomplete features are labelled and disabled.
4. No full historical download on routine runs; no reliance on temporary runner storage for persistent data.
5. No prediction from invalid or insufficient data; no unsupported claims of accuracy or superiority.
6. No bypassing the six-pillar architecture and no extra pillars or infrastructure without documented justification and approval.
7. No claim that an integration or test works until it has been implemented and verified.
8. Fixed formulas, interface contracts and documented defaults are allowed when explicit, tested and version-controlled.

## 6. Indian market coverage and stock universe

The MVP covers NIFTY 100 cash equities plus NIFTY 50 index options for payoff education; coverage then widens to NIFTY 500, all liquid BSE-only stocks, and stock derivatives as data licences and validation allow.

## 6.1 Coverage tiers

| Tier | Universe | Instruments | Analytics enabled | Phase |
|---|---|---|---|---|
| T1 | NIFTY 100 (NIFTY 50 + NIFTY Next 50), which includes most SENSEX constituents | Cash equities; NIFTY 50 and SENSEX index levels | All modules; forecasts in internal beta | MVP (Phase 1) |
| T2 | NIFTY 500 and SENSEX, BSE 500 | Cash equities; sectoral indices | All modules; forecasts only where validation passes per stock | Phase 2 |
| T3 | Stocks in the F&O segment | Stock futures and options | Derivatives analyzer, OI, IV | Phase 2/3 |
| T4 | Remaining NSE and BSE mainboard stocks meeting liquidity floor | Cash equities | Fundamentals and technicals; valuation only if data complete; no forecasts | Phase 3 |
| T5 | ETFs | Equity ETFs | Price, tracking difference, iNAV where licensed; no fair value | Phase 3 |
| Excluded | SME platform, suspended, illiquid below floor | Any | Profile and filings only, with warning banner | All |

Index constituents are taken from the licensed index data feed on each rebalancing date, not hardcoded, and stored with effective dates so historical tests use the constituents that existed at the time (prevents survivorship bias).

## 6.2 Classification attributes stored per security

- Identifiers: ISIN (master key), NSE symbol, BSE scrip code, company CIN where available.
- Sector and industry: the index provider's industry classification as primary [D]; a mapping table for any vendor classification.

- Market-cap bucket: [D] use the AMFI half-yearly categorisation (large cap = top 100, mid cap = 101 to 250, small cap = 251 onward by full market cap), with the effective list date stored. [A] Confirm the current AMFI list source and licence during Phase 0.
- Liquidity: 20-day median daily traded value (INR) and median bid-ask spread where depth data is licensed.
- Volatility: 20-day and 250-day annualised historical volatility; beta to NIFTY 50 over 250 trading days.
- Status flags: active, suspended, delisted, under surveillance (exchange ASM/GSM lists where licensed), in F&O ban period.

## 6.3 Edge-case handling rules

| Situation | Rule |
|---|---|
| Dual listing (NSE and BSE) | One company record keyed by ISIN. Primary price venue = NSE for NSE-listed stocks [D]; BSE for BSE-only. Volumes reported per exchange and combined. Show which exchange each price comes from |
| Different symbols or share classes | Separate security records under one issuer (e.g. DVR shares, partly paid shares), never merged |
| Suspended security | Freeze last price with "suspended since" date; disable technical signals, forecasts and stop calculators; show the exchange notice |
| Delisted company | Keep full history for backtests (survivorship control); remove from screener defaults; label "delisted on DATE, reason" |
| Trading halt / circuit hit | Show circuit band status and halt flag; stop calculator warns that exits may not be possible at any price while locked in a circuit |
| Illiquid stock (below floor) | [D] Floor = 20-day median traded value below INR 1 crore (to be calibrated). Show liquidity warning; no forecasts; slippage assumptions widened |

| Situation | Rule |
|---|---|
| Corporate actions (split, bonus, rights, dividend, merger, demerger, buyback) | Maintain an adjustment-factor table; store both raw and adjusted prices; recompute indicators from adjusted series; show the event on charts; per-share fundamentals restated on the same basis |
| Mergers and demergers | Link predecessor and successor ISINs; history of the demerged entity starts on its listing date; no synthetic history unless clearly labelled |
| Name or symbol change | Symbol history table keyed by ISIN and effective date |
| New listing (IPO) | No technical indicators until enough bars exist (e.g. 200 days for 200-day SMA); no valuation until two reported quarters exist; label "limited history" |
| Missing or late results | Mark the period as missing; never interpolate fundamentals |

# 7. Current-state and proposed-state workflows

Today a careful retail investor spends hours stitching together sources with no audit trail; the proposed workflow gives one report in minutes where each figure links back to its source and each conclusion lists what would prove it wrong.

## 7.1 Current state (researching one stock) [A]

1. Hear a stock name from news, social media or a friend.
2. Look up price and chart on a broker app or free website.
3. Open exchange filings and the latest results PDF; read selectively.
4. Check a free fundamentals website for ratios; definitions and periods often unstated.
5. Search news; no distinction between confirmed, reported and rumoured items.
6. Watch videos or read posts with price targets of unknown basis.
7. Pick an entry and a stop by feel or a round percentage.
8. Buy or sell. No record of the reasoning, so no learning loop.

Pain points: time (P1), unverified claims (P2), signals without limits (P3), no risk sizing, and no feedback on whether the reasoning worked (P6).

## 7.2 Proposed state: research workflow

1. **Discover.** User opens the Market Overview or runs a screen (FR-010 to FR-014). Screen results show the criteria that matched, not a "buy" label.
2. **Profile.** User opens the stock page. Header shows price, exchange, data timestamp and freshness badge (FR-061).
3. **Fundamentals.** Scorecard with each metric, its period, its peer percentile and its source filing (FR-020 to FR-026).
4. **Technicals.** Trend, momentum, volatility and relative-strength summary with formulas viewable (FR-030 to FR-035).
5. **Context.** Sector and market context plus events timeline with confirmation status (FR-040 to FR-045).
6. **Valuation and scenarios.** Fair-value range by method, bull/base/bear scenarios with assumptions, or an explicit insufficient-data result (FR-050 to FR-057).
7. **Risk.** User enters an assumed entry and risk budget; the calculator shows stop options, position size, reward-to-risk and warnings (FR-070 to FR-076).
8. **Explain.** Any term or number can be tapped to open the glossary entry or the evidence panel (FR-090 to FR-094, FR-062).
9. **Track.** User saves to a watchlist and sets alerts on invalidation conditions (FR-100 to FR-104).
10. **Learn.** The forecast ledger later shows how the scenario played out against the baseline (FR-110 to FR-113).

## 7.3 Proposed state: published research workflow (RA route only)

1. Engines produce a draft report from data snapshot S at time T.
2. AI layer drafts the narrative using only facts in snapshot S; the citation gate rejects any sentence with an unsupported number (FR-060).
3. A registered analyst reviews, edits and approves; conflicts of interest are checked against the analyst's declared holdings (REG-06).
4. Report is versioned, timestamped, published with required disclosures, including the extent of AI use (REG-03).
5. Forecast entries are written to the immutable ledger at publication time (FR-110).

6. Corrections follow the material-error procedure (REG-10).

### 7.4 Proposed state: data pipeline (summary)

1. Vendor and official feeds land in a raw zone with receipt timestamps.
2. Validation rules run (section 14.3); failures raise incidents and mark affected outputs stale.
3. Normalised, adjusted series and point-in-time fundamentals are written to the warehouse.
4. Engines recompute on schedule or on event (results filed, corporate action).
5. Outputs carry the snapshot ID, model version and data timestamps through to the UI.

## 8. Detailed functional requirements

The Platform has 124 functional requirements in 13 groups, each owned by one of the six pillars; 93 are Must for MVP, and every requirement that shows a named-stock price target, stop level or ranking is additionally gated by REG-01 (regulatory route, OD-01).

**Conventions.** Priority uses MoSCoW for the MVP: Must, Should, Could, Won't (for MVP). "Gated" means the feature is built but released publicly only after the named decision or control is closed. Every requirement inherits the global rules below, so they are not repeated in each row:

- G-1: Every numeric output carries source ID, data timestamp (IST, ISO 8601) and calculation version (FR-061).
- G-2: Missing inputs produce an explicit "insufficient data" state with the missing field named; values are never imputed silently.
- G-3: All calculations run in the deterministic engine; the LLM may only quote engine outputs (FR-060).
- G-4: Every user-visible output is logged with snapshot ID for audit (FR-123).
- G-5: Data entitlement checks run before display (FR-125).

**Pillar ownership of requirement groups** (architecture in section 15)

| Pillar | Requirement groups |
|---|---|
| P1 Data ingestion and external integrations | FR-001, FR-042, FR-043, FR-080, FR-142 to FR-144 |

| Pillar | Requirement groups |
|---|---|
| P2 Persistent storage and data quality | FR-002, FR-004, FR-020, FR-024, FR-110, FR-123, FR-125, FR-130 to FR-141 |
| P3 Quantitative analytics and model selection | FR-010 to FR-014, FR-021 to FR-023, FR-025, FR-026, FR-030 to FR-036, FR-040, FR-041, FR-046, FR-050 to FR-058, FR-063, FR-070 to FR-076, FR-081 to FR-086, FR-111, FR-180 to FR-184 |
| P4 Research, interpretation and reporting | FR-044, FR-045, FR-060 to FR-062, FR-064, FR-087, FR-090 to FR-094, FR-112, FR-113, FR-120, FR-121, FR-124, FR-165 |
| P5 Orchestration, automation and delivery | FR-150 to FR-159, FR-160 to FR-164 |
| P6 Interface, configuration, security and observability | FR-003, FR-005, FR-100 to FR-104, FR-122, FR-126, FR-170 to FR-176 |

User-facing screen requirements (FR-003 accounts, FR-100 to FR-104 watchlists and portfolio, and the screen parts of other FRs) are deferred with the web interface (OD-21); their calculations still run in the pipeline and appear in reports.

## 8.1 Platform foundation and security master

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-001 Security master | One identity per instrument prevents duplicate or merged histories. As a data engineer, I need every NSE and BSE instrument keyed by ISIN | In: exchange security lists, vendor reference data. Proc: map ISIN to NSE symbol, BSE code, series, issuer, listing dates; keep symbol history with effective dates. Out: security master table and API | Vendor reference feed. Conflicting mappings go to a review queue; the instrument is hidden from analytics until resolved | Must | 100% of T1 ISINs map to exactly one issuer; symbol changes in the last 5 years resolve to the same ISIN in tests |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-002 Corporate action adjustment | Unadjusted series break indicators and returns. As a trader, I need charts that do not show false crashes on split dates | In: corporate actions (split, bonus, rights, dividends, mergers). Proc: compute cumulative adjustment factors (Appendix A.1); generate adjusted OHLCV and per-share fundamentals. Out: raw and adjusted series, event markers | Corporate action feed with ex-dates. Missing ratio or ex-date: mark series "adjustment pending", disable indicators for that stock | Must | Golden tests on at least 20 historical splits/bonuses reproduce reference adjusted prices within 0.01%; no false gap greater than 20% on ex-dates |
| FR-003 Accounts and roles | Different rights for users, analysts, compliance and admins. As compliance, I need to see who approved each report | In: sign-up, SSO. Proc: role-based access (user, pro user, analyst, compliance, admin). Out: sessions, permissions | Identity provider. Lockout after repeated failures | Must | Role matrix tests pass; privilege escalation tests fail as expected |
| FR-004 Coverage and status flags | Users must know when analytics are unreliable. As a user, I need to see if a stock is suspended, illiquid or newly listed | In: exchange status, liquidity metrics, listing date. Proc: apply rules in 6.3. Out: flags and banners; module enable/disable map | Status feeds. Unknown status = treat as restricted | Must | All 6.3 scenarios covered by tests; suspended stocks show no forecasts |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-005 Language preference | Hindi-first users (persona Asha). As a user, I want explanations in Hindi | In: user preference. Proc: serve translated UI strings and glossary text; numbers in Indian digit grouping (lakh, crore) with toggle. Out: localised UI | Translation memory. Missing translation falls back to English with a marker | Must | 100% of glossary MVP terms and disclosures available in Hindi, reviewed by a qualified translator |

## 8.2 Screening and discovery

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-010 Screener | Users need to narrow 100 to 500 stocks quickly. As Asha, I want to filter by ROE, debt and valuation | In: filter rules on any engine metric, universe, sector, market-cap bucket. Proc: evaluate on latest point-in-time snapshot. Out: result table with matched criteria and values | Engine outputs. Metric missing for a stock: excluded and counted in a "not evaluated" line | Must | Results match an independent SQL reference for 50 random filter sets; response under NFR-001 limits |
| FR-011 Preset screens | Novices need starting points. As Meera, I want a "low debt, consistent profit" screen | In: preset definition. Proc: same as FR-010. Out: results plus plain explanation of each criterion | As FR-010 | Should | Every preset shows its full rule set; no preset named with buy/sell language |
| FR-012 Match explanation | Users must see why a stock appears. As a user, I want to know which criteria matched | In: FR-010 result. Proc: per-criterion value and threshold. Out: explanation panel | None | Must | Every result row expandable to criteria values with sources |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-013 Saved screens and alerts | Repeat use. As Rohan, I want an alert when a stock newly passes my screen | In: saved screen. Proc: re-run after each EOD batch. Out: diff and alert | Notification service. Batch failure: no alert, incident raised | Should | Alerts fire within 30 minutes of EOD batch completion |
| FR-014 Daily research queue | Answers "which stocks deserve research today". As a user, I want a ranked list of stocks with notable changes | In: events, metric changes, price moves. Proc: rank by a transparent "notability" score (new results, large estimate changes, unusual volume), never by expected return. Out: list with reasons | All engines. Gated: REG-01 | Could (gated) | Each item lists its triggering facts; legal sign-off recorded before public release |

## 8.3 Fundamental research

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-020 Financial statements | Base for all fundamentals. As Asha, I want 5 years of P&L, balance sheet and cash flow | In: quarterly and annual results (standalone and consolidated). Proc: normalise line items to a standard chart of accounts; keep as-reported values and restatements with filing dates (point-in-time). Out: statements view and API | Fundamentals vendor plus exchange filings. Conflict between vendor and filing above 0.5%: flag, show filing value, raise data ticket | Must | 100 sampled line items match the filed PDF exactly; restatements retain the original value |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-021 Ratio engine | Standard metrics with stated formulas. As Meera, I want to see how ROCE is calculated | In: FR-020 data, prices. Proc: formulas in Appendix A.2 (growth, margins, EPS, P/E, P/B, ROE, ROCE, D/E, OCF, FCF, dividend yield, payout). Out: ratio values with period, basis (TTM, FY) and formula | FR-020. Negative or zero denominators return "not meaningful" with reason | Must | Golden test set of 30 companies x 15 ratios matches a reviewed spreadsheet to 4 decimal places |
| FR-022 Peer comparison | Metrics only mean something relative to peers. As a user, I want to see where a bank's ROE sits among banks | In: sector classification, ratios. Proc: peer set = same industry, with minimum 5 peers, else widen to sector and say so; compute percentiles and medians. Out: peer table, percentile bars | Classification feed. Fewer than 5 peers at sector level: show values, no percentile | Must | Peer set rules documented and test-covered; banks and NBFCs use sector-specific metrics (NIM, GNPA, CASA) not EBITDA |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-023 Trend detection | "Improving or deteriorating" must be defined. As Asha, I want to know if margins are getting better | In: 12 to 20 quarters of a metric. Proc: rule: improving if latest 4-quarter TTM value exceeds prior TTM by more than threshold and at least 3 of last 4 YoY changes are positive (Appendix A.3). Out: improving / stable / deteriorating with evidence | FR-020. Fewer than 8 quarters: "insufficient history" | Must | Labels reproduce on a hand-labelled test set of 100 company-metrics with at least 90% agreement with analyst labels |
| FR-024 Ownership | Promoter pledges and institutional flows are material risk signals | In: quarterly shareholding pattern filings. Proc: promoter %, pledged % of promoter holding, FPI, DII, public; quarter-on-quarter change. Out: ownership panel | Exchange shareholding filings via vendor. Missing quarter: gap shown, no interpolation | Must | Matches filed shareholding pattern for 50 sampled company-quarters |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-025 Governance and disclosure flags | Factual governance indicators only (C-07) | In: filings: auditor change or resignation, audit qualifications, related-party transactions above materiality, regulatory penalties, credit-rating actions. Proc: rule-based detection plus AI extraction verified against source text span. Out: flags with source excerpts | Filings text. Extraction confidence below threshold: queue for human review, not shown | Should | Precision at least 95% on a labelled set of 200 filings; every flag links to the source paragraph |
| FR-026 Fundamental scorecard | Summarise without hiding detail | In: FR-021 to FR-025. Proc: score pillars (growth, profitability, balance sheet, cash conversion, valuation, governance) by peer percentile; weights shown and documented. Out: pillar scores with drill-down; never a buy/sell label | All above. Pillar with more than 1 missing metric: shown as "incomplete" | Must | Each pillar score reproducible from displayed inputs; methodology page published |

## 8.4 Technical analysis

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-030 Charts | Visual price history. As Rohan, I want daily and weekly candles with volume | In: adjusted OHLCV; intraday bars where licensed. Proc: render with toggle raw/adjusted, corporate action markers. Out: interactive chart | Price feed. Stale data: chart shows "as of" time and stale badge | Must | Chart values equal stored series; markers on all corporate action dates |
| FR-031 Indicator engine | Standard indicators with fixed formulas | In: adjusted series, parameters. Proc: SMA, EMA, RSI (Wilder), MACD (12,26,9), ATR (Wilder 14), Bollinger, OBV; formulas in Appendix A.4. Out: indicator series | Price. Fewer bars than lookback: indicator absent, reason shown | Must | Matches reference library outputs (e.g. TA-Lib) within 1e-6 on golden series |
| FR-032 Support and resistance | Stops and targets need reference levels | In: daily highs/lows, volume. Proc: swing-point clustering (Appendix A.5) with touches counted. Out: up to 3 levels each side, with touch count and last-touch date | Price. Too little history: none shown | Should | Algorithm deterministic; levels reproducible; documented limitations shown |
| FR-033 Relative strength | Is the stock leading or lagging? | In: stock and benchmark adjusted total-return series. Proc: ratio and Mansfield RS vs NIFTY 50 and sector index over 63 and 252 days. Out: RS values and rank in universe | Index data. Benchmark missing: use NIFTY 50 and say so | Must | Calculations reproduce on golden set |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-034 Volatility and drawdown | Risk context | In: adjusted closes. Proc: annualised volatility (20, 60, 250 days), max drawdown and duration, gap frequency (opens beyond 1 ATR). Out: risk stats | Price | Must | Golden tests pass |
| FR-035 Technical summary | One readable view without false certainty | In: FR-031 to FR-034. Proc: rule set in Appendix A.6 (trend state, momentum state, volatility regime, volume confirmation). Out: states with rule text and historical hit-rate where measured; label "condition, not a recommendation" | All above | Must | Every state shows the rule that fired; no output uses "buy" or "sell" |
| FR-036 Pattern detection | Common chart patterns | In: price. Proc: detection of gaps, 52-week breakouts, inside bars; complex patterns (head and shoulders) deferred. Out: annotations | Price | Could | Detected events validated on labelled sample with precision at least 90% |

## 8.5 Market, macro, news and events

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-040 Market overview | Daily context. As a user, I want to see NIFTY, SENSEX, sector moves and breadth | In: index levels, constituents' prices. Proc: returns over 1D/1W/1M/YTD, advance-decline, sector relative performance, India VIX where licensed. Out: dashboard | Index licence. Missing index: tile shows unavailable | Must | Values equal licensed source at the stamped time |
| FR-041 Macro dashboard | Rates, inflation, growth and currency drive sectors | In: RBI policy repo rate and reference rates, CPI and GDP from MoSPI, INR/USD reference rate, Brent and other commodities where licensed. Proc: latest value, release date, next release date, change. Out: macro panel and sector sensitivity notes | Official releases. Release revised: keep vintage history | Should | Each figure cites the official release and its date |
| FR-042 Corporate announcements | Official disclosures are the primary event source | In: exchange corporate filings feed. Proc: ingest, classify (results, board meeting, dividend, order win, pledge, rating, litigation, etc.), link to ISIN. Out: event records with original document link | Exchange or licensed vendor feed. Duplicate filings on NSE and BSE merged by content hash | Must | 99% of filings for T1 appear within 15 minutes of exchange timestamp (subject to vendor SLA) |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-043 News ingestion | Broader context | In: licensed news feed. Proc: entity linking, deduplication, classification as Confirmed (official source), Reported (named credible outlet), or Speculative (unnamed sources, rumours). Out: news items with status | Licensed news API only (C-05). Unlinkable items are not attached to stocks | Should | Entity linking precision at least 95% on labelled sample; status rules auditable |
| FR-044 Event impact card | Answers the 8 event questions in the brief | In: event, price and volume around event, sector exposure. Proc: fill the 8 fields; "already priced in" assessed only as abnormal return vs market model over event window, with caveat. Out: event card | FR-042/043, prices. Insufficient data for any field: say so | Should | All 8 fields present or explicitly "not assessable"; no causal language without evidence rule (A.7) |
| FR-045 Event calendar | Plan around known dates | In: board meeting dates, results dates, ex-dates, AGMs, derivative expiries, RBI policy dates, macro releases. Out: calendar per stock and market | Feeds above; expiry dates from contract master (FR-080) | Must | Calendar matches official sources for a sampled month |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-046 Move attribution | "What is driving the price?" without false causation | In: stock, market, sector returns, events. Proc: decompose return into market, sector and residual (factor model); list same-day events as "coincident", not "cause". Out: attribution bar and coincident events | Prices, events | Could | Language passes the causation-guard test suite (TST-AI-04) |

## 8.6 Valuation and forecasting

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-050 Method selector | Not every method fits every company | In: sector, data completeness, cash-flow profile. Proc: rules in 10.2 (e.g. no EV/EBITDA or FCF-DCF for banks; DDM only for stable payers; no P/E when EPS is negative). Out: eligible methods with reasons for exclusions | FR-020/021 | Must | Rule table fully test-covered; excluded methods always show a reason |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-051 DCF | Intrinsic value from stated assumptions | In: FCFF history, growth, margins, reinvestment, WACC inputs (risk-free from Government of India 10-year yield, equity risk premium, beta), terminal growth. Proc: 2-stage FCFF DCF (Appendix A.8); sensitivity grid on WACC and terminal growth. Out: value per share, grid, assumption table | Fundamentals, yields. Terminal growth not below WACC by at least 2 points: reject | Should | Golden models reproduce reviewer spreadsheet within 0.1%; assumptions editable and logged |
| FR-052 Multiples (peer and historical) | Market-relative value | In: peer and own-history P/E, P/B, EV/EBITDA. Proc: apply peer median and own 5-year median/interquartile range to current fundamentals. Out: value ranges by multiple | FR-022 | Must | Outputs reproducible; negative-earnings cases excluded with reason |
| FR-053 DDM | Stable dividend payers | In: DPS history, payout, cost of equity, growth. Proc: Gordon or 2-stage DDM. Out: value | Dividend history | Could | Golden tests pass |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-054 Fair-value range | Combine methods transparently | In: FR-051 to FR-053. Proc: show each method's range; overall range = interquartile span of method midpoints when at least 2 methods are eligible; no averaging that hides dispersion. Out: range bar vs current price; upside/downside to range edges | Above. Fewer than 2 methods: show single-method range with warning | Must | Range always shows contributing methods; never a single point labelled "target" |
| FR-055 Scenario builder | Bull/base/bear with explicit assumptions | In: revenue growth, margin, multiple, and event assumptions per scenario (defaults from history; user-editable). Proc: compute scenario value at 12 months; record assumptions. Out: 3 scenarios with assumptions, value, change vs price, key catalysts and risks. Gated: REG-01 for named stocks in public mode | Above | Must (gated) | Changing an assumption recomputes deterministically; every scenario lists at least one invalidation condition |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-056 Statistical horizon forecasts | Quantified 1, 3, 6, 12-month distributions | In: return history, volatility, factor exposures. Proc: models in 10.4 (baseline random walk with drift and volatility; validated challenger models only). Out: median path and 50%/80% intervals, with backtest error and coverage. Gated: TST-FC acceptance and REG-01 | Prices, validation results | Should (internal beta) | Only published for a horizon and stock group that passed section 19 thresholds; otherwise "not validated" |
| FR-057 Insufficient-data result | No target without data (brief section 5) | In: data completeness checks. Proc: if required inputs are missing or method unsuitable, return structured result: status, missing fields, method, next data date. Out: explicit card | All | Must | 100% of synthetic missing-data cases return this card; none returns a number |
| FR-058 Catalysts and invalidation | What would change the view | In: events calendar, scenario assumptions. Proc: generate invalidation conditions as measurable statements (e.g. "operating margin below 12% for two quarters") from assumptions; catalysts from calendar. Out: lists, linkable to alerts | FR-045, FR-055 | Must | Every scenario has at least one measurable invalidation condition tied to a data field |

## 8.7 Citations, freshness, explainability and reports

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-060 Citation gate | LLM must not invent numbers or facts | In: AI draft, fact sheet from engines. Proc: every number and named fact in the draft must match a fact-sheet entry (value, unit, period) or a retrieved source span; unmatched sentences are removed or the draft is regenerated; final fallback is the template-only report. Out: verified narrative with inline citations | Engines, retrieval index | Must | On the hallucination test set (TST-AI-01) zero unsupported numbers reach output |
| FR-061 Timestamp and freshness | Every market-sensitive output discloses its data time | In: source timestamps. Proc: compute freshness vs SLA per tier (NFR-010). Out: "Data as of 11 Oct 2026 15:30 IST, NSE, delayed 15 min" style badge; stale banner when SLA breached | All feeds | Must | 100% of price-bearing components show timestamp; stale state triggers in tests |
| FR-062 Evidence inspector | Users can inspect reasoning behind any score | In: output ID. Proc: retrieve inputs, formula, source documents, model version, snapshot. Out: evidence panel | Audit store | Must | For a random 100 outputs, inspector reproduces the shown value |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-063 Confidence rating | Evidence quality, not certainty of profit (C-04) | In: data completeness, source tier, model validation status, peer count, history length. Proc: rubric in 10.7 (High / Medium / Low / Not rated). Out: rating with reasons | Above | Must | Rubric applied deterministically; reasons always shown |
| FR-064 Stock research report | The report specified in section 16 | In: all engines. Proc: assemble sections; PDF/HTML export with disclosures. Out: report | All | Should | Report contains all 13 required sections or explicit "not available" |

## 8.8 Risk management

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-070 Stop-loss calculator | Several defensible stop methods | In: assumed entry, direction, method (support, ATR multiple, percentage, volatility-adjusted, trailing), parameters. Proc: formulas in 11.2. Out: stop price per method, distance in INR and %, distance in ATR units | Price, ATR, S/R levels. Stop on wrong side of entry: reject | Must | All formulas pass golden tests including short positions |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-071 Position sizing | Risk-based sizing | In: capital, risk budget (% or INR), entry, stop, lot size for derivatives. Proc: shares = floor(risk budget / risk per share), capped by max position % and liquidity cap (e.g. 1% of 20-day median volume). Out: quantity, capital used, actual risk | User inputs, liquidity | Must | Correct rounding (down to whole shares or lots); caps applied; tests include zero and negative edge cases |
| FR-072 Targets and reward-to-risk | Compare reward to risk honestly | In: entry, stop, targets (resistance, R-multiples, fair-value edge). Proc: R:R = (target - entry)/(entry - stop) for longs. Out: table per target, with the source of each target | Above | Must | Values reproduce; targets always labelled with their method |
| FR-073 Gap and slippage warnings | Stops do not guarantee execution | In: gap frequency, circuit status, liquidity. Proc: estimate historical gap-through probability (share of days opening beyond the stop distance); slippage estimate by liquidity bucket. Out: warning text and "worst observed gap" | FR-034 | Must | Warning shown on every stop output; text reviewed by compliance |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-074 Trailing stop simulator | Show how trailing rules behave | In: rule (ATR trail, % trail, chandelier). Proc: historical simulation on the stock, with costs. Out: path chart, exits | Price | Should | Simulation free of look-ahead (uses prior-bar data only) |
| FR-075 Stop suitability checks | Prevent stops chosen only for attractive R:R | In: stop, ATR, S/R. Proc: warn if stop is within 0.5 ATR of entry (noise), sits just beyond a round number or obvious level, or is chosen only to raise R:R. Out: suitability notes | Above | Must | Rule tests pass; message never suggests a better stop for R:R reasons alone |
| FR-076 Portfolio risk | Concentration and market risk | In: holdings. Proc: weights, sector concentration, beta, historical VaR (95%, 1-day and 10-day), correlation clusters, drawdown. Out: risk dashboard | Holdings, prices | Should | VaR backtests within regulatory-style traffic-light tolerance on test portfolios |

## 8.9 Options and derivatives

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-080 Contract master | Lot sizes, expiries and eligibility change by circular (C-06) | In: daily exchange contract files. Proc: load lot size, strike intervals, expiry dates, eligible underlyings, ban list. Out: contract master with effective dates | Exchange files via vendor. Load failure: derivatives analytics show "contract data unavailable"; no fallback to stale lot sizes | Must | No hardcoded lot size or expiry in code (static scan); daily reconciliation passes |
| FR-081 Option chain | See prices, OI, volume, IV per strike | In: option quotes, OI. Proc: compute IV per strike (FR-082), change in OI. Out: chain table with timestamp | Licensed derivatives feed | Should | Values equal source at stamped time |
| FR-082 Greeks and IV | Understand sensitivity | In: spot or futures, strike, expiry, rate, dividend yield, price or volatility. Proc: Black-Scholes-Merton for European-style options (NSE index and stock options are European-style; [A] confirm per contract spec), IV by Newton-Raphson with bisection fallback. Out: delta, gamma, theta (per day), vega (per 1 vol point), rho (per 1% rate) | Rates (T-bill), contract master. No IV solution (price below intrinsic): "IV not computable" | Must | Matches reference implementation within 1e-6; put-call parity check passes |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-083 Strategy payoff builder | Payoff, break-even, max gain/loss before trading | In: legs (long/short, call/put/future, strike, qty, premium). Proc: payoff at expiry and today (BSM) across price grid; analytic max profit/loss where bounded; break-evens by root finding. Out: payoff diagram, table, "unlimited" labels where unbounded | FR-080, FR-082 | Must | All strategies in section 12.5 and Appendix C reproduce known payoffs; short call shows unlimited loss |
| FR-084 Margin estimate | Margin is a key risk | In: positions. Proc: use exchange SPAN/risk parameter files where licensed; otherwise show "check with your broker" and do not estimate. Out: estimated margin with source and date | Clearing corporation risk files | Could | Estimates within 2% of clearing corporation calculator on test positions |
| FR-085 OI and PCR analytics | Popular but often misread | In: OI by strike. Proc: PCR (OI and volume), max-pain, OI change; mandatory interpretation limits text. Out: charts with caveats | FR-081 | Should | Caveat text always present; no directional label |
| FR-086 Futures basis | Cost of carry | In: spot, futures, rate, dividends. Proc: basis, annualised basis, fair futures price. Out: basis panel | Prices | Should | Golden tests pass |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-087 Derivatives risk disclosure gate | Investor protection | In: first use of derivatives analytics. Proc: show risk disclosure, including SEBI's published findings on F&O losses where cited, and require acknowledgement. Out: acknowledgement record | Compliance content | Must | No derivatives screen accessible without logged acknowledgement |

## 8.10 Glossary and learning

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-090 Glossary content model | Consistent, sourced entries | In: editorial content. Proc: schema in section 13 (definition EN/HI, example, formula, use, mistakes, related terms, source). Out: entries via CMS | CMS. Entry missing required field: cannot publish | Must | 300 MVP terms published with all required fields and reviewer sign-off |
| FR-091 Contextual help | Explain terms where they appear | In: page text. Proc: term tagging at build time; tooltip with short definition and link. Out: tooltips | Glossary | Must | 100% of tagged terms in report templates have entries |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-092 Glossary search | Find terms in English, Hindi or Hinglish | In: query. Proc: lexical plus semantic search; transliteration support. Out: ranked entries | Search index | Must | Top-3 hit rate at least 95% on a 200-query test set |
| FR-093 Learning paths | Structured learning | In: curated sequences. Out: lessons and quizzes using lagged or hypothetical data only | CMS | Could | Lessons use only hypothetical data or data lagged per REG-08 |
| FR-094 Ask about a term | Natural-language explanations | In: question. Proc: retrieval over glossary and methodology pages only; refuse stock-specific advice. Out: grounded answer with sources | Glossary index | Should | Grounding test: at least 98% of answers cite a glossary entry; zero stock recommendations |

## 8.11 Watchlists, portfolio, forecast ledger and compliance

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-100 Watchlists | Track stocks of interest | In: user selections. Out: lists with key metrics and freshness | Engines | Must | CRUD tests; limits (e.g. 20 lists x 100 stocks) enforced |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-101 Alerts | React to changes | In: conditions on price, events, invalidation conditions (FR-058). Proc: evaluate on data update. Out: in-app, email, push | Notification service. Delivery failure retried 3 times then logged | Should | Alert latency within NFR-012 |
| FR-102 Manual portfolio | Risk analytics on holdings | In: manual entry or CSV (ISIN, qty, cost). Out: holdings | Security master. Unknown ISIN rejected with row number | Should | CSV import validates 100% of rows; errors listed |
| FR-103 Portfolio risk dashboard | Concentration and drawdown view | As FR-076 | As FR-076 | Should | As FR-076 |
| FR-104 Notification preferences | User control and DPDP consent | In: preferences. Out: channel and frequency settings | Consent store | Could | Opt-out honoured within 24 hours |
| FR-110 Forecast ledger | Honest track record (P6) | In: every published forecast or scenario. Proc: append-only record of stock, horizon, method, model version, inputs snapshot, interval, publish time. Out: ledger | Write-once storage | Must | Records cannot be edited or deleted (tested); hash chain verifies |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-111 Accuracy evaluation | Measure against baselines | In: ledger, realised prices. Proc: at horizon end compute errors, interval coverage, directional accuracy, vs no-change and index baselines (section 10.6). Out: accuracy reports by horizon, sector, model | Prices | Must | Metrics reproduce on synthetic ledgers; evaluation excludes no forecasts |
| FR-112 Public accuracy page | Transparency | In: FR-111. Out: page showing all forecasts, not only favourable ones. Gated: REG-09 (PaRRVA / performance-claim rules) | Compliance | Should (gated) | No selective periods; compliance sign-off |
| FR-113 Research history | Users revisit their research | In: user activity. Out: history of reports viewed with snapshot so the view can be reproduced | Audit store | Should | Historical report reproduces the original numbers |
| FR-120 Disclosure management | Required disclosures on every output | In: disclosure templates by output type. Proc: attach versioned disclosures. Out: disclosures in UI and exports | Compliance CMS | Must | Every report/export contains current disclosure version |
| FR-121 Research approval workflow | Human accountability (RA route) | In: draft report. Proc: analyst review, edits, approval, conflict check. Out: approved version with approver ID | Roles | Must (if RA) | No public research without approval record |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-122 Complaint handling | Regulatory obligation | In: complaints. Proc: ticket, SLA timer, escalation, SCORES reference if applicable. Out: resolution records | Ticketing | Must | Complaints tracked to the regulatory window (7 working days for RAs per the Jan 2025 guidelines) |
| FR-123 Audit log | Traceability | In: all outputs and admin actions. Proc: append-only log with snapshot and model version. Out: searchable audit trail | Log store | Must | Retention per NFR-031; tamper-evidence verified |
| FR-124 Material error correction | Correct errors fast and visibly | In: error report. Proc: triage, fix, re-issue, notify affected users, log. Out: correction notice | Notifications | Must | Correction notices issued within NFR-033 timeline in drills |
| FR-125 Entitlement enforcement | Data licence compliance | In: user tier, data source licence terms. Proc: check display, delay and export rights per source. Out: allow, delay or block | Licence registry | Must | Tests prove unlicensed real-time data never reaches unentitled users |
| FR-126 Operations console | Run the platform | In: pipeline, model and cost metrics. Out: dashboards and alerts | Monitoring | Should | All NFR metrics visible |

## 8.12 Persistent historical store and incremental updates (mandatory)

Design detail is in section 14.4. These requirements are core architecture: no release, internal or public, ships without them.

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-130 Initial historical download | Download history once. As an operator, I want the first run to build the full local dataset | In: universe, exchanges, datasets, lookback, provider. Proc: chunked download with checkpoints, validation, atomic CSV write (14.4.3). Out: CSV files, manifest rows, initial quality report | Licensed provider (OD-02). Partial failure: valid rows kept, `status = partial`, failed ranges queued | Must | TST-HIST-A |
| FR-131 Persistent CSV store | Data must survive restarts, reboots, redeploys and retraining | In: validated records. Proc: write to the 14.4.1 layout on persistent storage; schemas per 14.4.2. Out: CSV files | Persistent volume or versioned object store. Mount missing at start-up: refuse to run, never re-download silently | Must | TST-HIST-G |
| FR-132 Dataset manifest and index | Know exactly what is stored | In: write events. Proc: maintain `manifest.csv` with coverage, last confirmed date, checksum, schema version, status. Out: manifest | Updated only after a successful write | Must | Manifest matches files on disk in every test |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-133 Incremental gap detection | Fetch only what is missing | In: manifest, exchange trading calendar, provider's latest date. Proc: expected trading dates minus stored dates, plus failed ranges; excludes holidays, weekends, unpublished and incomplete sessions. Out: list of ranges to request | Trading calendar. Calendar missing: stop the update and warn | Must | TST-HIST-B, TST-HIST-C, TST-HIST-J |
| FR-134 Incremental fetch and merge | Merge new data safely | In: missing ranges. Proc: request ranges only; validate; merge on natural key; dedupe; atomic replace (14.4.4). Out: updated CSV, log rows | Rate limits and licence terms respected | Must | TST-HIST-C, TST-HIST-D |
| FR-135 Revision and corporate action handling | History changes; appends alone are wrong | In: look-back window, new corporate actions, restatements. Proc: precedence rules, revision rows, revisions log, rebuild of adjusted series and affected features (14.4.5). Out: revised data and log | Corporate action feed. Conflicting same-precedence sources: hold and flag for review | Must | TST-HIST-F |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-136 Failure recovery | Never lose or corrupt valid history | In: errors, crashes. Proc: backoff retries, rate-limit handling, checkpoints, backups, restore, failed-range queue (14.4.6). Out: preserved data, actionable log entries | None | Must | TST-HIST-E |
| FR-137 Full refresh control | Full re-download only on purpose | In: operator command with reason, or recovery procedure. Proc: backup, re-download, compare, swap. Out: refreshed files, audit entry | Requires admin role | Must | Without the command, no run requests more than the computed gaps (TST-HIST-J) |
| FR-138 Incremental feature recompute | Avoid rebuilding all features | In: changed symbols and dates. Proc: recompute from the earliest affected date minus the longest rolling window; full symbol rebuild after corporate actions; feature-set versioning. Out: updated feature files | Feature definitions registry | Must | Recomputed features equal a full rebuild on test data; TST-HIST-F |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria |
|---|---|---|---|---|---|
| FR-139 Prediction on local validated data | Predictions must not trigger full downloads or use stale inputs | In: request. Proc: incremental update, freshness and alignment check, approved model load, predict, record references (14.4.7). Out: prediction with model and data references, or "unavailable" with reason | Model registry | Must | TST-HIST-H, TST-HIST-I |
| FR-140 Retraining policy | Do not retrain on every candle | In: schedule, drift alerts. Proc: retrain only on schedule or trigger, with walk-forward validation and leakage checks; artifacts stored outside `data/raw`. Out: new model version in registry | Feature store, NFR-043 | Must | Training runs logged with trigger reason; no retrain on a single new candle unless the strategy requires it |
| FR-141 Storage and download metrics | Show efficiency and freshness | In: logs. Proc: aggregate metrics in 14.4.8. Out: operations dashboard and run report | Logs | Should | All 11 metrics visible per run |

## 8.13 Ingestion, orchestration, delivery, configuration and model governance (v1.0)

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria and verification |
|---|---|---|---|---|---|
| FR-142 Provider adapter interface (P1) | One contract for every provider, so providers can change without touching analytics | In: dataset, symbols, date range. Proc: authenticated request with timeouts, pagination, rate limits; normalise to 14.4.2 schemas. Out: records plus provenance | Credentials from secrets. Provider failure returns an error, never a fabricated response; stored data untouched | Must | Contract tests per adapter; failure injection shows no writes on error |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria and verification |
|---|---|---|---|---|---|
| FR-143 Provider capability and licence check (P1) | Use only verified, licensed sources | In: provider registry (capabilities, licence terms, freshness, cost). Proc: run refuses a dataset whose provider lacks a recorded licence for that use. Out: allow or block | Licence registry (FR-125) | Must | A dataset without a licence entry is never fetched (test) |
| FR-144 Provenance capture (P1) | Every value traceable | Proc: store source, source record ID, source timestamp, fetched_at and request ID on every row | None | Must | 100% of rows carry provenance fields |
| FR-150 Scheduled runs (P5) | Research three times each weekday | In: cron in UTC (defaults 03:30, 07:30, 10:30 UTC = 09:00, 13:00, 16:00 IST), configurable. Proc: start the core pipeline. Out: run | GitHub schedule may be delayed or dropped under load: the run records actual start time and, if later than a configured tolerance, labels the report accordingly | Must | Workflow file lints; runs observed at the three slots in Phase 1 |
| FR-151 Manual runs (P5) | On-demand research | In: `workflow_dispatch` inputs: scope, symbols, mode, horizons, report format, full_refresh (default false) plus reason. Proc: validate inputs against config and the universe; reject unknown symbols. Out: same pipeline as FR-150 | Invalid input: run fails fast with an actionable message, no data changes | Must | Manual and scheduled runs produce identical outputs for identical inputs and data |
| FR-152 Single core pipeline (P5) | No duplicated logic | Proc: one Python entry point runs the 14 steps in 15.5.3; scheduled and manual workflows call it with parameters | None | Must | Code review and test show one implementation of ingestion, formulas, inference and reporting |
| FR-153 Session-state awareness (P5) | Know what the data represents | In: exchange calendar, provider status, clock. Proc: classify each dataset as completed session, incomplete intraday interval, or delayed/provisional. Out: labels carried into report | Holiday or weekend: run produces a "no new session" report or skips per config (FR-159) | Must | Each of the three slots labelled correctly in tests, including a holiday |
| FR-154 Concurrency control (P5) | Overlapping runs must not corrupt data | Proc: workflow concurrency group with no cancellation of an in-progress data write; data-branch commit uses fetch, rebase-free fast-forward check and retry | Conflict: the later run re-reads and re-merges, or exits without writing | Must | Two simultaneous dispatches leave consistent data (test) |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria and verification |
|---|---|---|---|---|---|
| FR-155 Run record (P5) | Auditable runs | Out: `runs/<run_id>.json` and `metadata/run_log.csv`: trigger, inputs, start/end, steps status, records fetched, data snapshot (manifest checksums), model versions, report ID, email status, errors | None | Must | Every run has a record, including failed runs |
| FR-156 Persist data to the data branch (P5) | Data must outlive the runner | Proc: after validation, commit changed CSVs, manifests, predictions and run records to the private data branch; commit only when something changed | Push failure: retry with backoff, then mark run `data_not_persisted` and alert | Must | TST-GHA-03 |
| FR-157 Full refresh safeguards (P5) | Prevent accidental rebuilds | Proc: full refresh only via dispatch input plus reason; backup tag on the data branch first | Missing reason: refuse | Must | Refresh without reason is rejected |
| FR-158 Retry and idempotency (P5) | Safe re-runs | Proc: steps are idempotent; re-running a run ID never duplicates rows, predictions or emails | None | Must | Re-run test shows no duplicates |
| FR-159 Holiday handling (P5) | No false "missing data" | Proc: exchange holiday calendar from licensed source; on holidays send a short "market closed" note or skip (configurable) | Calendar missing: run continues with warning, no gap marking | Must | Holiday test passes |
| FR-160 Report email (P5) | Deliver research automatically | In: validated report. Proc: send HTML email with execution time, market overview, key findings, forecasts and assumptions, risks and data-quality warnings, data timestamp, model versions, run status, plus attachments or links | `GMAIL_SENDER`, `GMAIL_APP_PASSWORD`, `REPORT_RECIPIENT` from secrets/variables | Must | TST-MAIL-01 |
| FR-161 Failure notification (P5) | Never send stale results as current | Proc: on failure send a clearly labelled failure email with the failing step and next action where SMTP still works | SMTP itself failing: recorded in run record and workflow summary | Must | TST-MAIL-03 |
| FR-162 Delivery status (P5) | Report only what is verified | Proc: record SMTP acceptance ("submitted") separately from any confirmed delivery; v1.0 can verify submission only | None | Must | Status values limited to verified states |
| FR-163 Duplicate-send prevention (P5) | Retries must not spam | Proc: idempotency key = run ID + report hash; send log checked before sending | None | Must | Re-run does not send a second email for the same report |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria and verification |
|---|---|---|---|---|---|
| FR-164 Configurable recipients (P5) | No hardcoded addresses | Proc: recipients from `REPORT_RECIPIENT` (comma-separated allowed); validated format; recipient count limited by config | Empty or invalid: run completes, email step fails with a clear message | Must | Address never appears in code or logs (masked) |
| FR-165 Report generation (P4) | Email-ready research | Proc: render the 14-section report (16.4) to HTML, plus a CSV summary of forecasts, valuations and risk metrics. Out: `reports/<date>/<run_id>.html` and `.csv` | Missing section data: section shows "unavailable" with reason | Must | All 14 sections present or explicitly unavailable |
| FR-170 Configuration management (P6) | Business settings without code changes | Proc: versioned YAML config (universe, schedule tolerance, lookbacks, horizons, risk defaults, report options) validated against a schema at start | Invalid config: run stops before any fetch | Must | Schema validation test |
| FR-171 Secrets handling (P6) | Protect credentials | Proc: all credentials only in GitHub Actions secrets; masked in logs; never written to `data/`, reports or run records | Secret scan in CI | Must | Secret-scanning and log tests pass |
| FR-172 Least-privilege permissions (P6) | Limit blast radius | Proc: workflow `permissions` set to the minimum (contents write only for the data-commit job); pinned action versions | None | Must | Workflow review checklist |
| FR-173 Run summaries and structured logs (P6) | Troubleshooting | Out: JSON logs; GitHub job summary with freshness, records fetched, failures, email status | None | Must | Summary present on every run |
| FR-174 Freshness indicators (P6) | Users see data age | Out: data timestamp and freshness label per dataset in summary, report and email | None | Must | Present in 100% of reports |
| FR-175 Model and dataset change audit (P6) | Traceability | Proc: every model promotion and dataset schema change is a reviewed commit with a changelog entry | None | Must | Audit trail test |
| FR-176 Authorised dispatch (P6) | Only authorised users trigger runs | Proc: rely on repository permissions (write access needed to dispatch); future web trigger only via a server-side token, never in browser code | None | Must | Read-only collaborator cannot dispatch (verified) |
| FR-180 Task-specific model selection (P3) | No single algorithm for every task | Proc: candidate sets per task as in 10.9; selection by validated performance, complexity and cost | None | Must | Selection report lists every candidate and result |

| ID and name | Rationale and user story | Inputs, processing, outputs | Data dependencies and error handling | Priority | Acceptance criteria and verification |
|---|---|---|---|---|---|
| FR-181 Model registry and versioning (P3) | Reproducibility | Out: `models/model_registry.json` with model ID, version, task, training window, feature-set version, validation report, approval, artifact checksum | Unapproved model: never used in production runs | Must | Registry test |
| FR-182 Robust evaluation (P3) | Avoid lucky results | Proc: walk-forward, regime, sector and period breakdowns; baselines; costs; metrics per task (10.6, 10.9) | None | Must | Evaluation report generated per candidate |
| FR-183 Stock ranking (P3) | "Which stocks deserve research" | Proc: factor and model-based ranking evaluated with rank IC and decile spreads; gated by REG-01 for any shared output | Ranking not validated: not shown | Could (gated) | TST-FC-04 |
| FR-184 Forecast output contract (P3) | Consistent forecasts | Out: fields in 10.9.3; explicit "unavailable" with reason when data or validation fails | None | Must | Contract test on every forecast |

# 9. Fundamental, technical, and macroeconomic research methodology

Every metric is defined once, computed in one place, labelled with its period and source, and shown next to its peer and historical context; formulas are collected in Appendix A.

## 9.1 Fundamental metrics

Period conventions [D]: Indian companies mostly report on an April to March fiscal year (FY26 = Apr 2025 to Mar 2026). TTM = sum of last four reported quarters. Consolidated statements are the default; standalone is shown when consolidated is unavailable, and labelled. Point-in-time rule: a value is usable in analytics and backtests only from its filing timestamp onward.

| Metric | Formula (summary) | Why it matters | Key limitations | Source |
|---|---|---|---|---|
| Revenue growth | (Rev_t / Rev_t-1) - 1, YoY on quarters and TTM | Demand and scale | Acquisitions and accounting changes distort; seasonality makes QoQ misleading | Results filings |
| PAT and EPS growth | Same on PAT attributable to owners; EPS = PAT / weighted average shares (diluted where reported) | Earnings power | Exceptional items; tax one-offs. Show "adjusted PAT" only when adjustments are itemised in filings | Results filings |
| EBITDA and operating margin | EBITDA = operating profit + depreciation and amortisation; margin = EBITDA / revenue | Operating efficiency | Not meaningful for banks and insurers; Ind AS 116 lease accounting raises EBITDA | Results filings |
| P/E | Price / TTM diluted EPS | Price paid per unit of earnings | Meaningless for negative or tiny EPS; cyclical peaks make P/E look low | Price + filings |
| P/B | Price / book value per share | Asset-heavy and financial firms | Intangible-heavy firms look expensive; book value can be stale | Price + balance sheet |
| ROE | PAT / average shareholders' equity | Return to owners | Inflated by leverage and buybacks | Filings |

| Metric | Formula (summary) | Why it matters | Key limitations | Source |
|---|---|---|---|---|
| ROCE | EBIT / average (total assets - current liabilities) | Return on all capital | Cash-rich firms understate; definition varies, so ours is fixed and shown | Filings |
| Debt-to-equity | Total borrowings (incl. lease liabilities, shown separately) / equity | Solvency | Not comparable for lenders; use capital adequacy for banks | Balance sheet |
| Operating cash flow and FCF | OCF from cash-flow statement; FCF = OCF - capex | Earnings quality | Half-yearly cash-flow reporting for many companies; capex lumpy | Cash-flow statement |
| Cash conversion | OCF / EBITDA (3-year sum) | Detects profit not turning into cash | Working-capital-heavy sectors differ structurally | Derived |
| Promoter holding and pledge | Promoter %; pledged shares / promoter shares | Alignment and forced-sale risk | Pledges may be disclosed with lag | Shareholding filings |
| Institutional ownership | FPI %, DII %, quarterly change | Sponsorship | Quarterly only; index flows distort | Shareholding filings |
| Dividend history | DPS, payout = dividends / PAT, yield = DPS (TTM) / price | Cash return | Special dividends inflate yield | Corporate actions |
| Bank-specific | NIM, GNPA %, NNPA %, CASA %, CRAR, credit cost | Banks are not analysed on EBITDA | Disclosure formats vary | Results and Basel III disclosures |

Competitive position, industry growth and business risk are summarised by the AI layer only from cited documents (annual report MD&A, investor presentations), and every claim cites its page.

## 9.2 Technical methodology

Data interval [D]: daily bars for all indicators in MVP; weekly bars derived from daily; intraday (5-minute) only for display where licensed. All calculations use split- and bonus-adjusted prices; total-return series (dividend-adjusted) are used for relative strength and backtests.

| Indicator | Parameters (default) | Signal rule used in technical summary | Known limitations |
|---|---|---|---|
| SMA / EMA | 20, 50, 200 | Trend up if close > 50 SMA > 200 SMA and 200 SMA slope (20-day) > 0; down if the reverse; else sideways | Lagging; whipsaws in ranges |
| RSI (Wilder) | 14 | Momentum strong above 60, weak below 40; "overbought/oversold" shown only as a description, never as a sell/buy | Can stay extreme in strong trends |
| MACD | 12, 26, 9 | Momentum improving if histogram rising for 3 bars | Lagging; noisy on daily data |
| ATR (Wilder) | 14 | Volatility regime by percentile of ATR% over 1 year | Does not show direction |
| Bollinger Bands | 20, 2 sd | Volatility squeeze if bandwidth in bottom 10% of 1-year range | Normality assumption |
| Volume confirmation | 20-day average volume | Breakout confirmed if volume at least 1.5x average | Block deals distort |
| Relative strength | 63 and 252 days vs NIFTY 50 and sector index | Leading if RS ratio rising and above its 50-day average | Benchmark choice matters |

| Indicator | Parameters (default) | Signal rule used in technical summary | Known limitations |
|---|---|---|---|
| Support / resistance | Swing points, 2 to 3 ATR clustering | Levels with at least 2 touches in 1 year | Subjective by nature; shown as zones |
| Breakout | Close above 52-week high or resistance zone | Shown as event with volume flag | High false-breakout rate; base rate shown when measured |
| Drawdown | Peak-to-trough on adjusted close | Risk context | Past drawdown is not a limit on future drawdown |

Every technical summary state shows the historical base rate of that state for the stock universe (for example "after this condition, NIFTY 100 stocks rose over the next 20 days 54% of the time, 2015 to 2025, n = X") only when computed in the validation pipeline; otherwise no rate is shown. [D]

## 9.3 Market, sector and macro methodology

- Market regime: NIFTY 50 trend state (rule above), breadth (% of NIFTY 500 above 200-day SMA), and India VIX percentile where licensed.
- Sector rotation: 1-, 3- and 6-month relative returns of NSE sectoral indices vs NIFTY 50, shown as a ranked table and a relative-rotation chart.
- Macro panel: policy repo rate and stance (RBI), CPI inflation (MoSPI), GDP growth (MoSPI), INR/USD reference rate (FBIL/RBI), 10-year G-sec yield, Brent crude where licensed. Each value shows release date and next scheduled release.
- Sector sensitivity notes are editorial content, reviewed by the research lead, citing evidence (e.g. lenders' NIM sensitivity to repo changes as disclosed by the companies).
- Budget, tax and policy events are tagged to affected sectors through a curated mapping table, each mapping citing the official notification.

### 9.4 News and event methodology

Source tiers [D]: Tier 1 = exchange filings, SEBI, RBI, ministries, company websites. Tier 2 = licensed major news wires and established business publications. Tier 3 = other licensed sources. Social media is not a source in MVP.

Confirmation status: Confirmed = Tier 1 source; Reported = at least one Tier 2 source naming its source; Speculative = anything else, shown greyed with a warning and never used in scenarios.

Causation guard (Appendix A.7): the system may say an event coincided with a price move. It may say "likely contributed" only if (a) the abnormal return on the event window is statistically significant against a market model at the 5% level, and (b) no other material event for the stock occurred in the window. It never says "caused". This rule is enforced in the AI layer by a phrase classifier and in tests (TST-AI-04).

## 10. Valuation and price forecasting methodology

The Platform keeps two products apart: a fundamental fair-value range (what the business may be worth under stated assumptions, horizon 12 months or more) and a statistical price distribution (where the price may trade over 1 to 12 months, with calibrated intervals); neither is a guaranteed target, and either may return "insufficient data".

### 10.1 Output contract per stock

| Field | Content | Always present? |
|---|---|---|
| Price | Latest verified price, exchange, timestamp, delay | Yes |
| Historical performance | 1M, 3M, 6M, 1Y, 3Y, 5Y total return vs NIFTY 50 and sector | Yes, when history exists |
| Key metrics | From section 9.1 | Yes |
| Fair-value range | By method, plus combined range | Or insufficient-data card |
| Scenarios | Bull, base, bear with assumptions, 12-month value, change vs price | Or insufficient-data card |

| Field | Content | Always present? |
|---|---|---|
| Horizon distribution | 1M, 3M, 6M, 12M median and 50%/80% intervals, with backtest coverage and error | Only if validated (FR-056) |
| Catalysts and risks | Dated catalysts; top downside risks with evidence | Yes |
| Confidence | Evidence-quality rating with reasons (10.7) | Yes |
| Invalidation | Measurable conditions that would void the view | Yes, for every scenario |
| Label | "Fundamental estimate" or "Statistical range" or "Trading reference level"; never "target" alone | Yes |

## 10.2 Method eligibility rules

| Method | Use when | Do not use when |
|---|---|---|
| FCFF DCF (2-stage) | Positive, reasonably stable operating cash flows; at least 5 years of history | Banks, NBFCs, insurers (use residual income or P/B-ROE); negative FCF in 4 of last 5 years without a credible turnaround input; holding companies |
| Peer multiples | At least 5 peers with meaningful metric | Unique business model with no peers; negative denominators |
| Historical multiples | At least 5 years of listed history without major business change | After a merger, demerger or business model change |
| P/B-ROE (justified P/B) | Banks and financials | Firms with large intangibles |
| DDM | Stable payout for 5 years; mature business | Irregular or zero dividends |
| Sum of parts | Conglomerates with listed subsidiaries | Phase 3 only |

## 10.3 Core valuation formulas

FCFF DCF, stage 1 for N explicit years then terminal value:

```
V_{firm} = \sum_{t=1}^{N} \frac{FCFF_t}{(1+WACC)^t} + \frac{FCFF_{N+1}}{(WACC - g)(1+WACC)^N}
```

```
\text{Value per share} = \frac{V_{firm} - \text{Net debt} - \text{Minority interest} + \text{Non-operating assets}}{\text{Diluted shares}}
```

```
WACC = \frac{E}{D+E} k_e + \frac{D}{D+E} k_d (1 - t), \quad k_e = r_f + \beta \times ERP
```

Rules [D]: r_f = Government of India 10-year benchmark yield on the valuation date (source and date shown); ERP is a documented, versioned house assumption, reviewed quarterly and shown to the user; beta = 2-year weekly regression vs NIFTY 50, Blume-adjusted, with a sector-median fallback; terminal growth g must be at least 2 percentage points below WACC and not above a documented long-run nominal GDP growth assumption. The DCF always ships with a 5 x 5 sensitivity grid on WACC and g.

Justified P/B for financials: P/B = (ROE - g) / (k_e - g). Gordon DDM: P = DPS_1 / (k_e - g).

## 10.4 Statistical price forecasting

**Baseline models (always computed, used as benchmarks):**

- B0 No-change: forecast price at horizon h = current price.
- B1 Drift: log-price random walk with drift equal to the long-run equity risk premium assumption; interval from historical volatility.
- B2 Market: stock return = beta x NIFTY 50 expected return (from B1 on the index).

B1's interval for horizon h trading days, with sigma the daily log-return volatility (EWMA, lambda 0.94, or GARCH(1,1) if validated):

```
P_{lo,hi} = P_0 \exp\left( \mu h \pm z_{\alpha} \sigma \sqrt{h} \right)
```

**Challenger models (published only if they beat baselines out of sample):** factor-based expected-return models (value, quality, momentum, low volatility, earnings revisions where licensed), gradient-boosted models on point-in-time features, quantile regression

for interval estimation, and conformal prediction to calibrate intervals. LSTM, GRU and time-series Transformers are evaluated under the policy in 10.9 and deployed only if they pass the same publication gate; they carry no presumption of superiority [D].

## 10.5 Bias and leakage controls

- Point-in-time data only: fundamentals usable from filing timestamp; index membership as of each date; prices adjusted only with corporate actions known at that date for feature construction.
- Survivorship: delisted and merged stocks remain in the backtest universe.
- Walk-forward validation: rolling origin, expanding or 5-year rolling training window, purged gap equal to forecast horizon between train and test (no overlapping labels), final hold-out period never used for tuning.
- Multiple-testing control: record every model variant tried; report the deflated performance or apply a Bonferroni/False Discovery Rate correction when choosing among variants.
- Costs: strategy-style evaluations include brokerage, STT, exchange charges, GST, stamp duty and SEBI fees at the rates in force (from a versioned cost table), plus slippage by liquidity bucket (e.g. 5 bps NIFTY 50, 15 bps NIFTY Next 50, 30+ bps smaller caps) [A] to be calibrated on licensed trade data.

## 10.6 Forecast evaluation

| Measure | Definition | Use |
|---|---|---|
| MAE / RMSE of log price | Error of median forecast | Point accuracy vs B0, B1 |
| Skill score | 1 - (error_model / error_B0) | Above 0 means better than no-change |
| Diebold-Mariano test | Significance of error difference vs baseline | Publication gate |
| Interval coverage | Share of outcomes inside the 50% and 80% intervals | Calibration |
| Pinball loss / CRPS | Quality of the full distribution | Model selection |
| Directional hit rate | Share of correct sign | Reported, never used alone |

| Measure | Definition | Use |
|---|---|---|
| Information coefficient | Rank correlation of forecast and realised returns across stocks | Cross-sectional usefulness |

Publication gate [D]: a model, horizon and universe segment is published only if, on the final hold-out of at least 3 years, (1) skill vs B0 is positive with Diebold-Mariano $p < 0.05$, and (2) 80% interval coverage is between 75% and 85%. Otherwise only baselines (clearly labelled as naive) and fundamental scenarios are shown. These thresholds are proposals for the quant lead and compliance to confirm (OD-07).

## 10.7 Confidence rating rubric

Confidence rates the evidence, not the chance of profit.

| Factor | High | Medium | Low |
|---|---|---|---|
| Data completeness | All required fields, 5+ years | Minor gaps, 3 to 5 years | Gaps or under 3 years |
| Source tier | All Tier 1 | Mixed | Mostly Tier 2/3 |
| Method agreement | Method midpoints within 15% | 15 to 35% apart | Over 35% apart |
| Model validation | Passed gate for this segment | Passed for broader segment | Not validated |
| Liquidity | Above 10 crore median daily value | 1 to 10 crore | Below 1 crore |

Overall = lowest factor rating, with a one-line reason per factor [D]. "Not rated" when any factor cannot be assessed.

## 10.8 Hypothetical example (illustrative data only)

HYPOTHETICAL company "ABC Industries Ltd", price INR 1,000. Methods: DCF midpoint INR 1,080 (range 940 to 1,250 across the sensitivity grid); peer P/E midpoint INR 1,020 (range 900 to 1,150); own 5-year P/E median gives INR 960 (range 850 to 1,100). Combined range = interquartile span of midpoints (960, 1,020, 1,080) = INR 990 to 1,050. Displayed: "Fundamental estimate range INR 990 to 1,050 (-1% to +5% vs price). Method agreement: within 15%, so Medium-to-High depending on other factors." Scenarios: bull INR 1,250

(revenue growth 18%, EBITDA margin 21%, exit P/E 30), base INR 1,040 (13%, 19%, 26), bear INR 780 (6%, 15%, 20). Invalidation for base: "EBITDA margin below 17% for two consecutive quarters" or "net debt/EBITDA above 2.5x". None of these figures refer to a real company.

## 10.9 Model selection policy (Pillar 3)

Each analytical task gets its own candidate set, and a model is chosen only on validated out-of-sample evidence, data quality, horizon, complexity and running cost; the simplest model that passes the gate wins ties.

### 10.9.1 Candidates by task

| Task | Candidates evaluated | Selection metrics | Notes |
|---|---|---|---|
| Short-term price and return forecasting (1 and 3 months) | Naive no-change and historical-average baselines; ARIMA/ETS and GARCH-type volatility models; linear and regularised regression (ridge, lasso, elastic net) on point-in-time features; XGBoost and LightGBM; LSTM or GRU where data volume justifies; time-series Transformers only if they beat simpler models on the hold-out | MAE, RMSE, skill vs baselines, directional accuracy, 50%/80% interval coverage, pinball loss | Deep models need enough history across many stocks; a pooled cross-sectional model is preferred over per-stock deep models |

| Task | Candidates evaluated | Selection metrics | Notes |
|---|---|---|---|
| Medium-term forecasting (6 and 12 months) | Baselines; factor models (value, quality, momentum, low volatility); gradient boosting on fundamentals and factors; quantile regression with conformal calibration | Same as above plus information coefficient | Overlapping horizons require purged, embargoed validation |
| Long-term valuation | DCF, comparable companies, historical multiples, earnings and cash-flow models, DDM where appropriate, bull/base/bear scenarios | Method eligibility rules (10.2), sensitivity, method agreement | Deterministic; no ML replaces valuation |
| Screening and ranking | Fundamental filters; quality, value and momentum factor scores; sector-relative ranks; learning-to-rank or classification models (gradient boosting) | Rank IC, decile spread after costs, turnover, maximum drawdown, risk-adjusted return | Ranking output gated by REG-01 |
| Derivatives | Black-Scholes-Merton / Black-76, IV solver, deterministic payoffs on verified contract specs | Exactness tests | No ML |
| Risk | Deterministic stop and sizing formulas; historical VaR and expected shortfall | Backtest exceptions | No ML |

### 10.9.2 Validation requirements

- Chronological train, validation and test splits; walk-forward or expanding-window validation with a purge gap equal to the horizon.
- Leakage prevention and point-in-time data (10.5); corporate actions applied as known at each date.
- Comparison with baselines on the same windows; evaluation by period, sector and market regime (trend up, trend down, high volatility).
- Transaction costs and slippage in any strategy-style test.
- Selection never on historical returns alone or on a single attractive test.
- Every candidate and its result recorded in the selection report, including those not chosen.

### 10.9.3 Forecast output contract

Each forecast record holds: symbol, exchange, ISIN; verified market price and its timestamp; forecast target (price or return) and horizon; model ID and version; baseline forecast and the model's skill versus it; forecast value or range (median and 50%/80% intervals); bull, base and bear scenarios where suitable; assumptions and uncertainty notes; out-of-sample performance summary for that segment; data snapshot and source references; major risks and invalidation conditions; and a status of `available`, `insufficient_data`, or `model_not_validated`. Reports always label the market price, the model forecast and the fundamental fair value separately.

### 10.9.4 Retraining

Retraining runs on a documented schedule (default monthly, outside the three research slots) or on a drift trigger (NFR-043), produces a validation report, and a new version enters production only after approval in the registry (FR-181). A new daily candle alone never triggers retraining.

## 11. Stop-loss and risk management methodology

The risk module is a calculator over user-supplied entry, capital and risk budget: it shows several stop methods side by side, sizes the position from the chosen stop, and always warns that a stop-loss order does not guarantee execution at the stop price, especially through gaps, circuit locks or thin liquidity.

## 11.1 Inputs and outputs for every trading scenario

| Output | Definition |
|---|---|
| Entry | User-entered or "assumed entry = last price" (labelled) |
| Stop price and method | One row per method in 11.2 |
| Risk per share | Entry - stop (long); stop - entry (short) |
| Position size | floor(risk budget / risk per share), then caps (11.3); lots for derivatives |
| Targets | Resistance zone, R-multiples (1R, 2R, 3R), fair-value range edge; each labelled with its method |
| Reward-to-risk | (Target - entry) / (entry - stop) for longs |
| Volatility and liquidity notes | ATR %, stop distance in ATR units, median traded value, position as % of daily volume |
| Gap and slippage risk | Historical frequency of opens beyond the stop distance; worst observed overnight gap; liquidity-based slippage estimate |
| Unsuitability reasons | From 11.4 |

## 11.2 Stop methods (long position; short is the mirror image)

| Method | Formula | Parameters (default) | When it is weak |
|---|---|---|---|
| Support-based | Stop = lower edge of nearest support zone - buffer; buffer = 0.25 x ATR | Zone from FR-032 | No clear support; zone too far (over 3 ATR) |
| ATR-based | Stop = entry - k x ATR(14) | k = 2 (range 1.5 to 3) | Volatility regime shift; event days |
| Percentage | Stop = entry x (1 - p) | p = 5% | Ignores the stock's volatility; 5% is noise for volatile small caps |

| Method | Formula | Parameters (default) | When it is weak |
|---|---|---|---|
| Volatility-adjusted | Stop = entry x exp(-z x sigma_d x sqrt(h)) | z = 2, h = expected holding days, sigma_d = EWMA daily vol | Fat tails; assumes normal returns |
| Trailing (chandelier) | Stop_t = max(stop_t-1, highest high since entry - m x ATR) | m = 3 | Gives back gains in fast reversals |
| Trailing (percentage) | Stop_t = max(stop_t-1, highest close since entry x (1 - p)) | p = 8% | Same as percentage |

## 11.3 Position sizing

```
Q = \min\left( \left\lfloor \frac{R_{budget}}{|E - S|} \right\rfloor,\ \left\lfloor \frac{w_{max} \cdot C}{E} \right\rfloor,\ \left\lfloor \lambda \cdot V_{20} \right\rfloor \right)
```

Where R_budget = risk budget in INR (capital C x risk %), E = entry, S = stop, w_max = maximum position weight (default 20%, user-editable), V_20 = 20-day median daily volume in shares, lambda = liquidity participation cap (default 1%). For derivatives, Q is rounded down to whole lots using the lot size from the contract master (FR-080), and risk per lot = |E - S| x lot size. If Q = 0, the calculator says the trade cannot be sized within the risk budget.

## 11.4 Suitability checks (FR-075)

- Stop within 0.5 ATR of entry: "likely inside normal daily noise".
- Stop distance over 4 ATR: "risk per share is large; position will be small".
- Stop placed exactly at a round number or the edge of a widely visible level: "common clustering point; consider that many orders may sit here".
- Stop below the lower circuit band or the stock is circuit-locked: "exit may be impossible until the lock clears".
- Upcoming results, board meeting or ex-date before the holding horizon: "event gap risk".

- Stop chosen produces R:R above 3 while sitting inside 1 ATR: "attractive ratio may come from an unrealistically tight stop". The calculator never proposes moving a stop to improve R:R.

## 11.5 Worked example (HYPOTHETICAL values)

HYPOTHETICAL stock "XYZ Ltd": entry INR 500, ATR(14) INR 12, support zone INR 478 to 482, resistance zone INR 546 to 550, daily volatility 2%, capital INR 5,00,000, risk budget 1% = INR 5,000, max position weight 20%.

| Method | Stop (INR) | Risk/share (INR) | Uncapped qty | Final qty (20% cap = 200 shares) | Actual risk (INR) |
|---|---|---|---|---|---|
| ATR, k = 2 | 476.00 | 24.00 | 208 | 200 | 4,800 |
| Support-based | 478 - 3 = 475.00 | 25.00 | 200 | 200 | 5,000 |
| Percentage 5% | 475.00 | 25.00 | 200 | 200 | 5,000 |
| Volatility, z = 2, h = 5 days | 500 x exp(-2 x 0.02 x 2.236) = 457.22 | 42.78 | 116 | 116 | 4,962 |

Reward-to-risk using the ATR stop and the lower edge of resistance (546): (546 - 500) / 24 = 1.92. Gap illustration: if XYZ opens at INR 465 after bad news, the ATR stop fills near 465, not 476, and the loss on 200 shares is INR 7,000 (1.4% of capital) instead of INR 4,800. All values are hypothetical.

## 11.6 Portfolio-level risk (FR-076)

- Historical VaR (95%, 1-day) from 500 days of portfolio returns using current weights; 10-day by square-root scaling only as an approximation, labelled.
- Expected shortfall (97.5%) shown beside VaR.
- Concentration: largest position, top-5 weight, sector weights vs NIFTY 50.
- Beta and correlation clusters; stress replays of named historical episodes (e.g. March 2020) using actual historical returns.
- No suggestion to buy or sell any holding; outputs are descriptive.

# 12. Options and derivatives requirements

Derivatives analytics compute payoffs, Greeks and risk from the exchange's current contract specifications loaded daily; nothing about lot sizes, expiry days or eligibility is hardcoded, because SEBI and the exchanges changed all three between Nov 2024 and Jan 2026.

## 12.1 Current rules the engine must respect (verified 11 Oct 2026)

| Rule | Current position [F] | Source | Engine requirement |
|---|---|---|---|
| Index derivative contract size | Raised to INR 15 to 20 lakh at introduction, from 20 Nov 2024 | SEBI circular 1 Oct 2024, Zerodha summary | Lot size from contract master per contract, by effective date |
| Lot sizes change over time | NSE cut NIFTY lot from 75 to 65 and Bank Nifty from 35 to 30 for contracts from Jan 2026 expiries (NSE circular of 3 Oct 2025) | Business Standard | Historical analytics use the lot size valid for each contract |
| Weekly expiries | Each exchange may offer weekly expiry on only one benchmark index | Zerodha summary | Expiry calendar from contract master only |
| Expiry day | Expiry limited to Tuesday or Thursday (SEBI circular 26 May 2025); NSE equity F&O expire on Tuesdays and BSE on Thursdays from 1 Sep 2025 | Taxmann; Moneylife | Holiday-adjusted expiry dates from exchange files |
| Option premium | Buyers pay full premium upfront | Zerodha summary | Capital required for long options = premium x quantity |

| Rule | Current position [F] | Source | Engine requirement |
|---|---|---|---|
| Expiry-day margins | Extra 2% ELM on short index options on expiry day; no calendar-spread margin benefit on expiry day (from 10 Feb 2025) | Same | Margin warnings on expiry day |
| Position limits | Delta-based Future Equivalent OI; PAN-level index option limits of INR 1,500 crore net and INR 10,000 crore gross, fully from 6 Dec 2025; MWPL revised from 1 Oct 2025 | Probe42 summary | Show FutEq OI where data allows; ban-period flags from exchange |

[A] Index options are cash-settled and European-style; stock derivatives are physically settled at expiry. The contract master and exchange product specifications are the source of truth and must be checked during build.

### 12.2 Payoff formulas at expiry (per unit; multiply by quantity)

With S_T = underlying at expiry, K = strike, p = premium paid or received:

| Position | Payoff at expiry | Max profit | Max loss | Break-even |
|---|---|---|---|---|
| Long call | max(S_T - K, 0) - p | Unlimited | p | K + p |
| Short call | p - max(S_T - K, 0) | p | Unlimited | K + p |
| Long put | max(K - S_T, 0) - p | K - p (if S_T falls to 0) | p | K - p |
| Short put | p - max(K - S_T, 0) | p | K - p | K - p |
| Long future at F_0 | S_T - F_0 | Unlimited | F_0 (to zero) | F_0 |

| Position | Payoff at expiry | Max profit | Max loss | Break-even |
|---|---|---|---|---|
| Short future | F_0 - S_T | F_0 | Unlimited | F_0 |

Terminology guard [D]: a long put is the right to sell; its loss is limited to the premium. Shorting the stock or a future is an obligation with unlimited loss. A short put is an obligation to buy at K, not a bearish trade. The glossary and AI layer must never conflate these; TST-DER-03 tests it.

## 12.3 Values, Greeks and implied volatility

Intrinsic value: call = max(S - K, 0), put = max(K - S, 0). Time value = premium - intrinsic. Moneyness: ITM, ATM (strike nearest spot), OTM.

```
C = S e^{-qT} N(d_1) - K e^{-rT} N(d_2), \quad P = K e^{-rT} N(-d_2) - S e^{-qT} N(-d_1)
```

```
d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2) T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T}
```

Greeks: delta (call) = e^(-qT) N(d1); gamma = e^(-qT) n(d1) / (S sigma sqrt T); vega = S e^(-qT) n(d1) sqrt T (shown per 1 volatility point); theta shown per calendar day; rho per 1% change in r. r = 91-day T-bill yield (source and date shown); q = expected dividend yield for stocks, or index dividend yield. For index options, the engine may use the synthetic forward from the nearest future (Black-76) and must say which.

IV is solved per strike by Newton-Raphson on vega, with bisection fallback; no solution returns "IV not computable". Historical volatility = annualised standard deviation of daily log returns (x sqrt(252)) over 20 and 60 days. IV percentile and IV - HV spread are displayed with their windows.

## 12.4 Open interest and put-call ratio

- PCR (OI) = total put OI / total call OI for an expiry; PCR (volume) likewise.
- Mandatory interpretation note: "PCR mixes hedgers, writers and speculators. A high PCR does not mean the market will rise or fall. There is no reliable threshold." [D]
- OI change by strike, with long build-up / short build-up labels only when price and OI change are both available, and labelled as "possible interpretation".

## 12.5 Strategy library (MVP)

| Strategy | Market assumption | Max gain | Max loss | Break-even(s) | Key risks |
|---|---|---|---|---|---|
| Long call | Rise beyond K + p before expiry | Unlimited | Premium | K + p | Time decay; IV crush |
| Long put | Fall below K - p | K - p | Premium | K - p | Time decay |
| Covered call | Flat to mildly up | K - S_0 + p | S_0 - p (stock to zero) | S_0 - p | Caps upside; physical settlement |
| Protective put | Hedge downside | Unlimited | S_0 - K + p | S_0 + p | Cost of hedge |
| Bull call spread (buy K1 call, sell K2 call, K1 < K2) | Moderate rise | K2 - K1 - net debit | Net debit | K1 + net debit | Expiry-day margin on short leg |
| Bear put spread | Moderate fall | K2 - K1 - net debit | Net debit | K2 - net debit | As above |
| Short straddle | Low volatility | Net credit | Unlimited | K +/- credit | Gap risk; large margin; expiry-day ELM |
| Long straddle | Large move either way | Unlimited | Net debit | K +/- debit | Time decay |
| Iron condor | Range-bound | Net credit | Wing width - credit | Short strikes +/- credit | Gap through wings |

Each strategy page shows: legs, payoff diagram at expiry and today, max gain/loss (or "unlimited"), break-evens, days to expiry, estimated margin only where FR-084 is enabled, liquidity of each strike (bid-ask, OI), and risks including assignment and physical settlement for stock options, expiry-day volatility, and the 2% ELM and calendar-spread rules on expiry day.

## 12.6 Derivatives scope by phase

- Phase 1 (MVP): payoff builder and Greeks calculator for NIFTY 50 and SENSEX options using delayed or end-of-day data; education-first; risk acknowledgement gate (FR-087).
- Phase 2: option chains, OI, IV analytics for index options.
- Phase 3: stock options and futures; margin estimates if risk-parameter files are licensed. Any strategy suggestion for a named underlying is gated by REG-01.

# 13. Financial glossary requirements

The glossary launches with 300 reviewed terms in English and Hindi across eight domains, each entry following one schema, and every term used in a report template links to its entry.

## 13.1 Entry schema

| Field | Required | Rule |
|---|---|---|
| Term (EN), aliases, abbreviation | Yes | e.g. "Price-to-earnings ratio", "P/E", "PE multiple" |
| Term (HI) and transliteration | Yes | e.g. मूल्य-आय अनुपात; Hinglish alias for search |
| Plain-English definition | Yes | At most 40 words, reading age about 12 |
| Hindi explanation | Yes for MVP set | Written (not machine-only) and reviewed by a bilingual finance editor |
| Indian-market example | Yes | Hypothetical numbers clearly labelled, or a real case citing a filing with data lagged per REG-08 when shown in learning content |
| Formula | If applicable | Same formula ID as Appendix A, so glossary and engine never diverge |
| When it is useful | Yes | One or two sentences |
| Common mistakes | Yes | At least two |
| Related terms | Yes | At least two links |
| Official source | Where relevant | SEBI, NSE, BSE, RBI, ICAI or Ind AS reference |
| Domain tags, difficulty | Yes | Fundamentals, technicals, options, futures, portfolio risk, market structure, corporate actions, valuation |
| Review metadata | Yes | Author, reviewer, review date, version |

## 13.2 Coverage for MVP (300 terms) [D]

| Domain | Count | Examples |
|---|---|---|
| Fundamentals and accounting | 70 | EBITDA, ROCE, TTM, consolidated vs standalone, exceptional item, contingent liability |
| Valuation | 35 | DCF, WACC, terminal value, EV/EBITDA, margin of safety, justified P/B |
| Technical analysis | 45 | SMA, EMA, RSI, MACD, ATR, support, breakout, drawdown, relative strength |
| Options | 50 | Call, put, strike, premium, intrinsic value, theta, IV, OI, PCR, assignment, ITM/ATM/OTM |
| Futures | 20 | Lot size, mark-to-market, basis, cost of carry, rollover, physical settlement |
| Portfolio and risk | 30 | Stop-loss, position sizing, R-multiple, VaR, beta, correlation, gap risk |
| Market structure | 30 | Circuit limits, T+1 settlement, pre-open session, ASM/GSM, F&O ban, ISIN |
| Corporate actions | 20 | Split, bonus, rights issue, record date, ex-date, buyback, demerger |

## 13.3 Example entry (abridged)

**Put option (पुट ऑप्शन).** Definition: a contract that gives the buyer the right, but not the obligation, to sell the underlying at the strike price on expiry. Hindi: खरीदार को तय कीमत (स्ट्राइक) पर बेचने का अधिकार, बाध्यता नहीं। Example (HYPOTHETICAL): a NIFTY put with strike 24,000 bought for premium 120 points; at expiry with NIFTY at 23,700 the payoff is 24,000 - 23,700 - 120 = 180 points per unit. Formula: max(K - S_T, 0) - p. Useful for: hedging a portfolio against a fall. Common mistakes: thinking buying a put is the same as short selling (loss on a long put is limited to the premium); ignoring time decay. Related: call option, strike price, protective put, theta. Source: exchange product specifications.

## 13.4 Contextual behaviour

- Tagged terms in any report show a dotted underline; tap opens a short card (definition, formula, "how it's used on this page").

- "Explain this number" on any metric opens the evidence inspector (FR-062) plus the glossary card.
- AI explanations (FR-094) retrieve only from glossary and methodology pages, and refuse stock-specific advice with a fixed message.

# 14. Data source and licensing matrix

Exchange market data is owned by the exchanges and may be redistributed only under a signed licence, and NSE's website terms prohibit scraping, so the Platform will buy all prices, corporate actions and derivatives data through exchange licences or an exchange-authorised vendor; free public pages are used only for reading official documents by people, never as a data feed.

## 14.1 Verified licensing facts

- [F] NSE's Data Sharing and Usage Policy covers live, snapshot, delayed, end-of-day, historical and tick data; ownership stays with NSE/NSE Data, redistribution is allowed only as the signed agreement permits, commercial use is priced by NSE Data, and data may not be used to create indices or supplied to gaming, virtual trading or simulation platforms without separate permission (NSE Data Sharing and Usage Policy).
- [F] NSE's website terms of use prohibit systematic or automated data collection, scraping, data mining and harvesting, and reproduction or commercial distribution of content without NSE's written permission (NSE Terms of Use).
- [F] BSE licenses market data to domestic clients directly; international licensing moves in-house from Deutsche Börse on 1 Jan 2027 (Kotak Neo report).
- Implication [D]: the virtual "paper trading" or simulation features some competitors offer would need separate NSE permission, so they are out of scope.

## 14.2 Matrix

| Data domain | Primary source | Key fields | Update frequency and latency target | Licensing and redistribution | History needed | Validation rules | Fallback |
|---|---|---|---|---|---|---|---|
| Equity prices (EOD) | NSE and BSE via licence or authorised vendor | OHLC, close, prev close, volume, value, trades, delivery % | Daily, within 60 minutes after the exchange publishes final close data | Licence required for display to end users; terms decide display, export and derived-data rights | 10+ years | V1 to V6 (14.3) | Second vendor or direct exchange EOD files under licence |

| Data domain | Primary source | Key fields | Update frequency and latency target | Licensing and redistribution | History needed | Validation rules | Fallback |
|---|---|---|---|---|---|---|---|
| Equity prices (intraday) | Licensed vendor | Last trade, bid/ask, volume, 1-min bars | Real-time or 15-min delayed per licence tier (OD-04) | Real-time display has higher fees and per-user reporting [A] | 2 years of 1-min bars | V1, V2, V7 | Fall back to delayed with banner |
| Corporate actions | Exchange filings via vendor | Type, ratio, record date, ex-date, amount | Daily plus intraday on announcement | Vendor licence | Full history | V8 cross-check with price gap on ex-date | Manual curation queue |
| Index levels and constituents | Index provider (NSE Indices, Asia Index for BSE indices) via licence | Levels, constituents, weights, rebalance dates | Daily; intraday levels per licence | Index data licences are separate from exchange price licences [A] | 10+ years with historical constituents | V9 | None; index tiles show unavailable |
| Derivatives | NSE/BSE F&O via licence | Contract master, quotes, OI, settlement prices, ban list, risk parameters | Daily contract files; delayed or real-time quotes | Licence required | 5 years | V10 to V12 | Analytics disabled if contract master missing |
| Financial statements | Fundamentals vendor plus filings | Standardised and as-reported statements, notes, segment data | Within 24 hours of filing [A] | Vendor licence; check derived-data and display rights | 10 years quarterly where available | V13 to V15 | Parse filing XBRL/PDF in-house (Phase 3) |
| Shareholding patterns | Exchange filings via vendor | Promoter, pledge, FPI, DII, public | Quarterly | Vendor licence | 10 years | V16 | Filing PDFs |
| Corporate announcements | NSE and BSE announcements via licensed feed | Category, timestamp, attachment | Near real time | Licence for redistribution of text; linking to official PDF is preferred [A] | 5 years | V17 dedup | Link-only mode |
| Macro data | RBI (policy rates, DBIE), MoSPI (CPI, GDP; eSankhyiki portal), FBIL (reference rates) | Rate, CPI, GDP, INR reference rate | On official release | Government data terms; attribution required [A] check each portal's terms | 20 years | V18 vintage tracking | Manual entry with second-person check |
| Bond yields and T-bills | RBI/CCIL or licensed vendor | 10-year G-sec, 91-day T-bill | Daily | Check terms [A] | 10 years | V1 | Previous day with stale flag |
| News | Licensed news API | Headline, body or abstract, time, source, entities | Near real time | Display rights as per licence; many allow headline + link only [A] | 2 years | V19 entity linking, dedup | Headline-only mode |
| Regulatory publications | SEBI, RBI, exchange circulars | Circulars, regulations | Daily check | Public documents; cite and link | n/a | Manual review by compliance | n/a |
| Glossary and methodology | In-house editorial | Entries | On change | Owned | n/a | Editorial review | n/a |

Candidate vendors for the Phase 0 RFP are to be shortlisted by data engineering and must demonstrate exchange authorisation, redistribution terms for a retail web app, SLA, and historical point-in-time fundamentals. No vendor is assumed in this BRD (OD-02).

## 14.3 Data quality rules

| ID | Check | Rule [D] | Action on failure |
|---|---|---|---|
| V1 | Staleness | Price timestamp older than tier SLA (NFR-010) | Stale badge; suppress forecasts and stop calculators that depend on it |
| V2 | Missing values | Null in required field | Mark field missing; never forward-fill fundamentals; forward-fill prices only for display of non-trading days, labelled |
| V3 | Range | Close outside the day's circuit band or high/low | Quarantine bar; incident |
| V4 | Jump detection | Absolute return over 20% without a corporate action or circuit event | Hold for review |
| V5 | OHLC consistency | Low <= open, close <= high; volume >= 0 | Reject bar |
| V6 | Cross-source | Close differs between primary and secondary source by more than 0.05% | Use exchange-official value; log |
| V7 | Intraday sequence | Out-of-order or duplicate ticks | Drop and log |
| V8 | Adjustment check | Adjusted series shows a gap on ex-date beyond the expected ratio +/- 2% | Recompute factor; review |
| V9 | Index membership | Constituent weights sum to 100% +/- 0.1% | Hold index update |
| V10 | Contract master | Every listed contract has lot size, expiry and strike | Disable analytics for incomplete contracts |
| V11 | Option price sanity | Premium below intrinsic beyond tick tolerance; put-call parity breach above threshold | Flag strike; exclude from IV surface |

| ID | Check | Rule [D] | Action on failure |
|---|---|---|---|
| V12 | Lot size change | Lot size differs from previous day for the same contract | Verify against exchange circular before accepting |
| V13 | Accounting identity | Assets = liabilities + equity within 0.1%; cash-flow reconciliation | Flag statement |
| V14 | Period consistency | Fiscal period labels and lengths (3 or 12 months); restated periods stored separately | Reject mislabelled period |
| V15 | Conflicting figures | Vendor vs filing difference over 0.5% | Use filing; ticket to vendor |
| V16 | Shareholding totals | Categories sum to 100% +/- 0.01% | Flag |
| V17 | Duplicate announcements | Same document on NSE and BSE | Merge, keep both links |
| V18 | Macro vintages | Revised release replaces value but keeps previous vintage | Store both |
| V19 | News entity links | Confidence below threshold | Unlinked |

Timestamps and time zones: all times are stored in UTC with the source's original timestamp and time zone, and displayed in IST (UTC+05:30). Trading calendars come from exchange holiday lists. Each value carries source ID, ingestion time and as-of time.

## 14.4 Persistent historical data store and incremental updates (mandatory)

Historical data is downloaded once at initial setup, saved as validated CSV files, and reused for every later analysis; each later run fetches only missing or newly published records, and the full history is never re-downloaded unless an operator explicitly requests it or a documented recovery needs it. This is a core architectural requirement set by the business sponsor on 11 Oct 2026, not an optimisation.

**Design decision [D].** The CSV store is the persisted system of record for raw and processed market data. Any query engine or warehouse in section 15 reads from it and can be rebuilt from it; it never becomes a second, divergent copy.

**Licensing constraint.** [F] NSE's policy keeps ownership of market data with NSE, allows use only as the signed agreement permits, and needs prior approval for offline media (NSE Data Sharing and Usage Policy). [A] The vendor contract (OD-02) must therefore permit persistent local storage, say how long data may be kept after the licence ends, and state whether stored CSVs may be shared with users. Until then, CSV files are internal only and never offered for download.

### 14.4.1 Directory layout

The layout separates exchange, frequency, data type and adjustment convention, so incompatible data is never mixed in one file.

```
data/
├── raw/                          # as received, never edited in place
│   ├── daily/NSE/<SYMBOL>.csv    # unadjusted daily OHLCV, one file per ISIN-
symbol
│   ├── daily/BSE/<SCRIPCODE>.csv
│   ├── intraday/<EXCH>/<INTERVAL>/<SYMBOL>/<YYYY-MM>.csv
│   ├── corporate_actions/<EXCH>.csv
│   ├── fundamentals/<ISIN>/statements.csv
│   ├── news_events/<YYYY-MM>.csv
│   └── derivatives/<EXCH>/contract_master/<YYYY-MM-DD>.csv and
bhavcopy/<YYYY-MM>.csv
├── processed/
│   ├── adjusted/split_bonus/<EXCH>/<SYMBOL>.csv
│   ├── adjusted/total_return/<EXCH>/<SYMBOL>.csv
│   ├── features/<FEATURE_SET_VERSION>/<SYMBOL>.csv
│   └── model_inputs/<DATASET_ID>.csv
├── metadata/
│   ├── manifest.csv
│   ├── download_log.csv
│   ├── data_quality_log.csv
│   ├── revisions_log.csv
│   ├── failed_ranges.csv         # retry queue
│   ├── trading_calendar/<EXCH>.csv
│   └── schemas/<dataset>_v<N>.json
├── checkpoints/                  # resumable large downloads
├── backups/<YYYY-MM-DD>/         # snapshot before any bulk rewrite
└── models/
    ├── model_registry.json
    └── artifacts/<MODEL_ID>/<VERSION>/
```

Credentials, tokens and passwords are never written under `data/`; they live in a secrets manager. The `data/` root sits on persistent storage (a mounted volume or object store with versioning), never on ephemeral container disk, so data survives restarts, reboots and redeploys.

### 14.4.2 CSV schemas (version 1)

Common rules: UTF-8, comma-separated, header row, one record per line; dates as ISO 8601 `YYYY-MM-DD` in the exchange's trading date; timestamps as ISO 8601 UTC with `Z`, and IST shown only in the UI; prices as decimals in INR with no thousands separators; volumes as integers; empty field means missing, never zero. Every file records its schema version in the manifest. Example values below are placeholders, not market data.

| Dataset | Columns | Natural key |
|---|---|---|
| Daily equity (raw) | date, isin, symbol, exchange, series, open, high, low, close, prev_close, volume, traded_value_inr, trades, currency, adjustment (always `raw`), source, source_record_id, fetched_at, revision | exchange, isin, series, date |
| Daily equity (adjusted) | Raw columns plus adj_factor_cum, adj_open, adj_high, adj_low, adj_close, adj_volume, adjustment (`split_bonus` or `total_return`), ca_version | exchange, isin, series, date, adjustment |
| Intraday candles | ts_start_utc, ts_end_utc, isin, symbol, exchange, interval, open, high, low, close, volume, is_complete, source, fetched_at, revision | exchange, isin, interval, ts_start_utc |
| Corporate actions | event_id, isin, symbol, exchange, action_type, ratio_num, ratio_den, amount_inr, announcement_date, ex_date, record_date, details, source, source_url, fetched_at, revision | exchange, isin, action_type, ex_date |

| Dataset | Columns | Natural key |
|---|---|---|
| Financial statements | isin, statement (`pl`, `bs`, `cf`), basis (`standalone`, `consolidated`), period_end, period_months, fiscal_label, line_item_code, line_item_name, value_inr, as_reported (`true`/`false`), filing_date, filing_ts_utc, restated_from_filing_id, source, fetched_at, revision | isin, statement, basis, period_end, line_item_code, filing_ts_utc |
| News and events | event_id, published_ts_utc, isin_list, source_name, source_tier, status (`confirmed`, `reported`, `speculative`), category, headline, url, licence_display_rights, fetched_at | event_id |
| Derivatives (EOD) | date, exchange, instrument (`FUTIDX`, `OPTIDX`, `FUTSTK`, `OPTSTK`), underlying, expiry, strike, option_type, open, high, low, close, settle_price, volume_contracts, open_interest, change_in_oi, lot_size, source, fetched_at, revision | exchange, instrument, underlying, expiry, strike, option_type, date |
| Contract master | as_of_date, exchange, instrument, underlying, expiry, strike, option_type, lot_size, tick_size, settlement_type, source, fetched_at | exchange, instrument, underlying, expiry, strike, option_type, as_of_date |
| manifest.csv | dataset, exchange, isin, symbol, frequency, adjustment, file_path, schema_version, coverage_start, coverage_end, last_confirmed_date, row_count, sha256, status (`complete`, `partial`, `stale`, `failed`), last_successful_update_utc, source, missing_ranges_count | dataset, exchange, isin, frequency, adjustment |

| Dataset | Columns | Natural key |
|---|---|---|
| download_log.csv | run_id, started_utc, ended_utc, mode (`initial`, `incremental`, `repair`, `full_refresh`), dataset, exchange, symbol, requested_from, requested_to, api_calls, records_received, records_new, records_duplicate, records_revised, http_status, error_code, error_message, retries | run_id, dataset, exchange, symbol, requested_from |
| data_quality_log.csv | run_id, dataset, exchange, symbol, rule_id (V1 to V19), severity, date_or_ts, detail, action_taken | run_id, rule_id, symbol, date_or_ts |
| revisions_log.csv | revision_id, dataset, natural_key, field, old_value, new_value, old_source, new_source, precedence_rule, detected_run_id, applied_utc | revision_id |

A sample header with an illustrative row (placeholders only):

```
date,isin,symbol,exchange,series,open,high,low,close,prev_close,volume,traded
_value_inr,trades,currency,adjustment,source,source_record_id,fetched_at,revi
sion
2026-10-
09,INE000X00000,EXAMPLE,NSE,EQ,100.00,102.50,99.10,101.20,100.40,123456,12493
000.00,4321,INR,raw,provider_name,abc-123,2026-10-09T13:30:00Z,1
```

### 14.4.3 Initial download (first run)

1. Read configuration: universe (T1 for MVP), exchanges, datasets, lookback (default 10 years daily, 2 years intraday) and provider.
2. Load or fetch the trading calendar for each exchange.
3. Split each symbol's lookback into chunks within the provider's request limits; write a checkpoint after each chunk.
4. Validate each chunk (rules V1 to V19); valid rows go to a staging file, invalid rows are logged and not saved.

5. When a symbol's chunks complete, write its CSV atomically and add a manifest row with `status = complete`. If any chunk failed, keep the valid rows, set `status = partial`, and add the failed date ranges to `failed_ranges.csv`.
6. Build the dataset index (manifest plus row counts and checksums).
7. Produce the initial data quality report from `data_quality_log.csv`.
8. Compute adjusted series and features only for datasets whose required inputs are `complete`; models are trained or initialised only after their training datasets pass validation.

#### 14.4.4 Incremental update (every later run)

1. Load the manifest; verify each file exists and its checksum and schema version match.
2. Determine `last_confirmed_date`: the latest date that was a completed trading session and passed validation.
3. Compute the expected trading dates from the exchange calendar between the coverage start and the provider's latest published date; their difference from the stored dates gives the missing ranges, plus any entries in `failed_ranges.csv`.
4. Ask the provider for its latest available date. Do not assume today has a completed session: weekends, exchange holidays, publication delays and the current session's incomplete candle are excluded (`is_complete = false` intraday rows are never treated as final).
5. Request only the missing ranges and new records, respecting rate limits, pagination and licence terms.
6. Fetch a short look-back window (default the last 5 trading days) [D] plus all corporate actions announced since the last run, to catch late corrections.
7. Validate incoming records, then merge on the natural key: new keys are appended; identical keys are dropped as duplicates; changed values are revisions resolved by the precedence rules in 14.4.5.
8. Write to a temporary file in the same directory, fsync, verify row count and checksum, then rename over the old file (atomic replace). The previous version is kept in `backups/` until the next successful run.
9. Update the manifest only after the write succeeds; log the run.
10. Recompute adjusted series and features only for affected symbols and dates, extended back by the longest rolling window of any affected feature (for example 200 days for a 200-day SMA) and fully for a symbol after a corporate action changes its adjustment factors.

### 14.4.5 Revisions and source precedence

- Raw rows are never edited in place: a revision writes a new row version (`revision` + 1) and an entry in `revisions_log.csv`; the merged raw file keeps the latest revision, and earlier revisions stay in the log and backups.
- Precedence [D]: exchange official file > licensed primary vendor > secondary vendor. A lower-precedence source never overwrites a higher one.
- Corporate actions trigger a rebuild of the symbol's adjusted series and features; raw prices are unchanged.
- Restated financial statements are stored as new rows with their own filing timestamp; the original stays for point-in-time use.
- Symbol changes and mergers are resolved by ISIN; files are keyed to ISIN in the manifest so a symbol rename does not create a new history.
- Raw, split-and-bonus-adjusted and total-return series are always separate files and labelled in every output.

### 14.4.6 Failure handling

| Failure | Behaviour |
|---|---|
| Network timeout, HTTP 5xx | Retry with exponential backoff (2 s, 4 s, 8 s, 16 s, 32 s, with jitter); then record in `failed_ranges.csv` |
| Rate limit (HTTP 429 or provider equivalent) | Honour the provider's retry-after; throttle the run-wide request rate |
| Response fails validation | Discard the response, keep the last valid CSV, log rule IDs |
| Response shorter than the stored history for the same range | Treat as incomplete; never replace the stored data |
| Process crash mid-download | Resume from the last checkpoint; staging files not renamed are discarded |
| Corrupted CSV (checksum mismatch) | Restore from `backups/`; if none, mark `failed` and schedule a repair download of that file only |

| Failure | Behaviour |
|---|---|
| Partial update | Manifest `status = partial`; outputs show a stale or incomplete warning and the dataset is never reported as current |

A full re-download runs only with an explicit operator command (`mode = full_refresh`) or the documented recovery procedure, and both are logged with who ran them and why. Valid historical data is never deleted without a retention policy entry approving it.

### 14.4.7 Prediction and training use

- The prediction engine never starts a full-history download. It loads the local validated data, runs the incremental update, checks that all required features exist and are aligned to the same as-of date, loads the approved model version and predicts.
- If required data is stale or incomplete after the update, the prediction is marked unavailable with the reason; it is never shown as current.
- Each prediction stores model ID and version, feature-set version, the manifest checksums and `last_confirmed_date` of every input file, and the run ID, so it can be reproduced exactly (FR-110).
- Retraining follows a documented schedule (default monthly) or an evidence trigger (drift alert per NFR-043), not every new candle [D].

### 14.4.8 Operational metrics

Each run reports: records downloaded (initial and per update), API calls, failures and retries, duplicates removed, revisions applied, missing intervals remaining, data freshness per dataset, update duration, storage used, and prediction run time. These feed the operations console (FR-126).

Each run also reports reused historical records (rows read from the store rather than downloaded), API calls by provider, and repository storage growth.

### 14.4.9 Persistence across GitHub Actions runs

GitHub-hosted runners start empty and are discarded after each job, so `data/` lives on a dedicated `data` branch of a private repository and is checked out at the start of every run and committed at the end; nothing persistent is kept on runner disk.

**Verified platform facts** (read 11 Oct 2026):

- [F] Git warns on files over 50 MiB and GitHub blocks files over 100 MiB; GitHub recommends repositories stay ideally under 1 GB and strongly under 5 GB (GitHub Docs: large files).
- [F] Scheduled workflows run only on the default branch, may be delayed under high load (and queued jobs may be dropped), use UTC cron by default, and in public repositories are disabled after 60 days without repository activity (GitHub Docs: events that trigger workflows).
- [F] Private repositories on GitHub Free include 2,000 Actions minutes and 500 MB of artifact storage per month; Pro and Team include 3,000 minutes (GitHub Docs: Actions billing).

**Design [D]:**

1. Code and workflows live on the default branch `main`; data lives on the orphan branch `data`, so code history stays small and data history can be pruned or migrated separately.
2. Each run checks out `main`, then checks out `data` into `data/`. After validation the run commits only changed files with a message containing the run ID, and pushes. Nothing is committed when nothing changed.
3. To keep history growth small, daily files are partitioned by year (`raw/daily/NSE/<SYMBOL>/<YYYY>.csv`), so a routine run rewrites only the current year's file. This refines the layout in 14.4.1. Predictions, reports and run records are append-only, one file per run.
4. Concurrency: one workflow concurrency group (`research-data`) with `cancel-in-progress: false`; before pushing, the run fetches `data` and pushes only if its base is still the tip, otherwise it re-merges once and retries (FR-154).
5. The initial download may exceed one job's time budget; it runs as resumable chunks, committing checkpoints, until the manifest is complete. Research reports are not produced from a partial dataset.
6. Backups: a dated tag on `data` before any full refresh or bulk revision; restoring is a checkout of the tag.

**Size estimate [A] (to be measured after the first download):** NIFTY 100 daily equity data for 10 years is about 100 stocks x 2,500 sessions = 250,000 rows; at roughly 200 bytes per row that is about 50 MB of CSV before Git compression. One-minute intraday bars for 2 years would be about 100 x 500 sessions x 375 bars = 18.75 million rows, around 2 GB,

which does not fit the recommended repository size. Therefore intraday history is not stored in Git in v1.0; runs use the provider's delayed intraday snapshot for the current session only and store the derived 13:00 snapshot rows (OD-25).

**Thresholds and migration:**

| Measure | Warning | Action |
|---|---|---|
| Any single CSV | 25 MB | Split by month |
| `data` branch size (packed) | 500 MB | Review retention, squash old data-branch history into a dated archive tag |
| `data` branch size (packed) | 1 GB | Start the approved migration (OD-24): Git LFS or external object storage, which is new infrastructure and needs approval; CSV stays the storage and export format, and Parquet may be added only as an optional copy |
| Monthly Actions minutes used | 70% of allowance | Reduce universe or move heavy jobs out of the three research slots |

## 14.5 Provider integration (Pillar 1)

No provider is selected yet, so no integration exists; the pipeline will refuse to run production fetches until a licensed provider is configured, and it will never substitute mock or scraped data.

| Provider category | Typical data [A] | Fit for v1.0 | Licensing and access considerations |
|---|---|---|---|
| Exchange-authorised market data vendor API | EOD and delayed prices, corporate actions, indices, derivatives | Preferred for prices and corporate actions | Licence must cover storage in a private repository, automated access from GitHub-hosted runners, and the intended report recipients; cost by quote (OD-02) |

| Provider category | Typical data [A] | Fit for v1.0 | Licensing and access considerations |
|---|---|---|---|
| Fundamentals data vendor | Standardised statements, shareholding, point-in-time history | Required for fundamentals and valuation | Same checks; restatement history needed |
| Broker trading APIs offering historical data to account holders | Prices for the account holder | Possible only for strictly personal use | Usually licensed for the client's own use; redistribution to others normally not permitted; confirm terms in writing before use |
| Exchange websites and public pages | Filings, circulars | Human reading and citation links only | [F] NSE's terms prohibit automated collection and scraping (14.1); never used as a feed |
| Official macro sources (RBI, MoSPI, FBIL) | Rates, inflation, growth, reference rates | Yes, through published downloads or APIs where terms allow | Attribution; check each portal's terms |
| Licensed news API | Headlines, abstracts, metadata | Phase 2 | Display rights often headline-plus-link |

**Interface contract (FR-142).** `fetch(dataset, symbols, start, end) -> records, provenance, status`, where `status` is `ok`, `partial` (with missing ranges), `rate_limited`, `auth_failed`, `unavailable` or `invalid_response`. Adapters handle authentication, pagination, timeouts (default 30 s per request), retries (5 with exponential backoff and jitter) and the provider's published rate limits, all configurable.

**Credentials.** Stored only as GitHub Actions secrets, named per provider (for example `MARKET_DATA_API_KEY`, `FUNDAMENTALS_API_KEY`); the names are configuration, the values are never in code, logs, CSVs or reports.

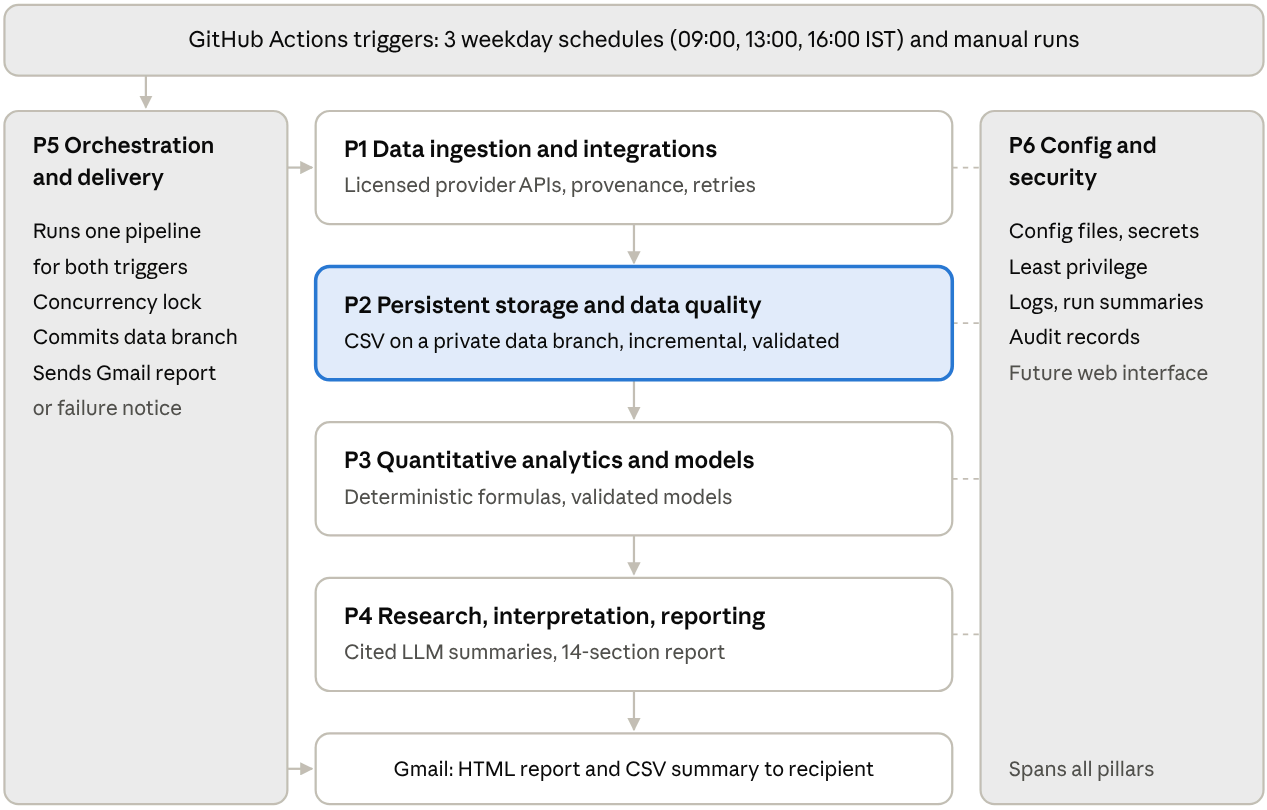
**Costs.** Unknown until vendor quotes arrive (OD-02); tracked per run as API calls and per month as licence cost for KPI-15.

**Failure behaviour.** On any provider failure the run keeps the last valid data, marks affected datasets stale in the manifest, labels the report, and sends a failure notice if no valid current report can be produced (FR-161).

## 15. Proposed system architecture

The architecture has six mandatory pillars run by one Python pipeline on GitHub Actions; the rule that makes it trustworthy is unchanged: only Pillar 3's deterministic engines and validated models produce numbers, and Pillar 4's language model may only explain them, through a citation gate that blocks any unsupported figure. No extra pillar or infrastructure is added without documented justification and approval.

**Six pillars, one pipeline: GitHub Actions runs it and Gmail delivers**




Six-pillar architecture · v1.0 GitHub Actions deployment

A trigger starts Pillar 5, which runs ingestion (P1), the persistent store (P2), analytics (P3) and reporting (P4) in order, then commits data and sends the email; Pillar 6 governs configuration, secrets, logs and audit across all of them.

## 15.0 Pillar responsibility matrix

| Pillar | Responsibilities | Main inputs | Main outputs | Owner | Key requirements |
|---|---|---|---|---|---|
| P1 Data ingestion and external integrations | Provider adapters, authentication, rate limits, pagination, timeouts, retries, normalisation, provenance | Config, credentials, missing ranges from P2 | Normalised records with provenance and status | Data engineering lead | FR-001, FR-042, FR-043, FR-080, FR-142 to FR-144; 14.5 |
| P2 Persistent storage and data quality | CSV store, manifests, incremental gap detection, merge and dedupe, revisions, corporate actions, quality rules, prediction and run records storage | P1 records, existing CSVs | Validated datasets, quality status, freshness | Data engineering lead | FR-002, FR-130 to FR-141; 14.3, 14.4 |
| P3 Quantitative analytics and model selection | Fundamentals, technicals, valuation, forecasting, ranking, risk, derivatives, model evaluation and registry | Validated datasets, approved models | Metrics, valuations, forecasts with uncertainty, risk outputs | Quantitative research lead | FR-020 to FR-087, FR-180 to FR-184; 9 to 12, 10.9 |
| P4 Research, interpretation and reporting | Retrieval-grounded summaries, news events, citation gate, glossary explanations, report rendering | P3 outputs, filings, news | Cited HTML report and CSV summary | Research head | FR-044, FR-060 to FR-064, FR-090 to FR-094, FR-165; 16 |
| P5 Orchestration, automation and delivery | Schedules, manual runs, step order, concurrency, data commits, email delivery, failure notices | Triggers, inputs | Run records, committed data, emails | Engineering lead | FR-150 to FR-164; 15.5, 15.6 |

| Pillar | Responsibilities | Main inputs | Main outputs | Owner | Key requirements |
|---|---|---|---|---|---|
| P6 Interface, configuration, security and observability | Config schema, secrets, permissions, logs, summaries, audit, future web interface | Config files, secrets | Validated config, logs, audit trail | Engineering lead with compliance | FR-170 to FR-176; 15.7 |

The component table in 15.1 lists the same capabilities at component level; each component belongs to the pillar shown in the matrix. Components that need a running server (the API gateway and web UI) are deferred to the future web interface (OD-21).

## 15.1 Components

| Component | Responsibility | Technology direction [D] | Key requirements |
|---|---|---|---|
| Market data ingestion | Vendor and exchange connectors, schedulers, raw landing with receipt timestamps | Event-driven connectors; object storage for raw files; idempotent loads | FR-001, FR-042, NFR-010 |
| Validation and normalisation | Rules V1 to V19, corporate action adjustment, ISIN mapping, period normalisation | SQL/dataframe rule engine with rule versioning | FR-002, 14.3 |
| Licence registry | Machine-readable terms per source: display, delay, export, derived-data rights | Config service read by API | FR-125 |
| Point-in-time store | Prices (raw and adjusted), fundamentals with filing timestamps, events, index membership with dates | Persistent CSV store as system of record (section 14.4); any columnar or query layer is rebuilt from it; revisions tracked by row version | FR-020, 10.5 |

| Component | Responsibility | Technology direction [D] | Key requirements |
|---|---|---|---|
| Feature store | Indicators, ratios, factors keyed by stock and as-of date | Materialised tables with lineage | FR-031, FR-056 |
| Fundamental engine | Ratios, peers, trends, scorecard | Python service with golden tests | FR-020 to FR-026 |
| Technical engine | Indicators, levels, technical summary | Vectorised library pinned to reference outputs | FR-030 to FR-036 |
| Valuation and forecasting engine | Methods, scenarios, statistical models, insufficient-data logic | Python; models registered with version, training window, validation report | FR-050 to FR-058 |
| Event processing | Filing classification, news linking, impact cards, calendar | Streaming consumer; classifiers with human review queue | FR-040 to FR-046 |
| Risk engine | Stops, sizing, R:R, portfolio risk | Stateless calculation API | FR-070 to FR-076 |
| Derivatives engine | Contract master, Greeks, IV, payoff | Stateless API; contract master from daily files | FR-080 to FR-087 |
| Glossary and retrieval | CMS, search, embeddings over glossary, methodology and filings | Hybrid lexical plus vector search | FR-090 to FR-094 |
| AI explanation layer | Draft narrative from a structured fact sheet; summarise filings; answer term questions | LLM via API with fact-sheet-only prompting, low temperature, structured output; citation gate validates every number and named fact | FR-060, NFR-040s |
| API and entitlements | Auth, rate limits, entitlement and delay enforcement | API gateway | FR-003, FR-125 |

| Component | Responsibility | Technology direction [D] | Key requirements |
|---|---|---|---|
| UI | Deferred: future web interface (Phase 4, needs approval, OD-21); v1.0 delivers HTML email reports | Component library with accessible charts | Section 16 |
| Forecast ledger | Append-only forecasts with hash chain | Write-once storage | FR-110 |
| Monitoring and evaluation | Pipeline health, freshness, model drift, forecast accuracy, LLM quality, cost | Metrics, logs, scheduled evaluation jobs | FR-111, FR-126, section 17 |
| Audit log | Every output with snapshot ID, model version, user, time | Append-only log store | FR-123 |

## 15.2 AI layer rules

1. Input to the LLM is a fact sheet (JSON) of engine outputs with IDs, plus retrieved source passages with document IDs and page numbers.
2. The LLM may not perform arithmetic that reaches the user. Derived comparisons ("margin up 2.1 points") are computed by engines and placed in the fact sheet.
3. Output is structured: sentences each tagged with fact IDs or source IDs.
4. The citation gate parses every number, date, percentage and named entity and matches it to a fact or source span; tolerance is exact match after unit normalisation.
5. Failing sentences are dropped; if more than 20% fail, the narrative is regenerated once, then replaced with the template report [D].
6. Prohibited-language filter: "guaranteed", "sure-shot", "will reach", "buy now", "multibagger" and similar in English and Hindi, plus the causation guard (9.4).
7. Every AI-assisted output is labelled with the extent of AI use, as SEBI's RA regulations require client disclosure of AI use (REG-03).
8. Model, prompt and retrieval-index versions are recorded per output.

## 15.3 Reproducibility

Each output stores: data snapshot ID (as-of times for each source), engine code version (git SHA), model version, parameter set, prompt version, and the rendered result hash. FR-062 and FR-113 reproduce any past output from these.

## 15.4 Shared data and execution contracts

Pillars talk only through these versioned contracts, and business logic lives in reusable Python modules called by the single pipeline entry point, so scheduled and manual runs never duplicate ingestion, formulas, inference or reporting.

| Contract | Producer to consumer | Format | Key fields | Failure behaviour | Owner |
|---|---|---|---|---|---|
| Market data retrieval | P1 to P2 | In-memory records + provenance | dataset, symbol, ISIN, interval, values, source, source_ts, fetched_at, status | Non-`ok` status: P2 writes nothing for that range and logs it | Data lead |
| Dataset manifest and CSV schemas | P2 to all | CSV + JSON schema files (14.4.2) | schema_version, coverage, last_confirmed_date, status, sha256 | Schema mismatch: run stops before analytics | Data lead |
| Data quality status | P2 to P3, P4, P5 | `metadata/data_quality_log.csv` + per-dataset status | rule_id, severity, dataset, status (`valid`, `stale`, `partial`, `failed`) | P3 skips forecasts on non-valid inputs | Data lead |
| Feature generation | P2/P3 to P3 | `processed/features/<version>/` CSV | as_of_date, symbol, feature values, feature_set_version | Missing features: affected outputs `insufficient_data` | Quant lead |
| Model input and output | P3 internal | Model input CSV + registry entry | dataset_id, model_id, version, predictions, intervals | Unapproved model: refuse | Quant lead |
| Forecast metadata | P3 to P2 and P4 | `predictions/<date>/<run_id>.csv` | Fields in 10.9.3 | Status field always present | Quant lead |
| Report generation | P4 to P5 | HTML + CSV summary in `reports/<date>/` | report_id, run_id, sections, data timestamps, model versions, warnings | Missing section: labelled unavailable | Research head |
| Email delivery | P5 to recipient | MIME email via SMTP | run_id, report_id, recipients (masked in logs), subject, idempotency key, smtp_status | SMTP error: recorded, failure visible in run summary | Engineering lead |

| Contract | Producer to consumer | Format | Key fields | Failure behaviour | Owner |
|---|---|---|---|---|---|
| Execution status and audit | P5/P6 to all | `runs/<run_id>.json`, `metadata/run_log.csv` | trigger, inputs, step statuses, timings, counts, snapshot checksums, errors | Always written, even on failure | Engineering lead |

## 15.5 GitHub Actions scheduling and on-demand execution

### 15.5.1 Schedule

| Run | IST | UTC cron (default, configurable) | What it covers |
|---|---|---|---|
| Morning | 09:00 | `30 3 * * 1-5` | Previous completed session, overnight filings, news and global context; market not yet open |
| Midday | 13:00 | `30 7 * * 1-5` | Delayed intraday snapshot, labelled as an incomplete session |
| Evening | 16:00 | `30 10 * * 1-5` | Today's session; prices labelled provisional unless the provider confirms final end-of-day data |

India does not observe daylight saving time, so the UTC times stay fixed. [F] GitHub runs scheduled workflows only from the default branch, may delay them under high load (queued jobs can be dropped), and supports intervals no shorter than 5 minutes (GitHub Docs). The pipeline therefore records the actual start time, labels a late run, and never assumes it ran at the nominal slot. Exchange holidays come from the licensed calendar (FR-159).

### 15.5.2 Workflow outline (illustrative; not yet implemented)

```
name: research
on:
  schedule:
    - cron: '30 3 * * 1-5'
    - cron: '30 7 * * 1-5'
    - cron: '30 10 * * 1-5'
  workflow_dispatch:
```

```
    inputs:
      scope:        { type: choice, options: [universe, watchlist, symbols],
default: universe }
      symbols:      { type: string, default: '' }
      mode:         { type: choice, options: [full_research, data_only,
report_only], default: full_research }
      horizons:     { type: string, default: '1M,3M,6M,12M' }
      report_format:{ type: choice, options: [html, html_csv], default:
html_csv }
      full_refresh: { type: boolean, default: false }
      refresh_reason: { type: string, default: '' }
concurrency:
  group: research-data
  cancel-in-progress: false
permissions:
  contents: write   # only for committing to the data branch
jobs:
  research:
    runs-on: ubuntu-latest
    timeout-minutes: 60   # configurable
    steps:
      - uses: actions/checkout@<pinned-sha>
      - uses: actions/checkout@<pinned-sha>
        with: { ref: data, path: data }
      - uses: actions/setup-python@<pinned-sha>
        with: { python-version-file: '.python-version' }
      - run: pip install -r requirements.lock
      - run: python -m research_platform.pipeline --trigger "${{
github.event_name }}"
        env:
          MARKET_DATA_API_KEY: ${{ secrets.MARKET_DATA_API_KEY }}
          GMAIL_SENDER: ${{ secrets.GMAIL_SENDER }}
          GMAIL_APP_PASSWORD: ${{ secrets.GMAIL_APP_PASSWORD }}
          REPORT_RECIPIENT: ${{ vars.REPORT_RECIPIENT }}
```

`REPORT_RECIPIENT` may be a variable or a secret (OD-27). Dispatch inputs are validated again inside the pipeline; a full refresh requires `full_refresh: true` plus a reason.

### 15.5.3 Pipeline step order

1. Check out the latest authorised code (default branch) and the `data` branch.
2. Set up Python with pinned dependencies.

3. Load and validate configuration (FR-170).
4. Load the manifest and CSV datasets; verify checksums and schemas.
5. Detect missing or new records against the trading calendar and the provider's latest date.
6. Fetch only those records (P1); validate, dedupe, apply revisions (P2).
7. Persist valid changes atomically to the working copy of `data/`.
8. Recompute affected adjusted series and features.
9. Load approved model versions from the registry.
10. Run research, valuation, forecasts and risk calculations (P3); mark outputs unavailable where inputs are stale or invalid.
11. Save dated prediction records and metadata.
12. Generate the HTML report and CSV summary (P4).
13. Send the report email, or a failure notice (P5), with an idempotency check.
14. Commit and push data, predictions, reports and the run record to `data`; write the job summary with final status and actionable errors.

Step 13 runs before step 14 so that a push conflict never blocks the email; if the push fails after the email was sent, the run record says so and the next run re-merges (FR-154, FR-156).

#### 15.5.4 Execution limits

[F] Private repositories on GitHub Free include 2,000 Actions minutes per month (GitHub Docs). Three runs on about 22 weekdays is roughly 66 runs a month, so the average run must stay under about 30 minutes to fit the Free allowance with no manual runs; NFR-101 sets the budget, and actual duration will be measured in Phase 1. Public repositories have free standard-runner minutes but are not allowed here because the data is licensed (A-09).

### 15.6 Gmail delivery

Each successful run sends one HTML email to the configured recipient; each failed run sends a clearly labelled failure notice where possible; no stale report is ever sent as current.

- **Method [D]:** Gmail SMTP with an App Password, from the runner, with no separate server. [A] Host `smtp.gmail.com`, port 587 with STARTTLS, set in configuration and confirmed during setup.

- **Account requirements [F]:** App Passwords work only on Google Accounts with 2-Step Verification; they may be unavailable for work, school or organisation accounts, Advanced Protection accounts, or accounts whose 2-Step Verification uses only security keys; and they are revoked when the account password changes (Google Account Help). Google also says App Passwords are not recommended in most cases.
- **Alternative if App Passwords are unavailable (OD-26):** the Gmail API with OAuth, storing a refresh token as a secret; this needs a Google Cloud project for OAuth credentials, which is a consent and setup step rather than hosted infrastructure, and requires approval.
- **Limits [F]:** Gmail shows sending-limit errors above 500 emails a day or 500 recipients in one message (Gmail Help); three reports a day is far below this.
- **Secrets and variables:** `GMAIL_SENDER`, `GMAIL_APP_PASSWORD`, `REPORT_RECIPIENT`; none hardcoded; addresses masked in logs.
- **Email contents:** execution date and time (IST), trigger and run ID; market overview; key research findings; forecasts with assumptions and uncertainty; major risks and data-quality warnings; data timestamps and freshness; model versions; execution status; the detailed HTML report inline or attached, and the CSV summary attached when configured; required disclosures (18.6).
- **Subject format [D]:** `[Research] {date} {slot} - {status}` for example `[Research] 2026-10-12 Evening - OK` or `- FAILED`.
- **Status semantics:** the run records `submitted` when the SMTP server accepts the message, or `smtp_error` with the code; it never records `delivered`, because SMTP acceptance does not confirm inbox delivery (FR-162).
- **Idempotency:** a send log keyed by run ID and report hash prevents duplicate emails when a run is retried (FR-163).

## 15.7 Security and configuration

| Control | Requirement |
|---|---|
| Repository | Private; branch protection on `main` (reviews required for workflow and code changes); `data` branch writable only by the workflow token |

| Control | Requirement |
|---|---|
| Secrets | Provider keys and Gmail credentials only in GitHub Actions secrets; never in code, CSVs, reports, logs or run records; rotated at least every 90 days [D] and immediately after any suspected leak |
| Workflow permissions | Least privilege (`contents: write` only where the data commit needs it); third-party actions pinned to commit SHAs; dependency versions locked |
| Inputs | Dispatch inputs validated against allow-lists and the configured universe; no shell interpolation of raw inputs |
| Configuration | `config/platform.yaml` (universe, schedule tolerance, lookbacks, horizons, risk defaults, report options, provider names) validated against a JSON schema; changes reviewed like code |
| Logs | Structured JSON; secrets and email addresses masked; job summary per run |
| Scanning | Secret scanning and dependency vulnerability alerts enabled; CI blocks commits containing credentials |
| Access | Only collaborators with write access can dispatch runs; read-only users cannot (FR-176) |
| Future web trigger | Through a server-side component holding a fine-grained token; never a token in browser code; needs approval (OD-21) |
| Personal data | Only recipient addresses are personal data in v1.0; stored as secrets or variables; DPDP controls in REG-12 |

## 16. UI and report specifications

Twelve screens share one pattern: the answer at the top, the timestamp and freshness badge beside every price, and an "evidence" control on every score, valuation and forecast that opens the inputs, formula, sources and model version.

In v1.0 there is no interactive interface: the screens in 16.2 are the design for the future web interface (OD-21), and their content reaches users through the email report specified in 16.4.

## 16.1 Global UI rules

- Header strip on any stock view: name, NSE symbol / BSE code, exchange of the shown price, price, change, "as of" time in IST, delay label (real-time / 15-min delayed / end of day), freshness badge (green within SLA, amber near, red stale).
- Labels: "Fundamental estimate", "Statistical range", "Trading reference level", "Hypothetical example". Never "Buy", "Sell", "Target" as standalone labels.
- Numbers in Indian format (1,00,000; lakh/crore) with a toggle to international format.
- Language toggle EN/HI on every screen; glossary underlines.
- Disclosures footer per output type (FR-120), including the extent of AI use.
- Mobile-first breakpoints; all charts have data-table alternatives (accessibility).

## 16.2 Screens

| # | Screen | Primary question answered | Key components | Key FRs | Phase |
|---|---|---|---|---|---|
| 1 | Market Overview Dashboard | What is the market doing today? | NIFTY 50, SENSEX, sector heatmap, breadth, macro tiles, today's events | FR-040, FR-041, FR-045 | 1 |
| 2 | Stock Screener | Which stocks match my criteria? | Filter builder, presets, result table with matched criteria, save/alert | FR-010 to FR-014 | 1 |
| 3 | Stock Research and Company Profile | What is this company and how is it doing? | Header strip, business summary with citations, scorecard, technical summary, valuation range, events | FR-020 to FR-064 | 1 |
| 4 | Financial Statements and Ratios | What do the numbers say? | P&L, BS, CF (standalone/consolidated, quarterly/annual), ratio table with formulas, peer comparison | FR-020 to FR-024 | 1 |
| 5 | Technical Charts and Indicators | What is the price trend and volatility? | Candles, indicators, S/R zones, RS vs index, drawdown | FR-030 to FR-036 | 1 |

| # | Screen | Primary question answered | Key components | Key FRs | Phase |
|---|---|---|---|---|---|
| 6 | Valuation and Forecast Scenarios | What might it be worth and how wide is the uncertainty? | Method cards, range bar vs price, DCF sensitivity grid, scenario editor, horizon fan chart (if validated), insufficient-data card | FR-050 to FR-058 | 1 (scenarios gated), 2 (fan chart) |
| 7 | Stop-Loss and Position Sizing Calculator | How much could I lose and how big should the position be? | Inputs, method comparison table, warnings, R:R, gap illustration | FR-070 to FR-075 | 1 |
| 8 | Options and Derivatives Analyzer | What is the payoff and risk of this options position? | Strategy builder, payoff chart (expiry and today), Greeks, break-evens, chain (Phase 2), disclosure gate | FR-080 to FR-087 | 1 (payoff), 2 (chain) |
| 9 | News and Corporate Events | What happened, is it confirmed, and what could it affect? | Event feed with status chips, event impact cards, calendar | FR-042 to FR-046 | 1 (filings), 2 (news) |
| 10 | Watchlist and Portfolio Risk | How risky is what I hold or watch? | Lists, alerts, holdings import, concentration, VaR, beta | FR-100 to FR-104, FR-076 | 1 (watchlist), 2 (portfolio) |
| 11 | Financial Glossary and Learning Center | What does this term mean? | Search, entries, learning paths | FR-090 to FR-094 | 1 |
| 12 | Research History and Forecast Accuracy | How did past views and forecasts turn out? | Personal history; ledger; accuracy vs baselines by horizon | FR-110 to FR-113 | 2 |

## 16.3 Stock research report template

| Section | Content | Source of content |
|---|---|---|
| 1. Company overview | Business, segments, listing details, index memberships | Filings; AI summary with citations |
| 2. Current data and timestamp | Price, exchange, as-of time, delay, market cap, liquidity | Price engine |

| Section | Content | Source of content |
|---|---|---|
| 3. Fundamental scorecard | Six pillars with metric values, periods, peer percentiles | Fundamental engine |
| 4. Technical summary | Trend, momentum, volatility regime, RS, levels | Technical engine |
| 5. Sector and market context | Sector relative performance, market regime, relevant macro | Market engine |
| 6. News and upcoming events | Last 90 days of material events with status; next 90 days' calendar | Event engine |
| 7. Valuation range | Methods, ranges, combined range, sensitivity | Valuation engine |
| 8. Scenario-based forecasts | Bull/base/bear with assumptions; horizon ranges if validated | Valuation and forecast engine |
| 9. Risk assessment | Volatility, drawdown, gap frequency, liquidity, governance flags, leverage | Risk and fundamental engines |
| 10. Bull and bear thesis | Three to five evidence-backed points each | AI drafting from fact sheet, gated |
| 11. Source citations | Numbered list of every source with dates and links | Citation registry |
| 12. Model limitations | Standard limitations plus stock-specific ones (short history, few peers) | Templates plus engine flags |
| 13. Assumptions and invalidation conditions | Each scenario's assumptions and measurable invalidation triggers | Scenario engine |

Plus: confidence rating with reasons, disclosures, extent of AI use, report ID, snapshot ID and version.

A sample report with hypothetical data is in Appendix B.

## 16.4 Email research report (v1.0 primary delivery)

In v1.0 the screens in 16.2 are deferred (OD-21) and their content is delivered in the HTML report; each run produces one report with the 14 sections below, and any section without valid data says "unavailable" with the reason rather than being dropped.

| # | Section | Content | Source pillar |
|---|---|---|---|
| 1 | Executive market summary | NIFTY 50, SENSEX, sector moves, breadth, key events since the last run | P3, P4 |
| 2 | Market data timestamp and freshness | Per dataset: as-of time, session state (completed, incomplete, provisional), freshness label, late-run flag | P2, P5 |
| 3 | Company overview and financial metrics | For each covered or selected stock: profile and key metrics with periods | P3, P4 |
| 4 | Fundamental analysis | Scorecard pillars with peer percentiles and trends | P3 |
| 5 | Technical analysis | Trend, momentum, volatility, relative strength, levels | P3 |
| 6 | Sector and macroeconomic context | Sector rotation, macro panel with release dates | P3, P4 |
| 7 | Company news and corporate actions | Events with source, time, confirmation status, affected companies, implications and uncertainty | P1, P4 |
| 8 | Valuation range | Methods, ranges, combined range; clearly separated from market price | P3 |
| 9 | Forecast scenarios and assumptions | Horizons that passed validation; baseline comparison; intervals; bull/base/bear; or `model_not_validated` / `insufficient_data` | P3 |
| 10 | Stop-loss and risk calculations | Where configured: method comparison for configured entry assumptions, sizing for the configured risk budget, R:R, gap and slippage warnings | P3 |

| # | Section | Content | Source pillar |
|---|---|---|---|
| 11 | Risks and invalidation conditions | Top risks and measurable invalidation triggers | P3, P4 |
| 12 | Source citations | Numbered sources with dates and links | P4 |
| 13 | Model version and evaluation context | Model IDs and versions, validation window, out-of-sample skill vs baseline, calibration | P3 |
| 14 | Data-quality warnings and limitations | Failed rules, stale datasets, missing ranges, revisions applied, method limitations | P2, P4 |

Every material statement is tagged as Fact (cited), Interpretation, Assumption or Scenario. The CSV summary attachment has one row per stock and horizon with price, timestamp, fair-value range, forecast status, median, interval bounds, model version and data snapshot ID. Disclosures and the extent of AI use appear at the end of every report and email.

## 17. Non-functional requirements

Targets below assume the MVP context stated first; each is provisional until load tests and vendor SLAs confirm it, and each has an owner and a test in section 19.

**Operating context assumed [A]:** up to 50,000 registered users and 5,000 concurrent users in Phase 1; peak load at market open (09:15 IST) and close (15:30 IST); T1 universe of about 100 stocks, growing to about 2,000 in Phase 3; delayed (15-minute) intraday data in MVP; single cloud region in India with a second region for disaster recovery.

| ID | Area | Requirement | Target | Rationale / assumption |
|---|---|---|---|---|
| NFR-001 | Performance | Stock page first meaningful render (p95, 4G mobile) | 2.5 s | Common web performance budget |

| ID | Area | Requirement | Target | Rationale / assumption |
|---|---|---|---|---|
| NFR-002 | Performance | Screener query over T2 universe (p95) | 1.5 s | Precomputed metrics |
| NFR-003 | Performance | Calculator responses (stop, sizing, payoff, Greeks) (p95) | 300 ms | Stateless computations |
| NFR-004 | Performance | Full research report generation including AI narrative (p95) | 20 s, with progressive display | LLM latency dominates |
| NFR-005 | Availability | Service availability during market hours 09:00 to 16:00 IST on trading days | 99.9% monthly | Peak value window |
| NFR-006 | Availability | Outside market hours | 99.5% | Lower risk |
| NFR-007 | Recovery | RPO / RTO | RPO 15 min; RTO 1 hour | Data can be re-ingested from vendors |
| NFR-010 | Freshness | Delayed intraday price age at display | At most delay tier + 60 s for 99.5% of trading minutes | Vendor dependent (A-01) |
| NFR-011 | Freshness | EOD data and recomputed analytics ready | By 19:00 IST on trading days | Exchange EOD files plus compute |
| NFR-012 | Freshness | Corporate announcements to users | 99% within 15 min of exchange timestamp | FR-042 |
| NFR-013 | Freshness | Fundamentals after results filing | Within 24 h | Vendor dependent |
| NFR-020 | Scalability | Universe growth | Linear compute; 2,000 stocks EOD batch under 60 min | Phase 3 |

| ID | Area | Requirement | Target | Rationale / assumption |
|---|---|---|---|---|
| NFR-021 | Scalability | Concurrency | 5x headroom over assumed peak via autoscaling | Results-season spikes |
| NFR-030 | Security | Authentication | MFA for analyst, compliance and admin roles; OAuth/OIDC | Role risk |
| NFR-031 | Auditability | Retention of audit logs, published research, client interactions | At least 5 years, or longer if counsel advises | [F] RA/IA guidelines require records of KYC, interactions and consents for at least 5 years (FinSec Law tracker) |
| NFR-032 | Security | Encryption | TLS 1.2+ in transit; AES-256 at rest; keys in managed KMS | Standard |
| NFR-033 | Error correction | Material error correction notice | Within 1 business day of confirmation [D] | REG-10 |
| NFR-034 | Security | Vulnerability management | Critical fixes within 7 days; annual third-party penetration test | Standard |
| NFR-035 | Privacy | Personal data handling | Data minimisation; consent records; deletion on request within 30 days unless retention is legally required [D] | DPDP Act (REG-12) |

| ID | Area | Requirement | Target | Rationale / assumption |
|---|---|---|---|---|
| NFR-036 | Privacy | LLM data use | No personal data in prompts except where essential; vendor contract bars training on our data | A-06 |
| NFR-040 | Explainability | Outputs with evidence inspector available | 100% of scores, valuations, forecasts | FR-062 |
| NFR-041 | Accuracy (AI) | Unsupported numbers in published narratives | 0 on test set; under 1 per 10,000 sentences in production sampling | FR-060 |
| NFR-042 | Accuracy (calc) | Golden test pass rate | 100% per release | Deterministic code |
| NFR-043 | Model monitoring | Drift checks on forecast models | Weekly; alert when rolling 80% interval coverage falls outside 70% to 90% | Calibration |
| NFR-044 | Model monitoring | Forecast evaluation | Monthly accuracy report vs baselines | FR-111 |
| NFR-050 | Accessibility | Standard | WCAG 2.2 AA | Inclusive access |
| NFR-051 | Accessibility | Language | English and Hindi for all user-facing text in MVP | Personas |
| NFR-060 | Cost | Cost per completed research report | Tracked per report; target set after OD-02/OD-03; alert at 120% of budget | BO-09 |

| ID | Area | Requirement | Target | Rationale / assumption |
|---|---|---|---|---|
| NFR-061 | Cost | LLM spend | Per-user and global daily caps; cache narratives per snapshot | Cost control |
| NFR-070 | API reliability | Internal API error rate (5xx) | Under 0.1% of requests | Standard |
| NFR-071 | API reliability | Vendor feed failures | Automatic failover or degraded mode within 5 min; user-visible banner | 14.2 fallbacks |
| NFR-080 | Compliance | Disclosure presence | 100% of outputs carry current disclosure version | FR-120 |
| NFR-081 | Compliance | Complaint resolution | Within the regulatory window (7 working days for RA) | BO-08 |

### Historical store NFRs (mandatory, section 14.4)

| ID | Area | Requirement | Target | Rationale |
|---|---|---|---|---|
| NFR-090 | Persistence | Historical data survives process restart, machine reboot, redeploy and retraining | Zero records lost in TST-HIST-E and TST-HIST-G | Core rule |
| NFR-091 | Efficiency | Requests on a run with no new data | Zero historical range requests; only latest-date and calendar checks | TST-HIST-B, TST-HIST-J |

| ID | Area | Requirement | Target | Rationale |
|---|---|---|---|---|
| NFR-092 | Efficiency | Incremental daily update for the T1 universe | Completes within 15 minutes of the provider publishing EOD data [A] | Fits NFR-011 |
| NFR-093 | Integrity | Checksums of all manifest files verified | Every run, before any read for analysis | Detect corruption |
| NFR-094 | Recovery | Backups of replaced CSV versions | Last 7 successful versions kept [D]; restore tested monthly | FR-136 |
| NFR-095 | Write safety | Atomic replace of every CSV write | 100% of writes via temp file and rename; never partial files in place | Core rule |
| NFR-096 | Storage | Storage growth monitored | Alert at 80% of allocated volume | Ops |
| NFR-097 | Reproducibility | Any prediction reproducible from recorded data references and model version | 100% on a monthly sample of 50 predictions | FR-139 |

### Pipeline NFRs (v1.0 GitHub Actions deployment)

All targets are proposals; none has been measured yet. Each lists how it will be measured, and the verification result is recorded in 19.5 once runs exist. NFR-001 to NFR-004, NFR-021, NFR-030 and NFR-050 apply to the future web interface.

| ID | Area | Requirement | Proposed target | Assumption | Measurement |
|---|---|---|---|---|---|
| NFR-100 | Data freshness | Morning report uses the previous completed session; evening report uses today's session or labels it provisional | 100% of reports correctly labelled | Provider publishes EOD data on the same evening [A] | Run records vs provider timestamps |

| ID | Area | Requirement | Proposed target | Assumption | Measurement |
|---|---|---|---|---|---|
| NFR-101 | Execution duration | Incremental run time (p95), and hard timeout | p95 under 20 minutes; timeout 60 minutes | T1 universe; vendor API latency [A] | Job timings in run log |
| NFR-102 | Reliability | Scheduled runs completing successfully | 98% per month | GitHub schedule delays or drops are counted as failures if no report is produced | Run log |
| NFR-103 | Recovery | A failed run leaves data intact and the next run recovers without manual action | 100% data intact; 95% automatic recovery | Provider outages are transient | Failure-injection tests and run log |
| NFR-104 | Storage growth | Growth of the `data` branch | Under 50 MB per month for T1; thresholds in 14.4.9 | Year-partitioned files [A] | Packed size after each run |
| NFR-105 | Security | Secrets exposed in repository, logs, reports or run records | Zero | Secret scanning enabled | Scanner results, log review |
| NFR-106 | Auditability | Runs with complete run records | 100%, including failed runs | None | Run log completeness check |
| NFR-107 | API usage | Requests on a no-change run; monthly calls vs provider quota | Zero historical requests; under 70% of quota | Quota known after OD-02 | download_log.csv |
| NFR-108 | Email status | Report email submitted after report generation | 99% of successful runs within 5 minutes; status recorded for 100% | Gmail SMTP available | Run record `email_status` |
| NFR-109 | Maintainability | Single implementation of each business function; automated test coverage of core modules | 100% golden coverage of formulas; at least 80% line coverage of core modules | None | CI coverage report |

| ID | Area | Requirement | Proposed target | Assumption | Measurement |
|---|---|---|---|---|---|
| NFR-110 | Operating cost | Actions minutes and data licence cost per month | Under 70% of the plan's included minutes; cost per completed run reported | GitHub Free or Pro plan | GitHub usage report, invoices |
| NFR-111 | Forecast evaluation | Monthly out-of-sample evaluation vs baselines | Report produced every month | Enough matured forecasts | Evaluation job output |

# 18. Regulatory and ethical considerations

The recommended route is to register the operating company as a SEBI Research Analyst before any public release of named-stock scenarios, price ranges, stop levels or ranked lists, and to run the platform as a data-and-education tool with no recommendations until then; a disclaimer alone does not make a regulated service compliant, and external Indian securities counsel must confirm the route in writing before launch.

This section is a business analysis of the rules, not legal advice. Every [F] item cites a source read on 11 Oct 2026; counsel should verify against the primary SEBI texts.

## 18.1 What the current rules say

| Topic | Current position [F] | Source |
|---|---|---|
| Who is a research analyst | A person who, for consideration, provides research services. Amended RA Regulations notified 16 Dec 2024 added definitions of "research services" and "trading calls" and a part-time RA category (max 75 clients) | SEBI RA Regulations as amended 16 Dec 2024; SCC Online |
| AI use | RAs must disclose to clients the extent of AI tool use in research services (Reg. 19(vii)); responsibility stays with the RA regardless of AI use | SCC Online; AZB & Partners |

| Topic | Current position [F] | Source |
|---|---|---|
| Basis for recommendations | Recommendations must be corroborated by relevant data and analysis (Reg. 20(4)); FAQs say recommendations need adequate documentary basis | SCC Online; LKS |
| Methodology is no exemption | RA FAQs (23 Jul 2025): research on SEBI-regulated securities based on any methodology, including technical analysis, is not exempt; general commentary, broad market trends and statistical summaries are not research reports | LKS |
| Free users | Non-paying clients still count as clients because consideration includes non-cash benefits; research bundled as a value-added service is for consideration | LKS |
| Fees | Fixed fee cap of INR 1.51 lakh per annum for individual and HUF clients (RA and IA); advance fees for RAs limited to one quarter | AZB & Partners; FinSec Law |
| Deposit, website, audit | Deposit of INR 1 lakh to 10 lakh based on client count replaces net worth; functional website (Reg. 19A); annual compliance audit reported to RAASB | AZB & Partners; FinSec Law |
| Segregation | Client-level segregation of research and distribution (Reg. 26C) | SCC Online |
| Model portfolios | Treated as research services; performance must be validated by a SEBI-specified agency against benchmarks | FinSec Law |

| Topic | Current position [F] | Source |
|---|---|---|
| Performance claims | PaRRVA (CARE Ratings with NSE) launched on pilot 8 Dec 2025 to verify past risk and return claims of RAs, IAs and algo providers; selective favourable periods not allowed | Business Standard |
| Education-only route | Regulated entities may not associate with unregistered persons giving advice or return claims; a person solely in education must not give future price indications, advice or recommendations | Fox Mandal on SEBI FAQs of 29 Jan 2025 |
| Price data lag for education | Uniform 30-day lag for sharing and using market price data for education and awareness, effective 1 Jul 2026 (circular dated 8 May 2026), replacing the earlier 1-day and 3-month lags | TaxGuru; Taxmann |
| Investment advisers | IA advice covers SEBI-regulated securities; IAs cannot give trading calls | AZB & Partners |
| AI/ML governance | SEBI consultation paper of 20 Jun 2025 proposed AI/ML governance (model governance, testing records, accountability, disclosures); final guidelines with human oversight and kill switches were described as forthcoming in Aug 2026 | Taxmann; Moneylife |
| Personal data | DPDP Rules notified 13 Nov 2025; consent-manager provisions from 12 Nov 2026; notice, consent and other substantive duties from 12 May 2027 | AZB & Partners |

| Topic | Current position [F] | Source |
|---|---|---|
| Market data | Exchange data redistribution only under licence; NSE site scraping prohibited | Section 14.1 |

## 18.2 What the Platform's features trigger

| Feature | Likely classification [A, counsel to confirm] | Without RA registration |
|---|---|---|
| Raw data, statements, ratios, charts, glossary, calculators driven by user inputs | Data and tools; statistical summaries | Allowed, subject to data licence |
| Screener with user-defined criteria | Tool | Allowed |
| Preset screens and the daily research queue ranking named stocks | Possibly research (selection of securities) | Presets only with neutral naming; ranked queue gated |
| Fundamental scorecard on a named stock | Grey area: statistical summary vs research opinion | Allowed only as transparent percentile display with no overall verdict [D] |
| Fair-value ranges, scenarios, horizon forecasts on named stocks | Research services (future price indication) | Not allowed publicly |
| Stop-loss levels or targets suggested by the platform for a named stock | Research / trading calls | Not allowed; calculator over user inputs only |
| Personalised position sizing based on the user's risk profile | Could be investment advice | Generic calculator over user-entered numbers only |
| Options strategy suggestions for a named underlying | Research / trading calls | Not allowed; payoff calculator only |

| Feature | Likely classification [A, counsel to confirm] | Without RA registration |
|---|---|---|
| Public accuracy page | Performance claims | Only with PaRRVA-compliant method, after registration |
| Educational content using real stock prices | Education | Only with prices at least 30 days old |

## 18.3 Route options (decision OD-01)

| Option | What it allows | Main obligations | Assessment |
|---|---|---|---|
| A. Data and education only | Tools, data, glossary, calculators; no named-stock views | Data licences; education content with 30-day-lagged prices; no future price indications | Fast and cheap, but does not meet the business objective |
| B. Register as RA (body corporate) - recommended | Research reports, scenarios, ranges, ranked lists, with disclosures | Qualified principal officer and analysts, deposit, website, AI-use disclosure, fee caps for individuals, record keeping, audit, complaints, segregation from distribution | Meets the objective; 3 to 6 months of set-up [A] |
| C. Host research of a registered partner RA | Partner's research inside the platform | Contracts; avoid association with unregistered persons; who is responsible for AI output must be clear | Faster but less control; brand risk |
| D. Register as IA | Personalised advice | Risk profiling, suitability, fee caps; no trading calls | Not suited to trading features; Phase 4 at earliest |

## 18.4 Regulatory controls

| ID | Control | Implementation |
|---|---|---|
| REG-01 | Route gate | Feature flags keep all "gated" features off in public mode until compliance records the OD-01 decision and counsel opinion |
| REG-02 | Registration prerequisites | Principal officer, qualified analysts, deposit, website (Reg. 19A) before Phase 2 public release |
| REG-03 | AI-use disclosure | Every output states which parts were AI-drafted and that calculations are deterministic; client terms describe AI use |
| REG-04 | Research basis and disclosures | Each research output stores its data, methods and analyst approval (FR-121); standard disclosures attached (FR-120) |
| REG-05 | Fees | Pricing respects the per-family cap and advance-fee limits for individual clients; billing system enforces |
| REG-06 | Conflicts of interest | Analysts and company declare holdings; trading blackout around publication; conflicts shown on reports |
| REG-07 | Segregation | No distribution or referral revenue tied to research clients |
| REG-08 | Education data lag | Learning content uses hypothetical data or prices at least 30 days old; automated check on content publishing |
| REG-09 | Performance claims | No performance or accuracy marketing until a PaRRVA-compliant method is confirmed; accuracy page shows all forecasts |
| REG-10 | Material error correction | Triage within 4 business hours of report, fix, notice to affected users within 1 business day, root-cause log [D] |
| REG-11 | No unregistered associations | No paid promotion through unregistered finfluencers; affiliate checks |

| ID | Control | Implementation |
|---|---|---|
| REG-12 | Privacy (DPDP) | Consent notices, purpose limitation, retention schedule, breach response, data principal rights, ready before 12 May 2027 and earlier where practical |
| REG-13 | Data licence compliance | Licence registry, entitlement checks (FR-125), usage reporting to exchanges/vendors |
| REG-14 | Derivatives risk warnings | Risk acknowledgement gate (FR-087); prominent risk text on every derivatives screen |
| REG-15 | AI governance | Model inventory, validation reports, human review for published research, kill switch for AI narratives, monitoring; align with final SEBI AI/ML guidelines when issued |
| REG-16 | Complaints | Ticketing with regulatory SLA; SCORES integration where applicable |

## 18.5 Ethical controls

- No dark patterns pushing trading frequency; no gamification of trading.
- Warnings scale with risk: derivatives, small caps, illiquid stocks and leverage carry stronger warnings.
- Equal treatment: the same research is visible to all clients of a tier at the same time (no front-running by premium tiers of the same report).
- Bias review of AI summaries for sensational language; balanced bull and bear thesis required.
- Users can download and delete their data, subject to legal retention.

## 18.6 Emailed reports, personal use and distribution (v1.0)

Who receives the emailed report decides the regulatory risk: a report generated for the sponsor's own investing is very different from research sent to other people, whether or not they pay.

| Recipient model | Likely position [A, counsel to confirm] | Requirements |
|---|---|---|
| Sponsor only, for personal research | No clients, so no research service to others; data licence must still allow personal automated use and storage | Data licence (OD-02); private repository; no forwarding |
| A few named individuals (family, colleagues), free | The RA FAQs treat non-paying recipients as clients when consideration includes non-cash benefits; named-stock forecasts and levels may be research services | Counsel opinion before adding recipients; licence must allow redistribution to them |
| Subscribers or the public | Research services under the RA Regulations | RA registration and all REG controls (18.4) |

Controls [D]: `REPORT_RECIPIENT` is restricted by configuration to an approved list held by compliance; adding a recipient needs a recorded approval (OD-22); every email carries disclosures, the extent of AI use, data timestamps and the statement that estimates are not guaranteed outcomes. Because reports contain licensed data, recipients must be covered by the data licence. These controls do not by themselves make a regulated service compliant.

## 19. Testing and acceptance criteria

Release is blocked by any failure in calculation, citation, entitlement or disclosure tests; forecasting features additionally need a signed validation report showing they beat naive baselines out of sample with calibrated intervals.

## 19.1 Test suites

| ID | Suite | Method | Pass threshold | Justification |
|---|---|---|---|---|
| TST-CALC-01 | Financial formula correctness | Golden datasets: 30 companies x all ratios, reviewed by two analysts in a spreadsheet; property-based tests (e.g. ratio sign rules) | 100% match to 4 decimal places | Deterministic code |
| TST-CALC-02 | Technical indicators | Compare to a reference library on 50 series including gaps and corporate actions | Max abs. difference 1e-6 | Same formulas |
| TST-DATA-01 | Data accuracy | Daily sample of 50 random closes and 20 fundamentals vs exchange-official values or filings | 100% within tolerance (0.01 INR price; exact for filed values) | Licensed source of truth |
| TST-DATA-02 | Freshness | Synthetic delays injected; production monitors | Stale badges within 60 s of SLA breach in 100% of injected cases | FR-061 |
| TST-DATA-03 | Corporate actions | 40 historical events (splits, bonuses, rights, demergers) | Adjusted series match reference within 0.01%; no false gaps | FR-002 |
| TST-FC-01 | Backtest integrity | Leakage audit: features timestamp-checked; survivorship check that delisted stocks are present | Zero leakage findings | 10.5 |

| ID | Suite | Method | Pass threshold | Justification |
|---|---|---|---|---|
| TST-FC-02 | Out-of-sample skill | Walk-forward with purge; 3-year final hold-out; Diebold-Mariano vs B0 and B1 | Positive skill, $p < 0.05$, per published segment | 10.6 publication gate |
| TST-FC-03 | Interval calibration | Coverage of 50% and 80% intervals on hold-out; PIT histogram | 80% interval coverage in 75% to 85%; 50% in 45% to 55% | Calibration tolerance [D] |
| TST-FC-04 | Benchmark comparison | Cross-sectional IC vs zero; long-short decile spread vs NIFTY 50 after costs | Reported; IC significance at 5% for publication of rankings | Avoid return-only proof |
| TST-FC-05 | Transaction costs and slippage | Recompute with cost table and liquidity slippage; sensitivity at 2x costs | Conclusions unchanged at 2x costs | Robustness |
| TST-DER-01 | Options payoffs | All strategies in 12.5 vs analytic payoffs on a price grid | Exact match | Deterministic |
| TST-DER-02 | Greeks and IV | Reference implementation; put-call parity; IV round-trip | 1e-6 tolerance | Deterministic |
| TST-DER-03 | Terminology | AI and glossary tests: long put vs short stock, short put direction, unlimited loss labels | 100% correct on 100 prompts | Section 12.2 guard |
| TST-DER-04 | Contract master | Lot sizes, expiries vs exchange files daily; static scan for hardcoded values | Zero mismatches; zero hardcoded constants | C-06 |

| ID | Suite | Method | Pass threshold | Justification |
|---|---|---|---|---|
| TST-RISK-01 | Stop and sizing | Golden cases incl. short positions, zero/negative inputs, lot rounding, caps | 100% | FR-070 to FR-075 |
| TST-AI-01 | Hallucination | 500 fact sheets with adversarial prompts (missing data, conflicting numbers) | Zero unsupported numbers in output | FR-060 |
| TST-AI-02 | Citation validity | Every citation resolves to a source span that contains the cited fact | 100% on test set; 99.5% in weekly production sample | BO-02 |
| TST-AI-03 | Prohibited language | Red-team set in English and Hindi | Zero prohibited phrases | 15.2 rule 6 |
| TST-AI-04 | Causation guard | 200 event-move cases | Zero "caused" claims; "likely contributed" only when rule A.7 met | 9.4 |
| TST-AI-05 | Insufficient data | 300 synthetic missing-data cases | 100% return insufficient-data card | FR-057 |
| TST-SEC-01 | Security | SAST, DAST, dependency scan, penetration test, role escalation tests | No open critical or high findings at release | NFR-034 |
| TST-SEC-02 | Entitlements | Users at each tier request each data class | Zero unlicensed exposure | FR-125 |
| TST-REG-01 | Regulatory controls | Gated features off in public mode; disclosures present; AI-use label present; education content price-age check | 100% | REG-01, REG-03, REG-08 |

| ID | Suite | Method | Pass threshold | Justification |
|---|---|---|---|---|
| TST-REG-02 | Error correction drill | Simulated material error | Notice within NFR-033 | REG-10 |
| TST-UX-01 | Usability | Task-based testing with 12 users per persona | 80% task completion for core tasks | BO-01 |
| TST-A11Y-01 | Accessibility | Automated plus manual WCAG 2.2 AA audit | No AA violations | NFR-050 |
| TST-PERF-01 | Performance | Load test at 5x assumed peak | NFR-001 to NFR-004 met | NFR |

## 19.2 Acceptance gates

1. **MVP internal beta:** all Must FRs pass; TST-CALC, TST-DATA, TST-AI, TST-SEC, TST-REG green.
2. **Public MVP (tools mode):** gate 1 plus usability and accessibility; counsel's opinion on the tools-mode boundary.
3. **Research features public:** RA registration in place (REG-02), approval workflow live (FR-121), TST-FC-01 to TST-FC-05 signed off by quant lead and compliance for each published segment.
4. **Accuracy page public:** 12 months of ledger data, PaRRVA-compatible method confirmed (REG-09).

## 19.3 Historical store acceptance tests (mandatory, Tests A to J)

The historical store (FR-130 to FR-141) is accepted only when all ten tests pass on a test universe of at least 20 symbols across NSE and BSE, using a mock provider that records every request, plus one run against the real provider.

| ID | Test | Setup | Pass condition |
|---|---|---|---|
| TST-HIST-A | First run | Empty `data/`; lookback 10 years | All configured datasets written as CSV, validated, manifest rows `complete`, quality report produced; models initialised only after validation passes |
| TST-HIST-B | Second run, no new data | Run again with the provider returning no new dates | Zero historical range requests in `download_log.csv` (only latest-date checks); files and checksums unchanged |
| TST-HIST-C | New market data | Provider publishes 3 new trading days and one back-filled gap | Only those 3 days and the gap are requested; merged rows correct; manifest coverage advanced |
| TST-HIST-D | Duplicate provider records | Provider returns overlapping and repeated rows | No duplicate natural keys in any file; `records_duplicate` logged |
| TST-HIST-E | Interrupted download | Kill the process mid-run and corrupt one staging file | Previous valid CSVs intact; next run resumes from checkpoint; failed ranges retried; no file reported `complete` while partial |
| TST-HIST-F | Historical correction | Provider revises 2 past closes; a 1:2 split is announced | Revisions applied by precedence and logged; raw rows versioned; adjusted series and dependent features rebuilt and equal to a full rebuild |
| TST-HIST-G | Restart | Restart the process and the machine (or container with the same volume) | Existing CSVs and manifest are read; no initial download is repeated |
| TST-HIST-H | Prediction | Request a prediction after new data | Uses the local dataset plus newly fetched rows; stores model version, feature-set version, input checksums and as-of dates |

| ID | Test | Setup | Pass condition |
|---|---|---|---|
| TST-HIST-I | Stale data | Provider unavailable and stored data older than the freshness SLA | Prediction is blocked or flagged "unavailable: data stale"; never shown as current |
| TST-HIST-J | API efficiency | 30 simulated daily runs | Download logs show each historical range fetched once (plus the documented look-back window); total API calls within the computed minimum + 10% |

Acceptance gate 1 (19.2) is extended: TST-HIST-A to TST-HIST-J must all pass.

## 19.4 Pipeline, delivery and security tests (v1.0)

Unit tests may use isolated fixtures written outside `data/`; integration tests use the real provider, the real repository and the real Gmail account once credentials exist, and any test that cannot run is reported as blocked, not passed.

| ID | Test | Method | Pass condition |
|---|---|---|---|
| TST-GHA-01 | Persistent CSV reuse across separate runners | Two consecutive workflow runs on fresh runners | Second run reads the CSVs committed by the first; no repeated initial download (with TST-HIST-B, TST-HIST-G) |
| TST-GHA-02 | Scheduled execution | Observe the three slots for 5 weekdays | Each slot starts, records actual start time and late flag, completes or fails with a run record |
| TST-GHA-03 | Data commit | Run with and without changes | Commit only when changed; commit message has run ID; `data` branch consistent |

| ID | Test | Method | Pass condition |
|---|---|---|---|
| TST-GHA-04 | Manual execution | Dispatch with valid and invalid inputs | Valid inputs run the same pipeline; invalid inputs fail fast without data changes; full refresh without reason rejected |
| TST-GHA-05 | Concurrency | Dispatch two runs at once and one during a scheduled run | No corrupted or lost data; second run waits or re-merges |
| TST-GHA-06 | Idempotent re-run | Re-run a completed run | No duplicate rows, predictions or emails |
| TST-GHA-07 | Session state | Runs at the three slots, a weekend and an exchange holiday | Correct completed, incomplete and provisional labels; no false gaps |
| TST-GHA-08 | Feature recalculation | Add new data and a revision | Only affected features recomputed; results equal a full rebuild |
| TST-GHA-09 | Forecast reproducibility | Re-run a past prediction from its run record | Identical outputs |
| TST-GHA-10 | Model selection and validation | Run the selection job | Selection report lists all candidates, baselines, walk-forward results; only approved models used |
| TST-RPT-01 | Report generation | Normal run, partial-data run | All 14 sections present or explicitly unavailable; facts carry citations |
| TST-MAIL-01 | Gmail authentication and submission | Real account with App Password | Email accepted by SMTP; status `submitted` recorded |
| TST-MAIL-02 | Email content | Inspect received email | Contains all required items (15.6); no secrets; addresses not in logs |

| ID | Test | Method | Pass condition |
|---|---|---|---|
| TST-MAIL-03 | Failure notification | Force a validation failure | Failure email labelled FAILED; no stale report sent as current |
| TST-MAIL-04 | Wrong credentials | Invalid App Password | Run records `smtp_error`; job summary shows the fix; data still committed |
| TST-SEC-03 | Secret handling | Scan repository, logs, reports, run records | No secrets found |
| TST-SEC-04 | Access control | Read-only collaborator tries to dispatch | Not permitted |
| TST-SEC-05 | Workflow hardening | Review permissions and pinned actions | Least privilege; all third-party actions pinned |
| TST-PRIN-01 | Engineering principles | Static checks and review | No hardcoded prices, metrics, addresses or credentials; no fixtures under `data/`; no mock provider reachable in production configuration |

## 19.5 Test execution status

No test has been executed: the code, repository, data provider licence and Gmail credentials do not exist yet. This table is updated as tests run.

| Test group | Status | Blocked by |
|---|---|---|
| Calculation and indicator golden tests | Not run | Code not written; repository (OD-28) |
| Historical store Tests A to J | Not run | Repository (OD-28); provider (OD-02) for the real-provider run |
| GitHub Actions tests | Not run | Repository (OD-28) |

| Test group | Status | Blocked by |
|---|---|---|
| Gmail tests | Not run | Sender account and App Password (D-11) |
| Forecast validation | Not run | Licensed historical data (OD-02) |
| Security tests | Not run | Repository (OD-28) |

## 20. Business KPIs

KPIs are split into three families that are never combined into one score: product and business, analytical integrity, and predictive performance; engagement growth is never cited as evidence of forecasting skill.

| ID | KPI | Family | Definition | Target (provisional) | Source |
|---|---|---|---|---|---|
| KPI-01 | Source attribution completeness | Integrity | Numeric claims with valid citation / all numeric claims in published outputs | 100% | Citation gate logs |
| KPI-02 | Data freshness compliance | Integrity | Trading minutes with prices inside SLA / all trading minutes | 99.5% | Freshness monitor |
| KPI-03 | Calculation test pass rate | Integrity | Passing golden tests / all golden tests, per release | 100% | CI |
| KPI-04 | Material errors | Integrity | Confirmed material errors per 1,000 reports; median time to correction | Trend down; correction within 1 business day | REG-10 log |
| KPI-05 | Forecast skill vs baseline | Predictive | 1 - (MAE model / MAE no-change), per horizon, rolling 12 months, out of sample | Above 0 with significance for published segments | FR-111 |
| KPI-06 | Interval calibration | Predictive | Realised coverage of 80% intervals | 75% to 85% | FR-111 |

| ID | KPI | Family | Definition | Target (provisional) | Source |
|---|---|---|---|---|---|
| KPI-07 | Fair-value range hit rate | Predictive | Share of stocks whose 12-month price falls inside the published fundamental range; reported with base rate of a naive range of equal width | Report only; no target | FR-111 |
| KPI-08 | Report generation time | Product | p95 time to full report | Under 20 s | APM |
| KPI-09 | System availability | Product | Market-hours availability | 99.9% | Uptime monitor |
| KPI-10 | Task completion rate | Product | Users completing core research tasks in usability tests and in-product funnels | 80% | UX research, analytics |
| KPI-11 | Terminology comprehension | Product | Uplift in quiz score after glossary help vs control | +20 percentage points | A/B test |
| KPI-12 | Weekly active researchers | Product | Users viewing at least 3 research pages a week | Set after launch | Analytics |
| KPI-13 | Retention | Business | Week-4 and month-3 retention | Set after launch | Analytics |
| KPI-14 | Conversion to paid | Business | Free to paid conversion (if OD-03 picks subscription) | Set after launch | Billing |
| KPI-15 | Cost per completed research report | Business | (Data + compute + LLM + review cost) / completed reports | Budget after OD-02 | FinOps |
| KPI-16 | Complaint resolution within SLA | Compliance | Resolved within regulatory window / all | 100% | Ticketing |

KPI-05 to KPI-07 are published internally monthly and externally only under REG-09.

Operational KPIs added in v1.0 (product family; never evidence of predictive skill):

| ID | KPI | Definition | Target (provisional) | Source |
|---|---|---|---|---|
| KPI-17 | Run success | Successful and failed runs per month, by trigger | 98% scheduled success | Run log |
| KPI-18 | Records downloaded vs reused | New records per run; historical records reused from the store | Reuse ratio rises after initial load | download_log.csv |
| KPI-19 | Missing records and revisions | Open missing ranges; revisions applied per month | Missing ranges trend to 0 | Manifest, revisions log |
| KPI-20 | API calls and run duration | Per run and per month | Within NFR-101 and NFR-107 | Run log |
| KPI-21 | Report and email status | Reports generated; emails submitted; failure notices sent | 99% submitted for successful runs | Run records |
| KPI-22 | Repository storage growth | Packed size of `data` and growth per month | Under thresholds in 14.4.9 | Git metrics |
| KPI-23 | Cost per completed research run | (Actions minutes cost + data licence share + LLM cost) / completed runs | Budget after OD-02 | Usage reports |

## 21. Risks, dependencies, and mitigations

The two risks that can stop the product are regulatory classification and data licensing; both are addressed in Phase 0 before significant build spend.

## 21.1 Risk register

| ID | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| R-01 | Features classified as unregistered research or advice | High without RA route | Severe (enforcement, debarment) | OD-01 decided in Phase 0; counsel opinion; REG-01 feature gates; RA registration before research features | Compliance |
| R-02 | Data licence cost or terms make the product unviable (e.g. no redistribution for retail display) | Medium | Severe | Vendor RFP with 3+ vendors; start with delayed and EOD data; licence registry | Data lead |
| R-03 | Forecasts perceived as predictions; users over-rely | High | High | Ranges not points; calibration shown; baselines shown; publication gate; language filters | Quant lead |
| R-04 | LLM states invented figures or facts | Medium | High | Citation gate, fact-sheet-only prompting, template fallback, red-team tests | AI lead |
| R-05 | Data errors (bad corporate action adjustment, wrong fundamentals) | Medium | High | V1 to V19 checks, dual-source validation, error-correction procedure | Data lead |
| R-06 | Look-ahead or survivorship bias overstates backtest skill | Medium | High | Point-in-time store, leakage audits, independent validation | Quant lead |
| R-07 | Regulatory change (SEBI AI/ML guidelines, F&O rules, data-lag rules) | High | Medium | Monthly regulatory watch; configurable rules; contract master; quarterly BRD review | Compliance |
| R-08 | Personal data breach | Low | Severe | Minimisation, encryption, access control, DPDP programme, incident response | Security |

| ID | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| R-09 | LLM and data costs exceed revenue | Medium | Medium | Caching per snapshot, model tiering, per-user caps, cost KPI | Product, FinOps |
| R-10 | Vendor outage at market open | Medium | Medium | Fallback sources, degraded mode, status banner | Engineering |
| R-11 | Hindi translations misstate regulated content | Low | Medium | Professional translation, compliance review | Content lead |
| R-12 | Reputational harm from a high-profile wrong scenario | Medium | Medium | Balanced bull/bear thesis, invalidation conditions, ledger shows all outcomes | Research head |
| R-13 | Defamation risk from governance flags | Low | High | Factual, cited flags only; legal review of flag wording | Compliance |

## 21.2 Dependencies

| ID | Dependency | Needed for | Blocking? |
|---|---|---|---|
| D-01 | Signed market data licence(s) for prices, corporate actions, indices | All modules | Yes, for any external release |
| D-02 | Fundamentals vendor with point-in-time history | Fundamentals, valuation, backtests | Yes |
| D-03 | Licensed news feed | FR-043, FR-044 | No (Phase 2) |
| D-04 | Derivatives data licence and contract files | Section 12 | Yes for Phase 2 derivatives |
| D-05 | Counsel opinion on OD-01 | Research features | Yes for public research |
| D-06 | RA registration and qualified staff | Public research | Yes (Option B) |
| D-07 | LLM provider contract with data-protection terms | AI layer | Yes |

| ID | Dependency | Needed for | Blocking? |
|---|---|---|---|
| D-08 | Bilingual finance editors | Glossary | Yes for MVP glossary |
| D-09 | Cloud region in India and DR region | Hosting | Yes |

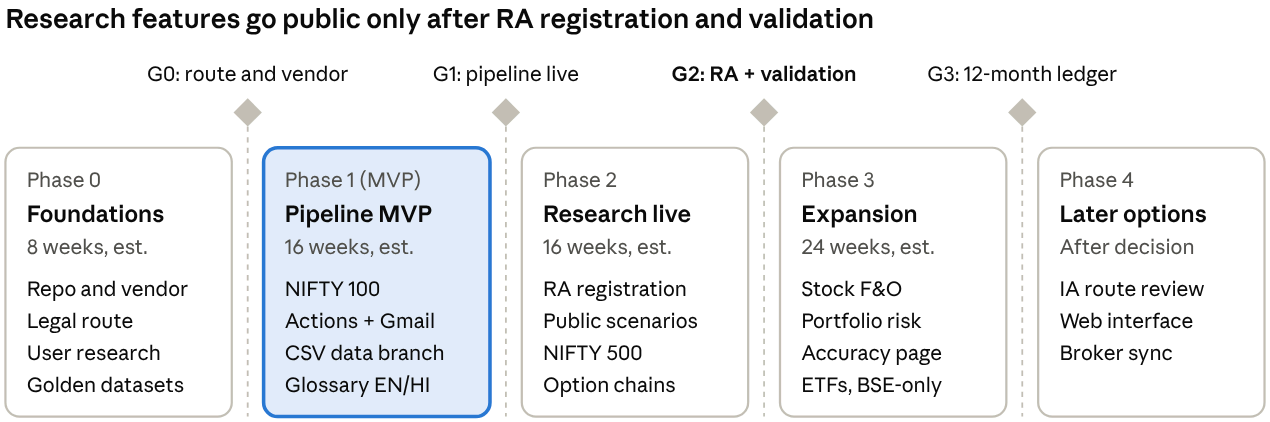
Risks and dependencies added in v1.0:

| ID | Risk | Likelihood | Impact | Mitigation | Owner |
|---|---|---|---|---|---|
| R-14 | GitHub delays or drops a scheduled run | Medium | Medium | Record actual start, label late runs, failure alert if a slot produced no report; manual re-run | Engineering |
| R-15 | Data branch outgrows practical Git limits | Medium (if intraday added) | High | Year-partitioned files, no intraday history in Git, thresholds and approved migration (14.4.9) | Data lead |
| R-16 | Gmail App Password unavailable or revoked (password change, policy) | Medium | Medium | Failure visible in run summary; documented re-issue; Gmail API alternative (OD-26) | Sponsor |
| R-17 | Secret leaked through logs or commits | Low | Severe | Masking, secret scanning, pinned actions, rotation | Engineering |
| R-18 | Overlapping runs corrupt data | Low | High | Concurrency group, fast-forward-only push with re-merge, idempotent steps | Engineering |
| R-19 | Provider blocks access from GitHub-hosted runners (IP allow-listing) | Medium | High | Confirm in RFP (A-13); fallback is a self-hosted runner, which is new infrastructure needing approval | Data lead |
| R-20 | Actions minutes exceed the plan allowance | Low | Low | Budget per run (NFR-101), monitoring (NFR-110) | Engineering |

| ID | Dependency | Needed for | Blocking? |
|---|---|---|---|
| D-10 | Private GitHub repository chosen and attached to this project (OD-28) | All implementation | Yes |
| D-11 | Gmail sender account with 2-Step Verification and an App Password stored as a secret | Email delivery | Yes for FR-160 |
| D-12 | Approved recipient list (OD-22) | Email delivery to anyone but the sponsor | Yes for extra recipients |
| D-13 | Provider API reachable from GitHub-hosted runners | Production runs | Yes |

## 22. MVP scope and phased roadmap

The MVP is a tools-mode product on NIFTY 100 that can launch without RA registration, while the forecasting and research features are built and validated in parallel and released at gate G2; durations are planning estimates [A] until vendor and legal timelines are known.




Phased roadmap · 5 phases, 4 gates

Each gate is a hard stop: work continues on the next phase internally, but nothing gated reaches the public until the gate's criteria in 22.3 are met.

## 22.1 MVP scope (Phase 1)

- Universe T1 (NIFTY 100), end-of-day plus 15-minute delayed prices under licence.
- Python pipeline on GitHub Actions: three weekday runs and manual runs, one core pipeline, concurrency control, run records (FR-150 to FR-159).
- Persistent CSV store on a private `data` branch with incremental updates and Tests A to J (FR-130 to FR-141).
- HTML email report with CSV summary through Gmail, and failure notices (FR-160 to FR-165).
- Analytics in the report: scorecard, technical summary, valuation ranges with insufficient-data handling, risk calculations for configured assumptions, index option payoffs; named-stock scenarios and horizon forecasts only for the sponsor′s own use until OD-01 and OD-22 are closed.
- Citation gate, timestamps, audit records, disclosures; glossary explanations of terms used in the report (English and Hindi).
- Deferred: interactive screens in 16.2 (OD-21).

## 22.2 Prioritised MVP backlog

Ordered by dependency then value. Size: S under 2 weeks, M 2 to 4 weeks, L over 4 weeks of a small squad [A].

| Rank | Epic | FRs | Priority | Size | Depends on |
|---|---|---|---|---|---|
| 1 | Regulatory route decision and counsel opinion | REG-01 | Must | M | OD-01 |
| 2 | Data vendor selection and licence registry | FR-125 | Must | M | OD-02 |
| 3 | Security master and ISIN mapping | FR-001, FR-004 | Must | M | 2 |
| 4 | Persistent CSV store, initial download, incremental updates, revisions, corporate action adjustment (mandatory) | FR-002, FR-030, FR-130 to FR-139 | Must | L | 3 |
| 5 | Point-in-time fundamentals store | FR-020 | Must | L | 2, 3 |
| 6 | Audit log, snapshot IDs, timestamps | FR-061, FR-123 | Must | M | 4 |
| 7 | Ratio engine and golden tests | FR-021, FR-022, FR-023, FR-024 | Must | L | 5 |

| Rank | Epic | FRs | Priority | Size | Depends on |
|---|---|---|---|---|---|
| 8 | Technical indicator engine and summary | FR-031, FR-033, FR-034, FR-035 | Must | M | 4 |
| 9 | Accounts, roles, disclosures, language | FR-003, FR-005, FR-120 | Must | M | none |
| 10 | Stock profile and statements screens | FR-020 to FR-026 UI | Must | M | 7 |
| 11 | Screener | FR-010, FR-012 | Must | M | 7, 8 |
| 12 | Market overview and event calendar | FR-040, FR-042, FR-045 | Must | M | 4 |
| 13 | Stop-loss, sizing, R:R calculators with warnings | FR-070 to FR-073, FR-075 | Must | M | 8 |
| 14 | Contract master, Greeks, payoff builder, risk gate | FR-080, FR-082, FR-083, FR-087 | Must | M | 2 |
| 15 | Glossary CMS, search, contextual help | FR-090, FR-091, FR-092 | Must | L | 9 |
| 16 | Fact sheet, AI drafting, citation gate | FR-060, FR-062, FR-063 | Must | L | 7, 8, 6 |
| 17 | Valuation methods and insufficient-data card | FR-050, FR-052, FR-054, FR-057 | Must | L | 7 |
| 18 | Scenario builder and invalidation conditions (internal beta) | FR-055, FR-058 | Must (gated) | M | 17 |
| 19 | Forecast ledger and evaluation jobs | FR-110, FR-111 | Must | M | 6 |
| 20 | Watchlists | FR-100 | Must | S | 9 |
| 21 | Complaints and error correction | FR-122, FR-124 | Must | S | 9 |
| 22 | Research approval workflow | FR-121 | Must (before G2) | M | 9 |
| 23 | Statistical baselines and challenger models (internal) | FR-056 | Should | L | 19 |
| 24 | Governance flags, ownership extras | FR-025 | Should | M | 5 |
| 25 | Alerts, saved screens | FR-013, FR-101 | Should | M | 11 |

Backlog additions in v1.0, placed by dependency ("after" names the rank above they follow):

| Rank | Epic | FRs | Priority | Size | Depends on |
|---|---|---|---|---|---|
| 0 | Private repository, branch protection, secrets, `data` branch, CI skeleton | FR-171, FR-172, FR-176 | Must | S | OD-28 |
| after 2 | Provider adapter interface, licence registry entries, provenance | FR-142 to FR-144 | Must | M | OD-02 |
| after 4 | Config schema and validation | FR-170 | Must | S | 0 |
| after 6 | Workflow schedules, dispatch inputs, core pipeline entry point, concurrency, run records, data commits | FR-150 to FR-159 | Must | L | 0, 4 |
| after 16 | HTML report renderer and CSV summary | FR-165 | Must | M | 16 |
| after 16 | Gmail delivery, failure notices, idempotency, status recording | FR-160 to FR-164 | Must | M | D-11 |
| after 19 | Model registry, selection job, evaluation reports | FR-180 to FR-182, FR-184 | Must | L | 19 |
| after 25 | Structured logs, job summaries, freshness indicators, change audit | FR-173 to FR-175 | Must | M | 6 |

Ranks 9, 10, 11, 12 and 20 now produce report sections rather than screens; their screen work moves to Phase 4.

## 22.3 Phase plan and exit criteria

| Phase | Goal | Main deliverables | Exit gate and criteria |
|---|---|---|---|
| 0 Foundations | De-risk regulation and data | Counsel opinion; route decision; vendor RFP and contract; user research baselines; golden datasets; architecture spike | G0: OD-01, OD-02, OD-03 closed; licence signed |
| 1 Pipeline MVP | Launch useful, compliant tools on NIFTY 100 | Backlog ranks 0 to 22 and the v1.0 additions: GitHub Actions pipeline, persistent CSV store, Gmail reports | G1: acceptance gate 2 (19.2) met |

| Phase | Goal | Main deliverables | Exit gate and criteria |
|---|---|---|---|
| 2 Research live | Publish research under RA registration | RA registration, approval workflow, public scenarios and ranges, NIFTY 500, news feed, option chains | G2: acceptance gate 3; RA certificate; validated segments only |
| 3 Expansion | Broaden coverage and risk tools | Stock F&O, portfolio risk, ETFs, BSE-only stocks, accuracy page | G3: acceptance gate 4 |
| 4 Later options | Decide on personalisation and channels | Web interface (16.2) with secured workflow trigger, IA route review, broker read-only sync | Business case and legal review |

## 23. Open questions requiring business decisions

Twenty-eight decisions are open; OD-01, OD-02, OD-03, OD-22 and OD-28 block Phase 1 build, and each has a recommended default so work can proceed while the business decides.

| ID | Decision needed | Options | Recommended default | Blocks | Owner |
|---|---|---|---|---|---|
| OD-01 | Regulatory route | A data/education only; B register as RA; C partner RA; D IA | B, with tools-mode launch first | All gated features | Founder + counsel |
| OD-02 | Market and fundamentals data vendor(s) | Direct exchange licences; one aggregator vendor; mix | One authorised aggregator for MVP, direct exchange licence evaluated for Phase 2; the contract must permit persistent local CSV storage and state retention rights after termination | Everything | Data lead |

| ID | Decision needed | Options | Recommended default | Blocks | Owner |
|---|---|---|---|---|---|
| OD-03 | Monetisation | Free with ads; freemium subscription; paid only; B2B API | Freemium subscription; no ads (conflict risk); fees within RA caps | Pricing, KPIs, RA fee rules | Founder |
| OD-04 | Price data tier | EOD only; 15-min delayed; real-time | 15-min delayed + EOD in MVP; real-time for paid tier later | Licence cost | Product |
| OD-05 | Derivatives in MVP | Education payoff tool only; plus chains and OI | Payoff and Greeks only | Phase 1 scope | Product + compliance |
| OD-06 | Primary exchange for dual-listed prices | NSE; BSE; best of both | NSE | Display rules | Data lead |
| OD-07 | Forecast publication thresholds | As in 10.6; stricter; looser | As in 10.6, reviewed after first validation run | FR-056 release | Quant lead + compliance |
| OD-08 | House equity risk premium and terminal growth policy | Fixed; quarterly review; user-only input | Versioned house value, reviewed quarterly, user can override in tools mode | DCF | Research head |
| OD-09 | Languages after Hindi | Hindi only; add Marathi, Gujarati, Tamil, Bengali, Telugu | Hindi only in MVP; next language by user data | Glossary scope | Product |
| OD-10 | Scorecard verdict | Show pillar percentiles only; add overall score | Pillars only until RA registered | FR-026 | Compliance |
| OD-11 | Daily research queue (FR-014) | Build; defer | Build internally, release after G2 | FR-014 | Product + compliance |
| OD-12 | LLM provider and data residency | Global API with contract terms; India-hosted model; self-hosted | Provider with no-training and data-protection terms; no personal data in prompts | AI layer | Engineering + security |
| OD-13 | Human review of AI narratives in tools mode | None; sampling; full | Sampling (5%) in tools mode; full review for published research | Ops cost | Research head |

| ID | Decision needed | Options | Recommended default | Blocks | Owner |
|---|---|---|---|---|---|
| OD-14 | Liquidity floor for analytics | INR 1 crore; INR 5 crore; dynamic | INR 1 crore 20-day median traded value, recalibrated | Coverage | Quant lead |
| OD-15 | Public accuracy page | Publish all forecasts; internal only | Internal until 12 months of ledger and REG-09 cleared | FR-112 | Compliance |
| OD-16 | Data retention periods | 5 years; longer | 5 years minimum for regulated records; shorter for analytics logs | NFR-031, DPDP | Compliance + legal |
| OD-17 | B2B offering to registered RAs (persona Vikram) | Yes; no | Phase 3 option | Roadmap | Founder |
| OD-18 | Portfolio import method | Manual/CSV; broker read-only sync | Manual/CSV in MVP | FR-102 | Product |
| OD-19 | Social or community features | None; moderated | None (finfluencer-association risk) | Scope | Founder + compliance |
| OD-20 | Brand naming and positioning words | Avoid "AI predicts"; allow | Avoid prediction language in all marketing | Marketing | Founder + compliance |

### Requirements that cannot be implemented without licensed data or a business decision

| Requirement | Needs |
|---|---|
| Any live or delayed price display, charts, screener on prices, technicals, risk calculators with market data | Exchange or vendor licence (D-01, OD-02) |
| Index levels, constituents and history | Index data licence (D-01) |
| Point-in-time fundamentals and shareholding | Fundamentals vendor (D-02) |
| Option chains, OI, IV analytics, margin estimates | Derivatives data licence and risk-parameter files (D-04) |
| News classification and event impact cards | Licensed news feed (D-03) |

| Requirement | Needs |
|---|---|
| Named-stock fair values, scenarios, horizon forecasts, ranked lists, platform-suggested stops | OD-01 (RA route) and counsel opinion |
| Public accuracy or performance claims | REG-09, PaRRVA method |
| Sentiment analytics | Licensed sentiment or social data (not scraping) |

### Decisions added in v1.0

| ID | Decision needed | Options | Recommended default | Blocks | Owner |
|---|---|---|---|---|---|
| OD-21 | Defer the interactive web interface | Defer to Phase 4; build a minimal static page; build now with a server | Defer; email reports only in v1.0 | Screens, accounts | Sponsor |
| OD-22 | Report recipients | Sponsor only; named individuals; subscribers | Sponsor only until counsel and the data licence allow more | Email to others | Sponsor + compliance |
| OD-23 | Repository visibility | Private; public | Private (licensed data) | Everything | Sponsor |
| OD-24 | Storage migration path beyond 1 GB | Git LFS; external object storage; reduce scope | Decide when the 500 MB warning fires | Phase 3 scale | Data lead |
| OD-25 | Intraday history | None in Git (current-session snapshot only); external store | None in Git for v1.0 | Intraday analytics | Data lead |
| OD-26 | Email method if App Passwords are unavailable | Gmail API with OAuth; other approved SMTP service | Gmail API with OAuth | Email delivery | Sponsor |
| OD-27 | `REPORT_RECIPIENT` as secret or variable | Secret (hidden); variable (visible to collaborators) | Secret | Configuration | Sponsor |

| ID | Decision needed | Options | Recommended default | Blocks | Owner |
|---|---|---|---|---|---|
| OD-28 | Which GitHub repository hosts code and data | A new private repository; an existing private repository | New private repository dedicated to this platform | All implementation | Sponsor |

**Implementation blockers (as of 11 Oct 2026)**

| Blocker | Why it blocks | What unblocks it |
|---|---|---|
| No GitHub repository attached to this project | Code, workflows, data branch and tests have nowhere to live | Sponsor names or creates a private repository (OD-28) and it is attached to the project |
| No licensed market and fundamentals data provider | Production runs must not use scraped or mock data | Vendor selected and contract signed (OD-02), credentials stored as secrets |
| No Gmail sender credentials | Email delivery cannot be tested or run | Sender account with 2-Step Verification, App Password stored as `GMAIL_APP_PASSWORD` (D-11) |
| Regulatory route not decided | Determines who may receive named-stock forecasts | OD-01 and OD-22 with counsel's opinion |

Until these clear, implementation can build the code, configuration schema, workflows and isolated unit tests, but no production run, integration test or report can be claimed as working.

# 24. Appendices

The appendices hold the formulas the engines implement (A), a sample report (B), calculation examples (C), the requirements traceability matrix (D) and the sources consulted (E); all example values are hypothetical.

# Appendix A. Formulas

**A.1 Corporate action adjustment factors.** Prices before the ex-date are multiplied by the factor; volumes are divided by it. The cumulative factor for a date is the product of the factors of all later events.

$$\text{Split (1 old into } n \text{ new)}: f = \frac{1}{n} \qquad \text{Bonus } (a \text{ new for every } b \text{ held}): f = \frac{b}{a+b}$$

$$\text{Rights}: TERP = \frac{N P_{cum} + R \cdot S}{N + R}, \quad f = \frac{TERP}{P_{cum}} \qquad \text{Dividend (total return only)}: f = 1 - \frac{D}{P_{cum}}$$

N = shares held, R = rights shares offered, S = subscription price, P_cum = last cum-rights or cum-dividend close.

**A.2 Ratios.**

$$ROE = \frac{PAT_{owners}}{\tfrac{1}{2}(Equity_{t} + Equity_{t-1})} \qquad ROCE = \frac{EBIT}{\tfrac{1}{2}(CE_{t} + CE_{t-1})}, \; CE = \text{Total assets} - \text{Current liabilities}$$

$$P/E = \frac{P}{EPS_{TTM,diluted}} \qquad P/B = \frac{P}{BVPS} \qquad D/E = \frac{\text{Borrowings}}{\text{Equity}} \qquad FCF = OCF - Capex$$

$$EV = \text{Market cap} + \text{Borrowings} + \text{Minority interest} - \text{Cash and liquid investments}$$

**A.3 Trend rule (FR-023).** Metric m is Improving if TTM_t / TTM_t-4 - 1 > theta and at least 3 of the last 4 YoY quarterly changes are positive; Deteriorating if the mirror conditions hold; otherwise Stable. Default theta = 5% for growth metrics, 1 percentage point for margins and returns [D].

**A.4 Technical indicators.**

$$SMA_n = \frac{1}{n} \sum_{i=0}^{n-1} C_{t-i} \qquad EMA_t = \alpha C_t + (1-\alpha) EMA_{t-1}, \; \alpha = \frac{2}{n+1}$$

$$RSI = 100 - \frac{100}{1 + \frac{AvgGain_{14}}{AvgLoss_{14}}} \quad \text{(Wilder smoothing: } Avg_t = \tfrac{13 \cdot Avg_{t-1} + X_t}{14})$$

$$MACD = EMA_{12} - EMA_{26}, \quad Signal = EMA_9(MACD), \quad Histogram = MACD - Signal$$

$$TR_t = \max(H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|), \quad ATR_t = \tfrac{13 \cdot ATR_{t-1} + TR_t}{14}$$

$$\sigma_{ann} = \sqrt{252} \cdot \text{stdev}(\ln(C_t / C_{t-1})) \qquad MaxDD = \min_t \left( \frac{C_t}{\max_{s \le t} C_s} - 1 \right)$$

**A.5 Support and resistance zones.** A swing high is a bar whose high is the maximum within 5 bars either side; swing lows likewise. Swing points from the last 252 bars are clustered when within 0.5 x ATR of each other. A zone = [min, max] of the cluster; strength = number of touches; zones with fewer than 2 touches are discarded. Up to 3 zones on each side of the current price, nearest first.

**A.6 Technical summary states.** As tabulated in section 9.2; each state stores the rule text that fired.

**A.7 Causation guard (event study).** Market model estimated on days -250 to -30 before the event; abnormal return on the event window [0, +1].

$$AR_t = R_{i,t} - (\hat\alpha + \hat\beta R_{m,t}), \quad CAR = \sum_{t=0}^{1} AR_t, \quad t_{CAR} = \frac{CAR}{\hat\sigma_{AR} \sqrt{2}}$$

"Likely contributed" is allowed only if |t_CAR| > 1.96 and no other material event for the stock is in the window; otherwise only "coincided with".

**A.8 Free cash flow to firm.**

$$FCFF = EBIT (1 - t) + D\&A - Capex - \Delta NWC$$

DCF, WACC, cost of equity, justified P/B and DDM are in section 10.3; Black-Scholes-Merton and Greeks in 12.3; position sizing in 11.3.

## Appendix B. Sample stock research report (HYPOTHETICAL data)

> **HYPOTHETICAL EXAMPLE.** "ABC Industries Ltd" is fictional. No figure below describes a real company or price.

| Section | Sample content |
|---|---|
| Header | ABC Industries Ltd · NSE: ABCIND · INR 1,000.00 · NSE · as of 10 Oct 2026 15:30 IST, end of day · Freshness: within SLA · Report ID R-000123, snapshot S-20261010-EOD, engine v1.4.2, AI narrative v0.9 |
| 1. Company overview | Manufacturer of industrial pumps; 3 segments (pumps 62% of FY26 revenue, valves 28%, services 10%) [source: FY26 annual report p.14]. Member of NIFTY 100. |
| 2. Current data | Market cap INR 52,000 crore; 20-day median traded value INR 85 crore; beta 0.92 |
| 3. Fundamental scorecard | Growth 78th percentile (TTM revenue +14% YoY) · Profitability 71st (ROCE 22%) · Balance sheet 85th (D/E 0.15) · Cash conversion 64th (OCF/EBITDA 0.81, 3 years) · Valuation 30th (P/E 32 vs peer median 27) · Governance: no flags. Peer set: 9 industrial machinery companies. |
| 4. Technical summary | Trend: up (close above 50-day SMA above 200-day SMA, 200-day slope positive) · Momentum: strong (RSI 63) · Volatility: normal (ATR 1.9% of price, 48th percentile) · RS vs NIFTY 50 (252 days): leading · Support zone 952 to 965; resistance 1,045 to 1,058. Conditions, not recommendations. |
| 5. Sector and market context | Capital goods index outperformed NIFTY 50 by 4.1% over 3 months; market regime: uptrend; 61% of NIFTY 500 above 200-day SMA. |
| 6. News and events | Confirmed: order win of INR 450 crore announced 2 Oct 2026 [exchange filing]. Upcoming: Q2 FY27 results board meeting 6 Nov 2026. |
| 7. Valuation range | DCF 940 to 1,250 (mid 1,080) · Peer P/E 900 to 1,150 (mid 1,020) · Own 5-year P/E 850 to 1,100 (mid 960) · Combined fundamental estimate range INR 990 to 1,050 (-1% to +5%). |

| Section | Sample content |
|---|---|
| 8. Scenarios (12 months) | Bull 1,250 (+25%): revenue growth 18%, EBITDA margin 21%, P/E 30 · Base 1,040 (+4%): 13%, 19%, 26 · Bear 780 (-22%): 6%, 15%, 20. Horizon ranges: not validated for this segment. |
| 9. Risk assessment | 1-year max drawdown -18%; overnight gaps beyond 1 ATR on 4% of days; promoter pledge 0%; customer concentration high (top 5 = 41% of revenue) [annual report p.62]. |
| 10. Bull and bear thesis | Bull: order book cover 1.6x revenue; margin expansion from services mix. Bear: valuation above peers; capex cycle sensitivity; customer concentration. Each point cites its source. |
| 11. Sources | [1] FY26 annual report; [2] Q1 FY27 results filing; [3] exchange filing 2 Oct 2026; [4] licensed price data; [5] peer data. |
| 12. Model limitations | DCF highly sensitive to terminal growth (±INR 140 per 1 point); peer set small for valves segment; scenarios are not probabilities. |
| 13. Assumptions and invalidation | Base case invalid if EBITDA margin below 17% for two consecutive quarters, or order inflow falls more than 20% YoY for two quarters, or net debt/EBITDA rises above 2.5x. |
| Confidence | Medium. Data completeness High; source tier High; method agreement High (midpoints within 12.5%); model validation Low for horizon ranges (not validated), Medium for valuation; liquidity High. Overall = lowest relevant factor. |
| Disclosures | Research disclosures (registration details, conflicts, analyst holdings), extent of AI use ("Narrative drafted with AI from verified data; all figures computed by deterministic engines; reviewed by [analyst]"), not a guarantee of returns. |

## Appendix C. Calculation examples (HYPOTHETICAL values)

Assumptions for all derivatives examples: index level S = 24,000; lot size L = 65 used only for illustration (the engine always reads the live contract master); premiums are hypothetical.

| Example | Legs | Max profit (points / INR per lot) | Max loss (points / INR per lot) | Break-even(s) | Payoff at a test expiry level |
|---|---|---|---|---|---|
| C.1 Long put | Buy 23,800 put at 95 | 23,705 points if index went to 0 (theoretical) | 95 / 6,175 | 23,705 | At 23,400: (23,800 - 23,400) - 95 = 305 points = INR 19,825 |
| C.2 Bull call spread | Buy 24,000 call at 180; sell 24,300 call at 70; net debit 110 | 190 / 12,350 | 110 / 7,150 | 24,110 | At 24,200: 200 - 110 = 90 points = INR 5,850 |
| C.3 Short straddle | Sell 24,000 call at 180 and 24,000 put at 160; credit 340 | 340 / 22,100 | Unlimited | 23,660 and 24,340 | At 24,800: 340 - 800 = -460 points = -INR 29,900 |
| C.4 Long call | Buy 24,000 call at 180 | Unlimited | 180 / 11,700 | 24,180 | At 23,900: -180 points = -INR 11,700 |
| C.5 Short put | Sell 23,800 put at 95 | 95 / 6,175 | 23,705 points (to 0) | 23,705 | At 23,400: 95 - 400 = -305 points = -INR 19,825 |

**C.6 Black-Scholes-Merton and Greeks.** S = 24,000, K = 24,000, T = 7/365, r = 6.5%, q = 1.2%, sigma = 13% (all hypothetical). Results: call = 184.71, put = 160.33; call delta = 0.526; gamma = 0.000921 per point; vega = 13.23 per volatility point; call theta = -14.08 points per calendar day. Put-call parity check: C - P = 24.38 = S e^(-qT) - K e^(-rT) = 24.38. Values computed with the formulas in 12.3.

**C.7 Futures fair value.** S = 24,000, r = 6.5%, q = 1.2%, 30 days to expiry: F = 24,000 x e^((0.065 - 0.012) x 30/365) = 24,104.8. A quoted future of 24,150 implies 45.2 points premium to modelled fair value.

**C.8 Stop-loss and sizing.** See section 11.5.

**C.9 Valuation range.** See section 10.8.

**C.10 Corporate action adjustment.** A 1:5 split (1 old share into 5) on ex-date E: all prices before E are multiplied by 0.2 and volumes by 5. A 1:1 bonus thereafter (1 new for every 1 held) adds factor 0.5; prices before both events carry cumulative factor 0.2 x 0.5 = 0.1.

# Appendix D. Requirements traceability matrix

## D.1 Business objectives to requirements, tests and KPIs

| Objective | Functional requirements | NFR / REG | Tests | KPIs |
|---|---|---|---|---|
| BO-01 Faster research | FR-010 to FR-014, FR-020 to FR-026, FR-030 to FR-035, FR-064 | NFR-001 to NFR-004 | TST-UX-01, TST-PERF-01 | KPI-08, KPI-10 |
| BO-02 Traceable claims | FR-060, FR-061, FR-062, FR-123 | NFR-040, NFR-041, REG-04 | TST-AI-01, TST-AI-02 | KPI-01 |
| BO-03 Correct calculations | FR-002, FR-021, FR-031, FR-051 to FR-054, FR-070 to FR-072, FR-082, FR-083 | NFR-042 | TST-CALC-01, TST-CALC-02, TST-DER-01, TST-DER-02, TST-RISK-01, TST-DATA-03 | KPI-03 |
| BO-04 Fresh, disclosed data | FR-061, FR-004, FR-042, FR-130 to FR-141 | NFR-010 to NFR-013, NFR-071 | TST-DATA-01, TST-DATA-02 | KPI-02 |
| BO-05 Honest forecasts | FR-056, FR-057, FR-063, FR-110, FR-111 | NFR-043, NFR-044 | TST-FC-01, TST-FC-03, TST-AI-05 | KPI-06 |
| BO-06 Forecasts add information | FR-056, FR-111, FR-112 | REG-09 | TST-FC-02, TST-FC-04, TST-FC-05 | KPI-05, KPI-07 |
| BO-07 Users understand terms | FR-090 to FR-094, FR-005 | NFR-050, NFR-051 | TST-DER-03, TST-A11Y-01 | KPI-11 |
| BO-08 Operate compliantly | FR-087, FR-120 to FR-125, FR-014 and FR-055 gating | REG-01 to REG-16, NFR-031, NFR-033, NFR-080, NFR-081 | TST-REG-01, TST-REG-02, TST-SEC-02 | KPI-04, KPI-16 |
| BO-09 Viable unit economics | FR-126 | NFR-060, NFR-061 | Cost monitoring checks | KPI-15 |

## D.2 Brief requirements to BRD coverage

| Brief section | Where addressed | Key IDs |
|---|---|---|
| 2 Business objective (10 questions) | 1, 3, 7.2 | BO-01 to BO-09 |
| 3 Indian market coverage | 6 | FR-001, FR-004 |
| 4A Fundamental analysis | 8.3, 9.1 | FR-020 to FR-026 |
| 4B Technical analysis | 8.4, 9.2, A.4 to A.6 | FR-030 to FR-036 |
| 4C Market, sector, economy | 8.5, 9.3 | FR-040, FR-041, FR-046 |
| 4D News and events | 8.5, 9.4, A.7 | FR-042 to FR-045 |
| 5 Valuation and forecasting | 8.6, 10 | FR-050 to FR-058, FR-110, FR-111 |
| 6 Risk and stop-loss | 8.8, 11 | FR-070 to FR-076 |
| 7 Options and derivatives | 8.9, 12, C | FR-080 to FR-087 |
| 8 Glossary | 8.10, 13 | FR-090 to FR-094 |
| 9 Data sources and quality | 14 | V1 to V19, FR-125 |
| 10 Architecture and explainability | 15 | FR-060, FR-062 |
| 11 UI and reports | 16, B | FR-064 |
| 12 Functional requirements | 8 | FR-001 to FR-126 |
| 13 Non-functional requirements | 17 | NFR-001 to NFR-081 |
| 14 Regulatory and ethical | 18 | REG-01 to REG-16 |
| 15 Testing | 19 | TST-xx |
| 16 KPIs | 20 | KPI-01 to KPI-16 |
| 18 Working instructions | Document control labels; 5.4; 23 | C-01 to C-08, OD-01 to OD-20 |

## D.3 Module to test coverage

| Module | FRs | Primary tests |
|---|---|---|
| Foundation | FR-001 to FR-005 | TST-DATA-03, TST-SEC-01 |
| Screening | FR-010 to FR-014 | TST-CALC-01 (SQL reference), TST-REG-01 |
| Fundamentals | FR-020 to FR-026 | TST-CALC-01, TST-DATA-01 |
| Technicals | FR-030 to FR-036 | TST-CALC-02 |
| Market and events | FR-040 to FR-046 | TST-DATA-01, TST-AI-04 |
| Valuation and forecasting | FR-050 to FR-058 | TST-CALC-01, TST-FC-01 to TST-FC-05, TST-AI-05 |
| Trust layer | FR-060 to FR-064 | TST-AI-01 to TST-AI-03 |
| Risk | FR-070 to FR-076 | TST-RISK-01 |
| Derivatives | FR-080 to FR-087 | TST-DER-01 to TST-DER-04 |
| Glossary | FR-090 to FR-094 | TST-DER-03, TST-A11Y-01 |
| Watchlist, ledger, compliance | FR-100 to FR-126 | TST-REG-01, TST-REG-02, TST-SEC-02 |

Historical store trace: the sponsor's mandatory requirement of 11 Oct 2026 maps to section 14.4, FR-130 to FR-141, NFR-090 to NFR-097, TST-HIST-A to TST-HIST-J, and KPI-02; it supports BO-04 (fresh data) and BO-05 (reproducible, honest forecasts).

### D.4 Consolidation specification (v1.0) to BRD sections

| Spec item | BRD location | Key IDs |
|---|---|---|
| Executive summary, problem, objectives | 1, 2, 3 | BO-01 to BO-14 |
| Scope, exclusions, assumptions | 5, 5.5, 5.6 | A-01 to A-13, C-01 to C-14 |
| Stakeholders and user journeys | 4 | J1 to J6 |
| Six-pillar architecture and responsibility matrix | 15, 15.0, 8 (pillar mapping) | P1 to P6 |
| Data sources and licensing | 14, 14.5 | FR-142 to FR-144 |
| Persistent CSV storage and incremental ingestion | 14.4, 14.4.9 | FR-130 to FR-141, TST-HIST-A to J |
| Research and model-selection methodology | 9, 10.9 | FR-180 to FR-184 |
| Valuation and forecasting | 10 | FR-050 to FR-058, FR-184 |
| Risk and derivatives | 11, 12 | FR-070 to FR-087 |
| Report specifications | 16.3, 16.4 | FR-165 |
| GitHub Actions scheduling and on-demand execution | 15.5 | FR-150 to FR-159 |
| Gmail delivery | 15.6 | FR-160 to FR-164 |
| Security and configuration | 15.7 | FR-170 to FR-176 |
| Shared contracts | 15.4 | Contracts table |
| Non-functional requirements | 17 | NFR-090 to NFR-111 |
| Regulatory and ethical | 18, 18.6 | REG-01 to REG-16 |
| Testing and acceptance | 19, 19.3 to 19.5 | TST-HIST, TST-GHA, TST-MAIL, TST-SEC, TST-PRIN |
| Monitoring and KPIs | 20 | KPI-01 to KPI-23 |
| Risks, dependencies | 21 | R-01 to R-20, D-01 to D-13 |
| Phased roadmap | 22 | Phases 0 to 4 |

| Spec item | BRD location | Key IDs |
|---|---|---|
| Open decisions and blockers | 23 | OD-01 to OD-28 |

New objectives trace as follows: BO-10 and BO-11 to FR-150 to FR-159, NFR-101 to NFR-103, TST-GHA-01 to TST-GHA-07, KPI-17; BO-12 to FR-160 to FR-164, NFR-108, TST-MAIL-01 to TST-MAIL-04, KPI-21; BO-13 to FR-155, FR-175, FR-181, NFR-097, NFR-106, TST-GHA-09; BO-14 to FR-133, NFR-091, NFR-107, TST-HIST-J, KPI-18.

## Appendix E. Sources consulted (opened 11 Oct 2026)

Regulation

- SEBI (Research Analysts) Regulations, 2014, last amended 16 Dec 2024
- SCC Online: SEBI (Research Analysts) (Third Amendment) Regulations, 2024
- AZB & Partners: Overhaul of the regulatory framework for IAs and RAs
- FinSec Law Advisors: tracker on the Jan 2025 guidelines for RAs and IAs
- Outlook Money: AI-use disclosure for RAs and IAs
- Lakshmikumaran & Sridharan: SEBI RA FAQs of 23 Jul 2025
- Fox Mandal: SEBI FAQs of 29 Jan 2025 on finfluencers and education
- TaxGuru: SEBI circular of 8 May 2026 on price data for education (30-day lag)
- Taxmann: SEBI revises time-lag norms for price data
- Business Standard: SEBI launches PaRRVA
- Taxmann: SEBI AI/ML consultation paper of 20 Jun 2025
- Moneylife: SEBI to mandate human oversight and kill switches for AI
- Taxmann: retail algo trading framework deadline extended to 1 Oct 2025
- AZB & Partners: DPDP Act phased rollout

Derivatives

- Zerodha: SEBI's 1 Oct 2024 index derivatives measures
- Taxmann: expiry limited to Tuesday or Thursday (circular 26 May 2025)
- Moneylife: NSE Tuesday and BSE Thursday expiries from 1 Sep
- Probe42: equity derivatives framework 2025 (FutEq OI, MWPL, position limits)
- Business Standard: lower NIFTY and Bank Nifty lot sizes from Jan 2026 expiries

Market data

- NSE Data Sharing and Usage Policy
- NSE website Terms of Use
- Kotak Neo: BSE brings international market data licensing in-house from Jan 2027

Platform (GitHub and Google)

- GitHub Docs: events that trigger workflows (schedule, workflow_dispatch)
- GitHub Docs: about large files on GitHub
- GitHub Docs: how use of GitHub Actions is measured
- Google Account Help: sign in with app passwords
- Gmail Help: limits for sending and getting mail

Secondary sources (law-firm notes and financial press) were used where the primary circular was not directly readable; counsel should confirm each point against the SEBI circulars before launch.